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Can LLMs Beat BERT in Biomedical Information Extraction?

Evaluating Prompting and Fine-Tuning Strategies for NER and Classification

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Abstract

The rapidly growing number of biomedical publications has made manual evidence synthesis increasingly challenging. This problem is addressed by the PsyNamic pipeline, which automatically extracts key information from clinical study abstracts and presents it in a living systematic review. For this purpose, BERT-based models were trained to perform 18 mostly multilabel classification tasks and one named entity recognition (NER) task. This thesis investigates whether decoder-based large language models (LLMs) can outperform the BERT baseline, exploring prompting and fine-tuning strategies. 11 models, including proprietary and open-source models as well as general-domain and medical-domain variants, are evaluated in zero-shot and few-shot settings using 1-, 3-, and 5-shot examples. An 8B Llama model is further instruction fine-tuned, and specifically for NER, label-supervised fine-tuned using parameter-efficient methods. The results show that in zero- and few-shot settings, BERT outperforms all LLMs in both classification and NER. Instruction fine-tuning enables an 8B Llama model to achieve performance comparable to BERT in classification and even surpass large proprietary models. For NER using LLMs, label-supervised fine-tuning yields the best results, outperforming in-context learning and instruction fine-tuning, but a performance gap to BERT remains. Across all experiments, fine-tuning proves to be a more reliable predictor of performance than either model size or domain adaptation. Overall, LLMs do not consistently outperform BERT on the PsyNamic dataset. They can match BERT in classification with fine-tuning but remain inferior for NER.

Zusammenfassung

Die wachsende Zahl biomedizinischer Publikationen erschwert die manuelle Aufbereitung von Forschungsergebnissen zunehmend. Die PsyNamic-Pipeline adressiert dieses Problem, indem sie automatisch wichtige Informationen aus Abstracts klinischer Studien extrahiert und in einem Living Systematic Review zusammenführt. Zu diesem Zweck wurden BERT-basierte Modelle trainiert, um 18, überwiegend Multilabel-, Klassifizierungsaufgaben sowie eine Named-Entity-Recognition-Aufgabe (NER) zu lösen. Die vorliegende Arbeit untersucht, ob Decoder-basierte große Sprachmodelle (LLMs) die BERT-Baseline übertreffen können, und untersucht dafür verschiedene Prompting- und Fine-Tuning-Strategien. Insgesamt wurden 11 Modelle, von proprietären bis zu Open-Source-Modellen, sowohl auch allgemeine und medizinische, in Zero-Shot- und Few-Shot Settings mit 1-, 3- und 5-Shot-Beispielen evaluiert. Ein 8B-Llama-Modell wird zusätzlich mittels Instruction Fine-Tuning und speziell für NER, mittels Label-Supervised Fine-Tuning unter der Verwendung parameter-effizienter Methoden optimiert. Die Ergebnisse zeigen: In Zero- und Few-Shot-Setting übertrifft BERT alle LLMs sowohl bei Klassifizierung als auch bei NER. Durch Instruction Fine-Tuning erreicht ein 8B-Llama-Modell bei Klassifizierungsaufgaben eine mit BERT vergleichbare Leistung und übertrifft sogar grosse proprietäre Modelle. Für NER mittels LLMs liefert Label-Supervised Fine-Tuning die besten Ergebnisse und schlägt sowohl In-Context Learning wie auch Instruction Fine-Tuning; es liegt jedoch unter der Leistung von BERT. In allen Experimenten erweist sich Fine-Tuning als zuverlässigerer Indikator für die Leistung als die Modellgröße oder die Domänenanpassung. Zusammengefasst übertreffen LLMs BERT auf dem PsyNamic-Datensatz nicht: Mit Fine-Tuning können sie bei Klassifizierungsaufgaben mit BERT mithalten, bei NER bleiben sie jedoch unterlegen.

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List of Acronyms

BERT Bidirectional Encoder Representations from Transformers. v, vi, viii, 2–18, 20, 24, 36, 37, 39–44, 46, 49–51, 53, 55–57, 92

BiLSTM Bidirectional Long Short-Term Memory. 5

BIO Begin-Inside-Out. vi, 17, 18, 20, 29, 32, 33, 35, 40, 55

CLM Causal Language Modeling. 5, 7–10, 32

CNN Convolutional Neural Network. 4

CRF Conditional Random Field. 5

DAPT Domain-Adaptive Pretraining. 14, 15

GPT Generative Pre-trained Transformer. v, 7, 8, 12

ICL In-context Learning. viii, ix, 7–9, 11, 12, 15, 16, 43, 49–51, 55, 57, 94

IFT Instruction Fine-Tuning. vii–ix, 2, 7–12, 14, 24, 26, 27, 31, 32, 48–51, 53, 56–59, 90, 91, 94

LLM Large Language Model. v–ix, 1–18, 24–27, 29, 31–33, 35–37, 39–41, 43, 46, 47, 49, 52, 53, 55–59, 93, 94

LoRA Low-Rank Adaptation. v, 10, 11, 31, 32, 55, 90, 91

LST Label-Supervised Fine-Tuning. vii–ix, 2, 10–12, 16, 24, 32, 51, 52, 55–59, 91

MLM Masked Language Modeling. 5

NER Named Entity Recognition. v, vi, viii, ix, 2, 3, 5, 7, 10, 14, 16–21, 23–27, 29–36, 39–41, 46–48, 50–53, 55–59, 79, 80, 92

NLP Natural Language Processing. 2, 4, 5, 7

PEFT Parameter-Efficient Fine-Tuning. 2, 10, 11, 31, 32, 56

QA Question-Answering. 5, 7, 15, 25

QLoRA Quantized Low-Rank Adaptation. v, 10, 11, 31, 55, 90

RLHF Reinforcement Learning from Human Feedback. 7, 8

RNN Recurrent Neural Network. 4

SFT Supervised Fine-Tuning. 8, 9

SOTA State-of-the-Art. 5, 7, 10

1. Introduction

1.1. Motivation & PsyNamic Project Context

The rapid growth of biomedical research and clinical trial publications has created a vast amount of unstructured information, making it increasingly difficult for researchers and practitioners to extract and synthesize relevant knowledge (Jimenez Gutierrez et al., 2022). In this context, Large Language Models (LLMs) offer a promising avenue: by leveraging their natural language understanding, they can potentially extract knowledge present in the research literature that has not yet been systematically organized (Nazi and Peng, 2024).

PsyNamic is a project developed by the STRIDE-lab, the medical data science group of the University of Bern¹, which specifically leverages this potential of language models within a niche in biomedical research: They aim at building a living systematic review of psychedelic therapy in psychiatric disorders, a rapidly growing research area. PsyNamic is designed to automatically retrieve new publications, identify relevant studies, and extract key information such as study characteristics, administered substances and dosages, as well as targeted psychiatric conditions. This information is then interactively presented on a freely available online platform (s. Figure 1.1).

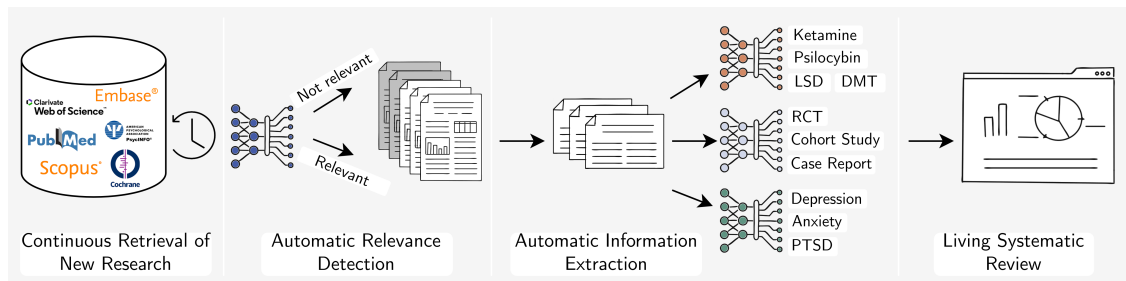


Figure 1.1.: PsyNamic Pipeline

¹ The author of this thesis is an employed member of this group and is also involved with the PsyNamic project as part of this employment. However, the research presented in this thesis was carried out independently, s. Appendix F for details.

To support this pipeline, a dataset consisting of approximately 1,000 clinical study abstracts was manually annotated by medical experts. The extraction of key information about the studies was modeled as 18 mostly multilabel classification tasks and one Named Entity Recognition (NER) task. Bidirectional Encoder Representations from Transformers (BERT)-based models trained on this dataset achieve strong performance, demonstrating the viability of the PsyNamic’s automated information extraction pipeline.

At the same time, recent advances in LLMs raise the question of whether larger, decoder-based models can further improve performance. While LLMs have shown strong zero- and few-shot capabilities in biomedical Natural Language Processing (NLP) (Wu et al., 2025), systematic comparisons with BERT-based models that go beyond evaluating large proprietary models remain scarce. In particular, few studies evaluate both classification and NER tasks under different prompting and fine-tuning strategies in realistic resource settings.

This thesis addresses this gap by systematically evaluating whether LLMs can outperform the established BERT-based baseline on the PsyNamic dataset.

1.2. Research Questions

Thus, the overarching research question that shall be answered in this thesis is the following:

RQ0. Can LLMs outperform BERT-based models in biomedical classification and NER?

Therefore, for both classification and NER, we will address the following research questions:

RQ1. Can LLMs outperform BERT-based models in zero-shot settings?

RQ2. To what extent does few-shot prompting improve LLM performance compared to BERT-based models?

RQ3. Can Instruction Fine-Tuning (IFT) and Label-Supervised Fine-Tuning (LST), implemented with Parameter-Efficient Fine-Tuning (PEFT),

enable LLMs to surpass BERT-based models?

RQ4. What roles do model size and medical domain adaptation play in the performance of LLMs?

1.3. Thesis Structure

The remainder of this thesis is organized as follows: Chapter 2 provides the theoretical background on BERT and LLMs, comparing their architectures, task adaptation paradigms and advantages. Chapter 3 reviews related work on biomedical LLMs, focusing on domain adaptation and comparisons with BERT for classification and NER. Chapter 4 introduces the PsyNamic dataset and defines the working definitions for classification and NER. Chapter 5 outlines the LLM experiments, models, and evaluation methodology. Chapter 6 presents the results and discussion of the experiments. Finally, Chapter 7 summarizes the contributions, answers the research questions, and discusses future work and limitations.

The Appendix contains supplementary tables, figures, and details on the dataset and LLM experiments, as well as the declaration of contribution and notes on LLM use. They are referenced in the relevant sections of the thesis.

The code repository for this thesis is available on GitHub:

github.com/vera-bernhard/PsyNamic-BERTvsLLMs

2. Background: BERT vs. LLMs

In this chapter, we introduce BERT and decoder-based LLMs, describe their architectures and task adaptation methods, and compare their advantages and limitations for NLP tasks.

2.1. Encoder-based Language Models

2.1.1. The Transformer and the Pre-Training + Fine-tuning Paradigm

Whereas research on how to model language can date as far back as the 1950s (Minaee et al., 2025), the current success can be traced back to the emergence of the *Transformer* architecture (Vaswani et al., 2017). Earlier neural architectures such as Recurrent Neural Network (RNN) or Convolutional Neural Network (CNN) processed tokens sequentially, resulting in slow model training and also in the failure to capture long-distance dependencies (Sajjadi Mohammadabadi et al., 2025). In contrast, the transformer processes all tokens simultaneously by replacing recurrence or convolution with the so-called self-attention, which computes the contextual relationship between all pairs of tokens in parallel. This means that, on the one hand, long dependencies can be easily modeled, and on the other hand, computation can be fully parallelized, enabling efficient large-scale training on modern GPU or TPU hardware (Vaswani et al., 2017). The original transformer was designed for sequence-to-sequence tasks such as machine translation, and hence the architecture included an encoder and decoder module. The encoder transforms the input into hidden representations using standard self-attention. The decoder is in charge of generating the output sequence token by token based on the previously generated tokens and the encoder output. As a consequence, it relies on two variants of self-attention: masked self-attention, which allows each token to attend only to the preceding tokens, and cross-attention, which allows each token in the decoder to attend to the hidden representation of the encoder.

Up until the advent of the transformer, solving an NLP task typically involved designing a task-specific architecture and training it on task-specific labeled data (Minaee et al., 2025). For example, Lample et al. (2016) proposed a former State-of-the-Art (SOTA) model for NER that combined a Bidirectional Long Short-Term Memory (BiLSTM) that captures the input text and a Conditional Random Field (CRF) layer that predicted the most likely sequence of entity labels. The efficient computation enabled by the transformer popularized a new paradigm in which a model for general language representation and understanding is trained on larger volumes of unlabeled text data. This so-called pre-trained model is then fine-tuned on a smaller task-specific labeled dataset. Before fine-tuning, the pre-trained model is task-agnostic, meaning that it can be the starting point to address many different NLP tasks (Radford et al., 2018; Sajjadi Mohammadabadi et al., 2025). This *Pre-training + Fine-tuning* paradigm offers key advantages: it eliminates the need to design task-specific architectures from scratch and allows a single model to adapt to many tasks efficiently, as linguistic knowledge is already captured in the pre-trained weights, and thus reducing the amount of labeled data and training required (Devlin et al., 2019; Radford et al., 2018).

2.1.2. BERT Model

One of the most influential language models following the pre-train + fine-tune paradigm is BERT, introduced in 2018 (Devlin et al., 2019), which is based solely on the transformer encoder. Devlin et al. (2019) depart from the traditional Causal Language Modeling (CLM) objective, in which a model predicts the next token given the previous tokens, and instead introduce Masked Language Modeling (MLM) as a pre-training objective: 15% of input tokens are randomly masked, and the model learns to predict them. This allows BERT to leverage bidirectional context, meaning each token is predicted using information from both its left and right neighboring tokens via self-attention. Combined with its second objective, Next Sentence Prediction, where the model learns to predict whether a given sentence follows another, BERT learns both rich token-level and sentence-level semantic understanding. The large, unlabelled pre-training corpus consists of the BooksCorpus (800M words) and English Wikipedia (2,500M words).

To apply BERT to a specific task, a small output layer is added and all parameters are jointly fine-tuned using a comparably small amount of labeled data from the task (s. Section A in Figure 2.1). With this approach, Devlin et al. (2019) achieved SOTA performance on 11 NLP tasks, including sentiment analysis, NER, extractive Question-Answering (QA), and several text classification, outperform-

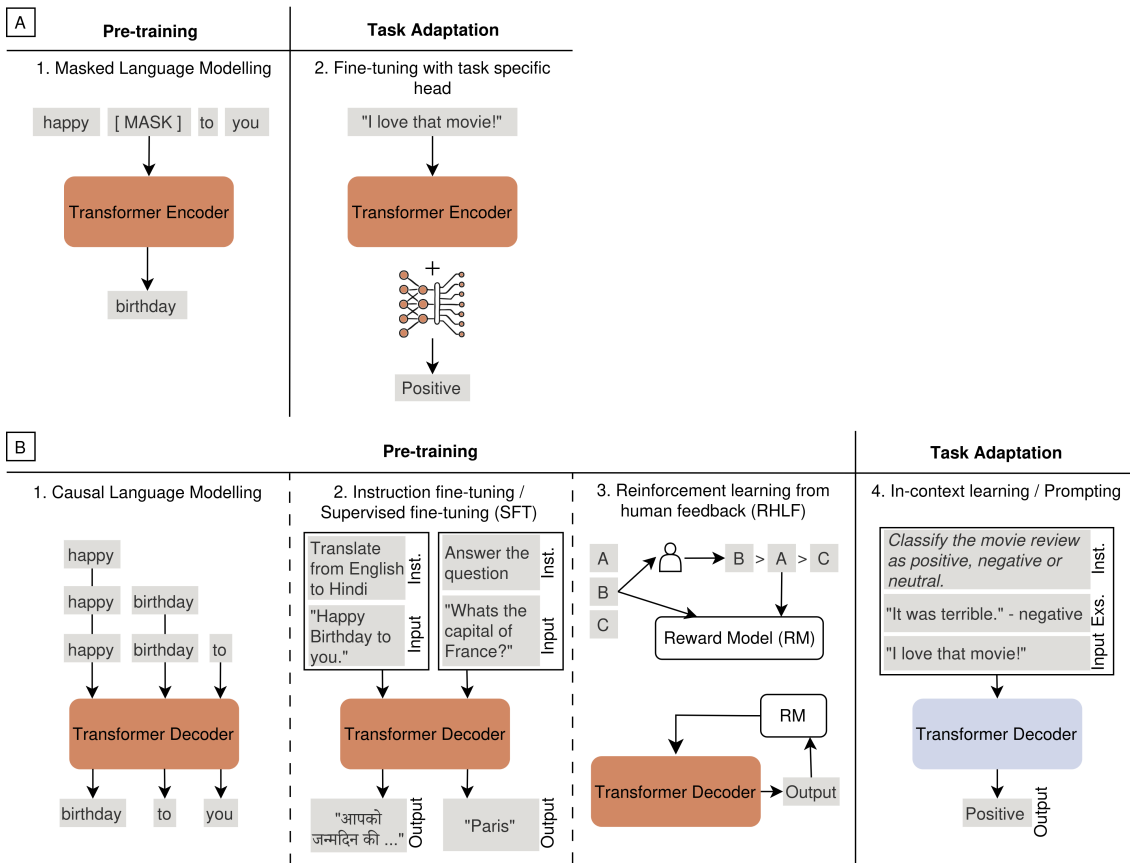


Figure 2.1.: Comparing BERT training and task adaptation in **A** with pre-training and task adaption of recent LLM such as GPT-4 or LLama3 in **B**, red marking weights that are being updated and blue marking frozen weights

ing task-specific architectures. As BERT models are well known for their strong performance in text classification and NER tasks (Devlin et al., 2019), as well as their manageable number of tunable parameters (a few hundred million), they were chosen as the baseline models for this thesis.

2.2. Decoder-Based LLMs

2.2.1. From Generative Pre-trained Transformer (GPT)-1 to modern LLMs

In parallel to research on encoder-only architectures such as BERT, another line of work explored decoder-only transformer architectures for solving NLP tasks. Like BERT, the first GPT is also pretrained in an unsupervised fashion using the more traditional CLM, predicting the next token based on the previously predicted token and applying masked self-attention (s. Section 2.1.1) (Radford et al., 2018). Whereas GPT-1 still relied on the pre-training + fine-tuning paradigm to adapt to a downstream task, the successor model GPT-2 discovered that simply scaling up the model and the training data by approximately a factor of 10 leads to strong, yet not SOTA, task performance without additional fine-tuning (Radford et al., 2019). Thus, Radford et al. (2019) introduce the concept of *Zero-shot* prediction, where the task input, such as the text and question from a QA dataset, is directly fed into the model. The prediction relies entirely on the knowledge acquired during pre-training, and the model weights are not updated. Building on this insight, GPT-3 further expanded the model and data scale by two orders of magnitude, reaching 175B parameters (Brown et al., 2020), while retaining the architecture and pre-training via CLM identical to GPT-2. Apart from its novel size, GPT-3 also systematically uses so-called In-context Learning (ICL) for the first time, where k examples of the task and the expected predictions are prepended to the input. With this additional context and without any changes to its weights, GPT-3 achieves strong performance on several downstream tasks, such as QA and translation, in some cases even surpassing the SOTA results of fine-tuned models.

To further improve the ability of language models to generalize to any unseen task, two additional pre-training stages after CLM were developed: IFT and Reinforcement Learning from Human Feedback (RLHF) (Sajjadi Mohammadabadi et al., 2025). The former entails collecting a corpus of instruction and response pairs covering various tasks (s. Section B 2. in 2.1 for some examples). This corpus is then

used to fine-tune the model, also using a CLM objective². Models like FLAN (Wei et al., 2022) or T0 (Sanh et al., 2022) demonstrated that this approach improves zero-shot performance on unseen tasks and can even compensate for smaller model sizes. An issue that remained is that language models can still produce “outputs that are untruthful, toxic, or simply not helpful to the user” (Ouyang et al., 2022). To address this issue, the RLHF process aligns a model with human preferences: Firstly, different model outputs are ranked by humans according to criteria such as helpfulness, harmlessness, or truthfulness, these rankings are then used to train a reward model. Subsequently, the language model is improved using a reinforcement learning algorithm that uses the reward model to guide it toward human-aligned outputs (Ouyang et al., 2022; Sajjadi Mohammadabadi et al., 2025).

In summary, the combination of pre-training on massive text corpora via CLM, the discovery of the zero-shot and ICL abilities of large transformer decoders, and the successive improvements through IFT and RLHF was the technical foundation that enabled the release of ChatGPT in 2022 (Minaee et al., 2025; Sajjadi Mohammadabadi et al., 2025). ChatGPT’s successor GPT-4, together with other popular LLMs like Google’s Gemini and Meta’s LLaMA, generally follow a three-step training process: CLM³, followed by IFT and RLHF (s. Section B in Figure 2.1).

2.2.2. Task Adaptation Paradigms

As discussed in the previous section, LLMs, such as the GPT family, have become famous for their ability to solve unseen tasks through text-based interaction without additional fine-tuning (s. Section B in Figure 2.1). However, to achieve consistently high performance on a specific task, techniques to fine-tune the model weights are still popular. The following section outlines different strategies to adapt an LLM to a task, which are also used in this thesis.

2.2.2.1. In-context Learning (ICL): Zero- and Few-Shot Prompting

ICL, as first introduced by Brown et al. (2020) in the GPT-3 paper (s. Section 2.2.1), summarizes all strategies where context for a task is provided in the input text but the weights of the LLM remain unchanged. The additional context might be a detailed description of the task and/or examples that illustrate the desired task.

² IFT is often also called Supervised Fine-Tuning (SFT), as instruction-response pairs are considered labeled data.

³ Also called *Self-supervised Training*, as no labeled data is provided.

When no examples are provided, the approach is referred to as zero-shot prediction; when a single example is included, 1-shot prediction; and when multiple examples are provided, it is referred to as few-shot prediction.

As a general rule, performance tends to improve with an increasing number of examples (Brown et al., 2020). However, a substantial performance gain can already be observed with a small number, such as 3-5, of examples (Brown et al., 2020; Ramesh et al., 2025; Wu et al., 2025). Since the size of the input naturally increases with the number of examples, the context window of an LLM is a limiting factor (Sajjadi Mohammadabadi et al., 2025). In this thesis, we explore 1-shot prompting with models that have small context windows, and 1-, 3-, and 5-shot prompting with all others.

Traditionally, the examples are sourced from the validation split of a dataset, an additional subset distinct from the test set, and parameters such as the number of examples or the exact prompt phrasing are typically optimized on this split. Perez et al. (2021) criticize this approach, as a validation split in a real-life scenario, where one relies on few-shot prompting instead of fine-tuning, is often not available. Beyond the source of the examples, one can also differentiate whether (1) a static set of examples representative of a task is used, or (2) the examples are dynamically selected for each test instance (Liu et al., 2022; Purohit et al., 2025). A popular approach for (2) is to select the k most similar examples to the test instance based on embeddings (Liu et al., 2022; Chen et al., 2025). As option (2) introduces additional complexity (e.g., selecting an embedding model or similarity metrics) and its effectiveness is highly task- and model-dependent (Chen et al., 2025), we only adopt option (1) in this thesis (s. Section 5.2.4).

2.2.2.2. Instruction Fine-Tuning (IFT)

As described in Section 2.2.1, IFT is a crucial step during pre-training of LLMs to make the model follow natural language instruction and thus, enabling zero-shot or few-shot prediction on an unseen task (Wei et al., 2022; Ouyang et al., 2022). If the zero-shot/few-shot performance is insufficient, it is common to apply further IFT on the specific task. In literature, this stage is often called SFT and, in contrast to ICL, the model weights are updated. An IFT dataset consists of instruction-response pairs, where the instruction describes the task and provides the input and the response is the desired output. The model is fine-tuned with a CLM objective, meaning it learns to predict each token of the desired output, the response, based on the previous tokens as well as the instruction. It is common

to compute the loss only on the tokens of the response since producing the correct output in the correct format is more important than modeling the exact phrasing of the instruction (HuggingFace, 2025). In this thesis, this method is explored with one fine-tuned model (s. Section 5.2.5).

2.2.2.3. Label-Supervised Fine-Tuning (LST)

Li et al. (2023) propose an alternative fine-tuning method particularly for sequence and token classification that closely resembles how BERT models are fine-tuned: A task-specific output layer, consisting of a fully connected feed-forward layer followed by a softmax layer that maps the model outputs to the set of possible labels, is added on top of the LLM. In what Li et al. (2023) calls a LST method, the LLM encodes the input text into latent representations, which are then passed through the output layer. Instead of a CLM objective, the model is then directly trained on the labeled data using cross-entropy loss. This method outperforms significantly larger LLMs and rivals BERT in text classification. Additionally, Li et al. (2023) criticize masked self-attention in LLMs for classification tasks, as it prevents each token from attending to its right-hand context (s. Section 2.1.1). They therefore replace masked-head attention with self-attention, which is able to capture all contextual dependencies. This unmasking is particularly successful for NER surpassing SOTA BERT models on selected datasets. Therefore, this method is explored for NER in this thesis (s. Section 5.2.6).

2.2.2.4. Parameter-Efficient Fine-Tuning (PEFT): Low-Rank Adaptation (LoRA) & Quantized Low-Rank Adaptation (QLoRA)

In the previously presented methods, IFT and LST, in theory all model parameters are updated during training. However, with model sizes reaching billions of parameters, computational resources become a challenge. PEFT methods such as LoRA or QLoRA address this challenge by freezing the pre-trained model weights and training only a small set of additional parameters. LoRA injects low-rank matrices into the attention layers of an LLM. During fine-tuning, only these matrices are updated while the original weights remain unchanged (Hu et al., 2021). Hu et al. (2021) show that LoRA matches the performance of full fine-tuning while drastically reducing the number of trainable parameters by a factor of 10,000 in the case of GPT-3. QLoRA (Quantized LoRA) extends LoRA by first quantizing the pre-trained model weights, which means representing them with lower-precision numbers (e.g., 4-bit instead of 16-bit), before applying LoRA (Dettmers et al., 2023). Quantization re-

duces memory usage, which allows fine-tuning large LLMs on a single GPU. As the resources for this thesis are limited, we apply QLoRA for the IFT experiment (s. Section 5.2.5) and, following Li et al. (2023), LoRA for the LST experiment (s. Section 5.2.6).

2.3. Advantages and Disadvantages of BERT and LLMs

In the previous sections, we discussed in detail how BERT and LLMs are pre-trained and fine-tuned for downstream tasks. Here, we summarize their respective advantages and disadvantages in practical use. We particularly focus on considerations relevant to the PsyNamic use case in this thesis.

Training Data Requirements BERT models require annotated datasets for fine-tuning, which are expensive and time-consuming to produce. In contrast, with ICL, LLMs can potentially perform tasks without any labeled data or with only a handful of examples. This makes them particularly useful in low-resource scenarios or in domains where expert annotators are hard to find or expensive (Jimenez Gutierrez et al., 2022; Keraghel et al., 2024; Nagar et al., 2025). For ICL to be successful, the prompts have to be carefully engineered, which often involves manual effort (Keraghel et al., 2024; Sajjadi Mohammadabadi et al., 2025). If an LLM is fine-tuned, e.g., with IFT or LST, it still requires labeled data. Depending on the task and the desired performance quality, the amount needed can be comparable to that needed for BERT.

Resource Requirements Compared to modern LLMs, BERT models are lightweight, with only approximately 100 million trainable parameters (Devlin et al., 2019). This allows fine-tuning to be performed on modest GPU hardware or, if time is not a critical factor, even on CPU-only systems. In contrast, LLMs demand significant computational resources, not only for fine-tuning but also for inference after fine-tuning or for ICL without additional fine-tuning (Keraghel et al., 2024; Sajjadi Mohammadabadi et al., 2025). Although PEFT techniques can mitigate some of the computational demands, LLMs remain significantly more resource-intensive compared to BERT. Apart from computational power, LLMs also need large storage space and exhibit slower inference times (Keloth et al., 2024). Hence, deploying LLMs in a pipeline, such as the project context of this thesis, poses greater chal-

allenges than integrating a BERT model. Instead of hosting a LLMs locally or in the cloud, one can access LLMs via commercial APIs (e.g., OpenAI’s GPT models). The resource requirements then shift from computational to monetary ones (Marks and Haupt, 2023).

Multitasking Ability BERT is typically trained for a single task only, and thus having x tasks usually means x fine-tuned BERT models. Furthermore, BERT cannot easily handle different task types (such as token and sequence classification) at the same time. On the other hand, a single LLM can perform different tasks through ICL. Also, IFT allows training the LLM on a mixed dataset containing several tasks (Keloth et al., 2024). This innate multitasking ability of LLMs makes them attractive for projects like PsyNamic, where several different tasks need to be addressed.

Reliability and Output Quality BERT models output probabilities over a predefined set of labels. This means that BERT predictions are always interpretable and valid. By adjusting the probability thresholds, it is even possible to exercise control over whether the model focuses on recall or precision. In contrast, LLMs may not match this level of reliability and controllability: Especially when used without fine-tuning, their output can be inconsistent, requiring hand-crafted postprocessing, or may include hallucinations (Ashok and Lipton, 2023; Wang et al., 2025). Especially, the latter can cause problems when an LLM is part of a pipeline, such as PsyNamic, which requires consistent output. Also, the outputs of LLMs can generally vary as LLMs are not fully deterministic due to factors like how requests are distributed on the compute resources ⁴ (He and Lab, 2025). To improve the consistency of LLMs, one can apply IFT or, for classification tasks, LST, which adapts the LLM to produce a probability distribution over a predefined label set, similar to BERT (Li et al., 2023).

Adaptability Due to their task-specific fine-tuning, BERT models need to be fully retrained whenever a task definition or label set changes (Keloth et al., 2024). The same applies to an LLM trained with LST, as it is also trained on a fixed label mapping. If one relies on ICL with LLMs only, it can, in principle, be easily adapted to label and task changes; the definitions and task descriptions in the prompt can be reformulated (Keraghel et al., 2024). However, the effectiveness of ICL can differ depending on the task and prompt formulation and might still require some IFT,

⁴ Even though the underlying computations are theoretically deterministic.

which is significantly more resource-intensive than retraining a BERT model. Considering that the PsyNamic platform should remain a relevant information source in the future, adaptability is important to consider.

Input Data BERT models have a fixed input size of 512 tokens, which requires truncating or splitting longer texts (Devlin et al., 2019). LLMs generally support much larger context windows, which enables processing entire documents. However, longer input comes at a cost of increased memory and computation requirements (Jimenez Gutierrez et al., 2022). Apart from input length, the sensitivity of the data can also limit LLMs usage: When an LLM is hosted on a cloud or provided through an API, sending sensitive information may pose a privacy or security risk (Marks and Haupt, 2023). As in this thesis, the data consist of scientific abstracts. Hence, neither input length nor data privacy are specifically taken into consideration.

3. Related Work on (L)LMs for Biomedical Text Classification and NER

In this chapter, we review related work on biomedical LLMs as well as prior comparisons between BERT and LLMs in biomedical text classification and NER.

3.1. Biomedical (L)LMs and Domain Adaptation

The original BERT model was trained on general-domain corpora, including Wikipedia and BooksCorpus (Devlin et al., 2019). Although effective on a wide range of tasks, the model transferred poorly to expert domains like biomedical text (Lee et al., 2019). This motivated so-called Domain-Adaptive Pretraining (DAPT) methods, where the weights of a generally pre-trained language model are used as initialization and then further pre-trained on domain-specific data (Jeong et al., 2024). BioBERT (Lee et al., 2019) and ClinicalBERT (Alsentzer et al., 2019) are canonical examples: the former continues pre-training BERT on PubMed abstracts and PubMed Central full-text articles, and the latter on clinical notes. Both BioBERT and ClinicalBERT demonstrate substantial improvements over general BERT in tasks like biomedical QA or relation extraction. Alongside DAPT, models like PubMedBERT (Gu et al., 2021) have explored training biomedical models from scratch. This has the advantage that the tokenizer vocabulary is also tailored to the biomedical domain, which leads to performance gains in tasks like NER (Gu et al., 2021; Jimenez Gutierrez et al., 2022).

With LLMs, medical domain adaptation is also a popular research topic. As training from scratch is much more resource-intensive than BERT, most open-source medical LLMs are created by continued pre-training and/or IFT of general-purpose LLMs on biomedical corpora (Wang et al., 2024). Examples include Meditron (Chen et al., 2023) and Me-LLaMA (Xie et al., 2025), both based on the Llama-2 model (Touvron

et al., 2023). Although each paper introducing a medically adapted LLM claims improvements over its general-domain backbone, the effectiveness of DAPT remains debated. For example, a systematic comparison by Jeong et al. (2024) of seven open-source medical LLMs and their corresponding base models across a variety of medical QA benchmarks found that domain-adapted LLMs do not consistently outperform their general-domain counterparts.

Like in the study by Jeong et al. (2024), the evaluation of medical LLMs is most often conducted using biomedical question-answering datasets, including medical licensing examination questions (Chen et al., 2024). However, Wu et al. (2025) criticizes this approach, as QA datasets alone do not reflect the wide range of biomedical tasks and are thus insufficient to judge an LLM’s medical knowledge. Instead, Wu et al. (2025) proposes the BRIDGE benchmark, which consists of a diverse set of 87 biomedical tasks and represents the largest benchmark for evaluating LLMs in the medical domain. They initially tested 52 LLMs⁵ with in-context methods only and found that specialized medical models did not consistently outperform general-purpose ones.

3.2. LLMs vs. BERT for Classification

For biomedical classification tasks, the literature that compares BERT and LLM often focuses on large proprietary models such as GPT-3.5 or GPT-4 using ICL strategies rather than fine-tuning. Across these studies, fine-tuned BERT-based models outperform proprietary LLMs: For example, Chen et al. (2024) show that fine-tuned BioBERT significantly outperforms GPT-3.5 or GPT-4 in multilabel document classification under zero-shot, 1-shot, and few-shot prompting. Similarly, Emilova Doneva et al. (2025) report that BERT surpasses GPT models by a large margin in multiclass study classification. They evaluated a range of prompting strategies, which did not significantly impact this performance gap.

A smaller number of studies include open-source (medical) models along with proprietary models. However, the conclusion that BERT demonstrates superior performance does not change. Labrak et al. (2024) compare BERT-based models to ChatGPT as well as open-source models such as Flan-T5 UL2, Tk-Instruct, and Alpaca on clinical and biomedical classification tasks. They find that fine-tuned PubMedBERT outperforms all models under both zero-shot and five-shot prompting. Similarly, Chen et al. (2025) evaluate GPT-3.5, GPT-4, LLaMA-2, and its

⁵ New models are continuously being tested, and the results are displayed on a public leaderboard: huggingface.co/spaces/YLab-Open/BRIDGE-Medical-Leaderboard

medical counterpart PMC LLaMA on multilabel biomedical document classification and show that BioBERT achieves the strongest performance in various ICL settings.

Beyond ICL, the conclusions are mixed. Chen et al. (2025) compare instruction fine-tuned Llama-2-13B and PMC Llama-13B with a BioBERT baseline on biomedical multilabel document classification. They conclude that BioBERT outperforms even fine-tuned LLMs. In contrast, evidence from the non-medical domains suggests that fine-tuning newer open-source LLMs can surpass both BERT and proprietary models: Zhao et al. (2025) demonstrate that fully fine-tuned LLaMA3-70B surpasses RoBERTa and GPT-4/GPT-4o across several classification benchmarks. Trust and Minghim (2024) even show that fine-tuning smaller LLMs, like Mistral-7B or Zephyr-7B, can exceed BERT’s performance on several text classification tasks.

3.3. LLMs vs. BERT for NER

In biomedical NER, many studies compare BERT-based models with proprietary LLM models such as GPT-3, GPT-3.5, or GPT-4 using ICL. These works consistently show that fine-tuned biomedical BERT variants outperform GPT-based models with zero-shot and few-shot prompting by a large margin (Jimenez Gutierrez et al., 2022; Hu et al., 2024; Keloth et al., 2024; Mullick et al., 2024; Chen et al., 2025). Although prompt engineering and different few-shot selection strategies improve the performance of GPT, they do not close the gap to BERT (Hu et al., 2024; Lu et al., 2024).

A smaller set of studies also includes general-domain open-source models such as Flan-T5, Tk-Instruct, Alpaca, Llama-2, or medical open-source models like PMC LLaMA or Meditron. These studies similarly find that PubMedBERT or BioClinicalBERT outperform both proprietary and open-source LLMs when evaluated with prompting only (Labrak et al., 2024; Chen et al., 2025; Lu et al., 2024).

Although LLMs perform poorly under ICL, fine-tuning appears more promising: Several studies show instruction fine-tuned LLaMA variants approach BERT performance. However, they typically only match the performance of BERT rather than surpassing it (Keloth et al., 2024; Lu et al., 2024; Chen et al., 2025). Only Obeidat et al. (2025) report small and dataset-dependent performance gains over BERT using an instruction-fine-tuned LLaMA-3.1-8B model. Lastly, in the non-medical domain, LST (s. Section 2.2.2.3) of LLMs such as LLaMA-2-7B has demonstrated consistent improvements over BERT-based models.

4. Dataset

In this chapter, we define the text classification and NER tasks and describe the PsyNamic dataset, including its annotation scheme and descriptive statistics.

4.1. Classification and NER: Task Definitions

Def. Classification Text classification is the task of assigning one or more predefined labels to a given text input. In binary classification, each input is assigned to one of two possible classes. Multiclass classification extends this to more than two classes, where each text is assigned to exactly one class from a set of classes. In contrast, multilabel classification allows each input to be associated with multiple labels simultaneously (Li et al., 2022). In the PsyNamic all three classification task types - binary, multiclass, and multilabel - are present.

Def. Named Entity Recognition NER describes the task of identifying meaningful spans, the named entities, in a given input text and classifying each span into predefined categories, so-called entity types (Keraghel et al., 2024). Entity types are defined based on the specific use case and may encompass general categories such as people and locations, or specialized categories such as dosages, as used in PsyNamic.

Traditionally with BERT models, NER has been cast as a token classification problem. For this, the training data is provided in the so-called Begin-Inside-Out (BIO) format: Each token is annotated with either O, not being part of a named entity, B-X being the start of an entity of entity type X, or I-X, being subsequent tokens belonging to the same entity (Obeidat et al., 2025). This format and problem formulation not only identifies the entity strings but also their exact position and boundaries in the input text, as each input token is explicitly assigned a BIO label (s. (a) in Figure 4.1).

In contrast, with the advent of LLMs, NER is sometimes also modeled as an ex-

traction task, meaning the model outputs entities directly as text and entity type, without predicting token-level labels (s. **(b)** in Figure 4.1). As the generated entities are not aligned to specific tokens in the input, the location of each entity within the original text is no longer explicitly encoded (del Moral-González et al., 2025; Obeidat et al., 2025; Wu et al., 2025).

Which problem formulation is appropriate depends on the project context and the structure of the available data. In the case of PsyNamic, the dataset is provided in BIO format. Moreover, because we compare LLMs to a BERT baseline, which inherently operates in the token classification problem definition **(a)**, we generally also adopt a location preserving approach with the LLMs (for details, s. Section 5.2.3). However, we also report an additional evaluation metric that ignores token-level alignment to approximate the extraction task definition **(b)** (s. Section 5.4.2.2).

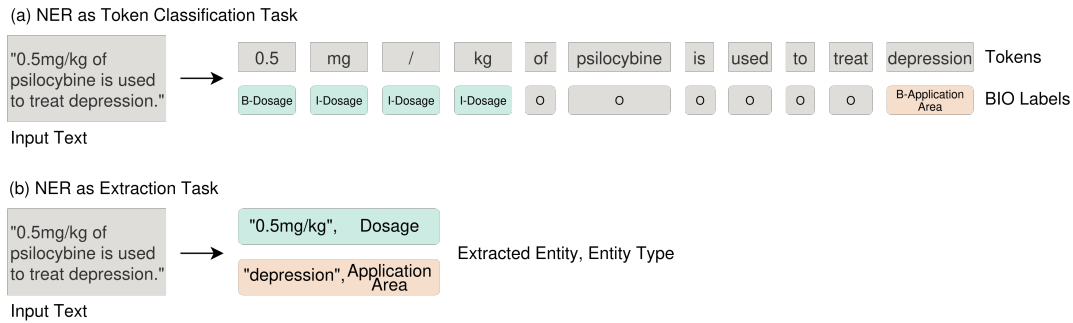


Figure 4.1.: Two different definitions of NER task **(a)** each token is assigned a BIO tag and positional information is preserved vs. **(b)** entity string is extracted along with entity type.

4.2. PsyNamic Dataset

Data Collection The PsyNamic dataset consists of titles and abstracts from clinical studies involving psychedelic substances, as full-text versions of publications are often not freely available. In January 2024, a comprehensive literature search was conducted across multiple (biomedical) platforms, including Embase, Medline via PubMed, Cochrane Trials (CENTRAL), PsycInfo, Scopus, and Web of Science, to identify clinical studies involving psychedelic substances. Manual screening of the titles and abstracts was then performed to exclude studies such as animal-only studies or those investigating non-classical psychedelic substances like THC or heroin. From the remaining 3,314 relevant studies, 1,000 studies were randomly chosen to be annotated. With this elaborate screening method, the selection was intended to best reflect the current state of research. In collaboration with mental

Comparison of Rapid Antidepressant and Antisuicidal Effects of Intramuscular Ketamine, Oral Ketamine, and Electroconvulsive Therapy in Patients With Major Depressive Disorder : A Pilot Study

PURPOSE/BACKGROUND: This study was devised to compare the antidepressant and antisuicidal effects of oral and intramuscular (IM) ketamine versus electroconvulsive therapy (ECT). **METHODS/PROCEDURES:** In our pilot study, 45 patients with major depressive disorder (based on Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, criteria) in the age range of 18 to 70 years who were determined suitable candidates for ECT got randomly divided into 3 equal groups. Each group received one of these treatment modalities: 0.5 mg/kg of IM ketamine; 1 mg/kg of oral ketamine; and ECT in 6 to 9 sessions during 3 weeks. Depression and suicidal ideation scores were recorded using the Hamilton Depression Rating Scale and the Beck Scale for Suicidal Ideation, respectively, at baseline, 24 hours, 1 week, 2 weeks, and 3 weeks within the intervention. The measurements were repeated 1 week and 1 month after the end of the intervention as well. Vital signs and adverse effects were noted. Finally, satisfaction levels of patients for each method were recorded and compared between groups. **FINDINGS/RESULTS:** The Hamilton Depression Rating Scale and the Beck Scale for Suicidal Ideation scores significantly improved in all groups compared with baseline with no significant differences between the 3 groups. The adverse effects for ketamine-consuming groups such as dissociative symptoms were brief and transient, whereas memory loss for the ECT group remained up to 1 month in some patients. Ketamine-receiving groups preferred it more than ECT. **IMPLICATIONS/CONCLUSIONS:** Oral and IM ketamine probably have equal antidepressant in addition to more antisuicidal effects compared with ECT but had less cognitive adverse effects and higher preference by patients. Thereby, ketamine can be an alternative method in the treatment of patients with severe and/or suicidal MDD.

Classification Labels

Relevance:	relevant
Study Type:	Randomized-controlled trial (RCT)
Study Purpose:	Efficacy endpoints, Safety endpoints
Study Control:	Active control
Data Type:	Longitudinal short
Data Collection:	Unknown
Number of Participants:	41–60
Sex of Participants:	Both sexes
Age of Participants:	Adult (≥ 18 years)
Substances:	Ketamine
Application Form:	Oral, Other
Regimen:	Multiple Applications
Setting:	Clinical
Substance Naivety:	Unknown
Condition:	Psychiatric condition, Depression
Outcome:	Suicide, Physiological Functions, Mental Functions
Clinical Trial Phase:	Unknown
Study Conclusion:	Positive

NER Labels

Application Area
Dosage

Figure 4.2.: Example of a sample (title + abstract) annotated with classification and NER labels.

health professionals and researchers⁶, the key information to be extracted from the studies was identified and annotation guidelines were developed. Annotation was conducted in two rounds, including calculation of inter-annotator agreement and resolution sessions. When the agreement scores fell below expectations (e.g., Cohen’s kappa <0.6), the annotation guidelines were refined. In the final dataset, all selected titles and abstracts were double-annotated by medical professionals to ensure high-quality annotations.

Tasks The annotation guidelines comprise 18 classification tasks (s. Table 4.1), in which labels are selected from a set of predefined options. In addition, there is 1 NER task with two entity types (s. Table 4.2), for which relevant tokens are marked within each study, consisting of concatenated title and abstracts. The first classification task is a binary classification task, resulting from the manual screening that determined whether a study is relevant or not. The remaining 17 classification tasks can be thematically grouped into study characteristics, including study type, control type, and participant information; substance-related information, such as application form and dosage regimen; and clinical measures, such as the condition being treated and the reported outcomes (s. thematic coloring in Table 4.1 and s. Appendix A.2 for a more detailed description of the classification tasks). From a technical perspective, the tasks can also be grouped into 6 multilabel classification tasks and 11 multiclass, multilabel classification tasks (s. subtables in Table 4.1). Lastly, two entity types annotated at the token level and provided in BIO format are the dosage of a psychedelic and the application area, the condition or disease studied. An example of a study annotated with all classification and the NER task can be found in Figure 4.2.

4.3. Descriptive Statistics

Sample Length Across all classification and NER tasks, one sample of the PsyNamic dataset corresponds to the concatenated title and abstract of a clinical study. On average, one sample is 358 tokens long⁷, with 90% of all samples being shorter than 488 tokens (s. Appendix A.1 for more details). This means that the vast majority of samples fit within BERT’s 512-token context window, while a small fraction may require truncation.

⁶ and thus potential future users of the PsyNamic platform.

⁷ The tokenization is provided by the annotation tool Prodigy that was used for annotation.

Datasets Size & Splits The largest dataset is the one for the binary relevance classification task, comprising 5,643 samples. These samples result from the manual screening process which determined whether a study met the inclusion criteria (i.e., is relevant for PsyNamic) or not (i.e., is irrelevant for PsyNamic). All other classification tasks, as well as the NER task, are based on the sampled subset of the relevant portion. For the classification tasks, samples with empty predictions were excluded, resulting in slightly varying dataset sizes of approximately 960 samples. These were split in a stratified fashion, allocating 80% for training and 10% each for validation and testing, ensuring that the label distribution was preserved in all splits. As stratified splitting is not easily defined for NER, samples were randomly allocated to the 80/10/10 split. The sizes per task and split are reported in Table 4.1 and Table 4.2.

Classification Characteristics The label sets for the multiclass task range from 4 up to 16 predefined labels. Even though 11 of the 17 classification tasks are multilabel, one sample is still only assigned 1-2 labels on average. Notably, the tasks *Condition* and *Outcome* are the only tasks that more frequently contain multiple labels per sample (s. Table 4.1). Since the studies were randomly sampled from the broader research corpus rather than carefully selected, class imbalance is a dominant characteristic across almost all tasks. Particularly, labels such as *unknown*, *other*, or *not applicable* often make up a substantial portion, and some minority classes only have 1 to a few instances in the test set (s. Appendix A.3 for more details on the label distribution). The class imbalance and the variety of tasks render the PsyNamic dataset a challenging benchmark, yet one that is close to a real-world scenario where large, carefully engineered datasets are not common.

NER Characteristics The NER task consists of two entity types, namely *Dosage* and *Application Area*. Despite the data splits being random, one can see in Table 4.2 that the splits exhibit similar characteristics and are thus suitable for model training and evaluation: About 2/5 of all samples do not contain any entities. In all others, entities of type *Application Area* are more frequently present than entities of type *Dosage*. The average entity length is short, 2-3 tokens overall, with *Dosage* phrased slightly longer than *Application Area* on average.

Binary Classification						
Task	Labels	Train	Val	Test		
Relevance	{relevant, irrelevant}	4,513	565	565		

Multiclass Classification						
Task	Labels	#L	Tr	V	T	
Study Type	{Randomized-controlled trial (RCT), Cohort study, Real-world study, Study protocol, Systematic review/meta-analysis, Qualitative Study, Case report, Case series, Other}	9	767	97	97	
Data Type	{Cross-sectional, Longitudinal long, Longitudinal short, Not applicable, Unknown}	5	766	96	96	
Data Collection	{Not applicable, Prospective, Retrospective, Unknown}	4	768	96	96	
Number of Participants	{1-20, 21-40, 41-60, 61-80, 81-100, 100-199, 200-499, 500-999, ≥ 1000 , Not applicable, Unknown}	11	768	96	96	
Sex of Participants	{Male, Female, Both sexes, Not applicable, Unknown}	5	768	96	96	
Study Conclusion	{Positive, Negative, Neutral, Mixed, Unknown, Not Applicable}	6	768	95	98	

Multiclass, Multilabel Classification						
Task	Labels	#L	Tr	V	T	#L/s
Study Purpose	{Efficacy endpoints, Safety endpoints, Pharmacokinetics, Mechanism of action, Disease model study, Demographic study}	6	758	92	97	1.50
Study Control	{Placebo, Active control, Pre-post, Other control, Non-controlled, Not applicable}	6	770	93	98	1.02
Age of Participants	{Adult (≥ 18 years), Pediatric (< 18 years), Not applicable, Unknown}	4	768	95	98	1.02
Substances	{Ketamine, S-Ketamine, R-Ketamine, MDMA, LSD, Psilocybin, Psychedelic mushrooms, Ayahuasca, DMT, 5-MeO-DMT, Mescaline, Ibogaine, Salvinorin A, Combination Therapy, Analogue, Unknown}	16	770	94	96	1.35
Application Form	{Oral, Nasal, Intravenous, Smoking, Subcutaneous, Other, Unknown, Not applicable}	8	771	96	94	1.02
Regimen	{Microdosing, Multiple Applications, Single Dose, Unknown}	4	762	95	96	1.05
Setting	{Clinical, Naturalistic, Party setting, Other setting, Unknown, Not applicable, Palliative Setting}	7	767	95	98	1.03
Substance Naivety	{Substance-naïve participants, Substance-non-naïve participants, Unknown, Not applicable}	4	770	94	97	1.11
Condition	{Psychiatric condition, Depression, Anxiety, PTSD, Alcoholism, Other addictions, Anorexia, Alzheimer's disease, Non-Alzheimer dementia, Substance abuse, (Chronic) Pain, Palliative Setting, Recreational Drug Use, Healthy Participants}	14	754	93	96	1.78
Outcome	{The ability to drive a car, Suicide, fMRI, MRI, PET/SPECT, Physiological Functions, EEG, Mental Functions, Psychedelic experience, Soluble biomarker, Surveys, Interviews, Neurotoxicity, Emergency department visits}	14	731	91	94	1.74
Clinical Trial Phase	{Phase 1, Phase 2, Phase 3, Phase 4, Unknown, Not Applicable}	6	769	95	96	1.00

Table 4.1.: Summary of classification tasks. **Labels:** their label sets, **#L:** number of labels, **Tr, V, T:** split sizes (train/val/test), and for multilabel tasks **#L/s:** the average number of labels per sample, row color indicate topic groups: study characteristics, substance, clinical measures.

Entity Type	Description	Train 772			Val 97			Test 97		
		#E/s	len(E)	#E=0	#E/s	len(E)	#E=0	#E/s	len(E)	#E=0
Dosage	Dosage(s) of psychedelic substance(s)	0.79	3.46	459	1.00	3.43	57	0.62	3.52	62
Application Area	Condition/disease studied	1.54	2.19	441	1.57	2.47	50	1.40	2.11	60
Overall entities		2.33	2.62	302	2.57	2.84	36	2.02	2.54	45

Table 4.2.: Summary of the NER task per entity type and over all entities across data splits. **#E/s**: average number of entities per sample, **len(E)**: average entity length in tokens, **#E=0**: number of samples without entities. Row colours indicate topic groups: substance, clinical measures.

5. Methods

This chapter describes our experimental setup for comparing LLMs with BERT, covering model selection, prompt design, and training configurations for zero-shot, few-shot, the IFT and LST experiments. We then describe the postprocessing steps and evaluation metrics used for classification and NER.

5.1. BERT-based Baseline

As a baseline, we fine-tuned six different BERT variants for each classification task and for the NER task⁸. As discussed in Section 3.1, domain-specific pre-training or continual pre-training of BERT models improves the performance in domain-specific tasks (Lee et al., 2019; Beltagy et al., 2019). As our dataset consists of medical research abstracts, BERT models pre-trained on scientific and/or biomedical literature were chosen. These include BioBERT (Lee et al., 2019), BiomedBERT (and its earlier version PubMedBERT), its abstracts-only variant BiomedBERT-abstract (Gu et al., 2021), SciBERT (Beltagy et al., 2019), and BioLinkBERT (Yasunaga et al., 2022). For each classification task and for NER, we then selected the best performing variant according to the evaluation metrics described in Section 5.4, resulting in a total of 19 task-specific fine-tuned BERT baselines.

5.2. LLM Experiments

5.2.1. Model Selection

To obtain representative results, we selected a diverse set of LLMs spanning general-purpose, medical-specialized, open-source, and closed-source models (s. Table 5.1 for an overview). As in previous literature, the usefulness of medical domain-specific

⁸ The fine-tuning of the BERT baseline was mostly conducted as part of professional employment within the PsyNamic project and not this thesis, s. Appendix F. The fine-tuning code can be found here: github.com/Ineichen-Group/PsyNamic/model

Family	Name	Model Size		OS	Me	Rel
		Params (B)	Context			
GPT	GPT-4o-mini	8-30(?)	128K			07-2024
	GPT-4o	200(?)	128K			08-2024
LLaMA	Llama-2-13B -chat	13	4096	x		07-2023
	↳ Me-LLaMA-13B -chat	13	4096	x	x	06-2024
	Llama-2-70B -chat	70	4096	x		07-2023
	↳ Me-LLaMA-70B -chat	70	4096	x	x	06-2024
	Llama-3-8B -Instruct	8	8192	x		04-2024
	↳ Med-LLaMA3-8B	8	8192	x	x	06-2024
Gemma	Gemma-3-27b -it	27	128K	x		03-2025
	↳ MedGemma-27b -text-it	27	128K	x	x	05-2025

Table 5.1.: Overview of the LLMs considered, OS=open-source, Me=medical-domain specific fine-tuning, Rel=release date, arrows representing the relatedness of models. The exact model version/repositories are linked in the model name. For simplicity, we refer to all models using their shortened names, marked in bold. The parameter sizes of the GPT model is not officially released; these numbers are mere speculations (Li, 2025).

fine-tuning has been contested (s. Section 3.1), so we paired each medical model with its general-purpose backbone. Only instruction fine-tuned models were considered to ensure a fair comparison in the prompting-only experiments.

As representatives of proprietary models, we used **GPT-4o** and **GPT-4o-mini** (OpenAI et al., 2024). Their parameter sizes are not officially disclosed, but **GPT-4o** is likely substantially larger than all other models considered, whereas the size of **GPT-4o-mini** may be in a similar range.

Among open-source medical models, we evaluated **Me-LLaMA** in two sizes, 13B and 70B (Xie et al., 2025). These Llama-2-based models are continually pre-trained and instruction-tuned on what Xie et al. (2025) claim to be the largest biomedical dataset available. They report that **Me-LLaMA** models outperform other open-source medical models in zero-shot settings and, after fine-tuning, even large proprietary models such as GPT-4. Unlike many medical LLMs, which are often evaluated on a single QA benchmark (a practice that has drawn skepticism, cf. Section 3.1), the models are assessed across a broad range of tasks, including summarization, NER, and relation extraction. Access to the models required completion of an ethics and data privacy training module⁹. For comparison, we evaluated their corresponding

⁹ Namely the CITI “Data or Specimens Only Research” provided by PhysioNet physionet.org/about/citi-course/

general-purpose base models, Llama-2-13B and Llama-2-70B (Touvron et al., 2023).

We further considered Med-LLaMA3¹⁰, the successor of Me-LLaMA, which is trained on the same biomedical corpus but based on the more recent Llama-3 model (Xie et al., 2025; Dubey et al., 2024). For comparison, we evaluate the corresponding general-purpose Llama-3 model as a non-domain-adapted baseline, and additionally assess Llama-3.1, the latest text-only model in a comparable size range (Dubey et al., 2024).

Finally, to capture the most recent advances, we included the medical open-source model MedGemma (Selligren et al., 2025) and its base model Gemma (Team et al., 2025). As of August 2025, MedGemma is reported to be the highest-performing model under 40B parameters in few-shot settings across various medical tasks on the BRIDGE benchmark¹¹(Wu et al., 2025).

Given the rapidly growing landscape of LLMs, a comprehensive evaluation of all available models was not feasible. We hence focused on the presented 11 models, selected for their strong reported performance and for representing key developments of the past two years. Due to limited computational resources (s. Section 5.2.2), we restricted our experiments to models of modest size, 8-27B parameter, with the exception of GPT-4o¹² and Me-LLaMA-70B.

5.2.2. General LLM Configurations

Reproducibility As touched upon in Section 2.3, LLMs are not fully deterministic. However, to maximize reproducibility, random seeds were set, and during inference, greedy decoding with temperature=0 was used for all LLM experiments.

Compute All experiments were conducted on NVIDIA A100 GPUs with 80GB of memory. For the majority of experiments, a single GPU was sufficient, whereas the largest 70B models required two A100 GPUs. Access to two GPUs at the same time was limited, and thus the few-shot experiments with Llama-2-70B and Me-LLaMA-70B for the NER task had to be omitted (s. Section 7.4).

¹⁰ Med-LLaMA3 is presented as a study extension to Xie et al. (2025) and comes with minimal standalone documentation. We therefore assumed that its continued pre-training and IFT follow the same procedure described for Me-LLaMA in the original paper.

¹¹ Leaderboard available at: huggingface.co/spaces/YLab-Open/BRIDGE-Medical-Leaderboard

¹² This model was accessed via API rather than self-hosted.

System prompt and chat templates For all LLM experiments, we used one fixed system prompt for the classification tasks and one for NER:

For classification: ‘‘You are a helpful medical expert who is helping to classify medical abstracts.’’

For NER: ‘‘You are a helpful medical expert who is helping to extract named entities from medical abstracts.’’

The system prompt provides the LLM with its general role and high-level instructions.

While the system prompt was the same across all models, each model requires the prompt to be wrapped in its specific chat template. The chat template is the model’s expected input format as learned during pre-training or IFT. Using the correct template is crucial for the model to follow instructions and produce valid outputs. For example, the Llama-2 chat template is (AI, 2023):

```
<s>[INST] <<SYS>>
{{ system_prompt }}
<</SYS>>
{{ user_message }} [/INST]
```

Here, `system_prompt` is replaced with the task-specific system prompt, and `user_message` is replaced with the task- and experiment- specific prompt, which includes the input text.

5.2.3. Zero-Shot Experiments

Classification Prompt For the classification tasks, we closely follow the prompting strategy proposed by Chen et al. (2025). The prompt template consists of five components: (1) a concise task description, (2) a description of the input data, (3) task-specific documentation including label definitions, (4) a description of the desired output format, and (5) the input text to be classified. Previous work has shown that providing task context to the data improves performance (Labrak et al., 2024; Chen et al., 2025). Figure 5.1 shows the task-agnostic prompt template, and in Appendix B.1.1, a specific example for the task *Regimen* can be found. To facilitate robust postprocessing, the LLMs were required to return their predictions in JSON format (Zhao et al., 2025). An example of the expected output structure is shown in Figure 5.3.

Classification Prompt Template

TASK

TASK DESCRIPTION

INPUT

The input is the title and abstract text.

DOCUMENTATION

There are NUMBER OF LABELS options. The followings are the options and their definitions:

LABELS+DEFINITIONS

OUTPUT

The output must be exactly one JSON object, and nothing else. Do not include reasoning, the input or multiple JSONs. The JSON should have relevant values for each option:

LABELS

Put value 1 if the option applies to the research paper, 0 if it does not apply.

Please note again that IS MULTILABEL can be selected for each research paper.

EMPTY LABEL JSON

0-shot

EXAMPLES

INPUT: TITLE+ABSTRACT

OUTPUT: LABEL JSON

...

FINAL INPUT TO CLASSIFY

INPUT: TITLE+ABSTRACT

OUTPUT:

few-shot

Figure 5.1.: Classification prompt template. The blue box is added for 0-shot and the orange box for few-shot.

NER Prompt For NER, different prompting strategies have been explored in prior work. Some approaches prompt LLMs to generate only a list of extracted entities (Jimenez Gutierrez et al., 2022; Wu et al., 2025). However, this formulation discards positional information and thus is not suitable for the PsyNamic dataset (s. Section 4.1). Other studies ask LLM to directly generate BIO labels (Obeidat et al., 2025; Lu et al., 2024), which performs poorly when the models are not explicitly fine-tuned for sequence labeling. As an alternative, Wang et al. (2025) propose reformulating NER as a generation task in which the model reproduces the input text while explicitly marking entity occurrences inline. This preserves positional information, reduces hallucinated entities, and improves overall NER performance. Building on the same idea, many works in the biomedical domain adopt inline annotation using HTML `<span>` tags. The entity type is specified via a class attribute (Keloth et al., 2024; Hu et al., 2024; Xie et al., 2025; Chen et al., 2025). As the input text is reproduced, the positions of the entities are explicitly encoded and can be converted to BIO labels.

Following this line of work and analogous to the classification setup, we add additional task context, including entity definitions and annotation guidelines, which have been shown to improve NER performance (Hu et al., 2024). Our final prompt template consists of five components: (1) a concise task description, (2) a description of the desired output format, (3) definitions of the entity types, (4) annotation guidelines for each entity, and (5) the input text, closely following Hu et al. (2024). Figure 5.2 illustrates the NER prompt template, and Figure 5.3 shows an example of the expected model output. In Appendix B.2.1, the full NER prompt can be found.

5.2.4. Few-Shot Experiments

We extend the zero-shot prompt templates to few-shot prompting by adding a small number of input-output examples while keeping the rest of the prompt structure unchanged. This design choice ensures comparability between zero-shot and few-shot settings. Reflecting a realistic scenario where annotated data is scarce, which motivates the use of LLMs (s. Section 2.3), examples are sourced from the test splits. We carefully ensure that no input instance is ever used both as an in-context example and as an input sample. The examples are fixed per task and do not vary across inputs. Thus, we adopt a static example selection strategy in contrast to sample-dependent example retrieval (Section 2.2.2.1). Rather than selecting examples at random, they were manually chosen to approximately reflect the label distribution of each task (s. Appendix A.3). For example, the most frequent label is used in

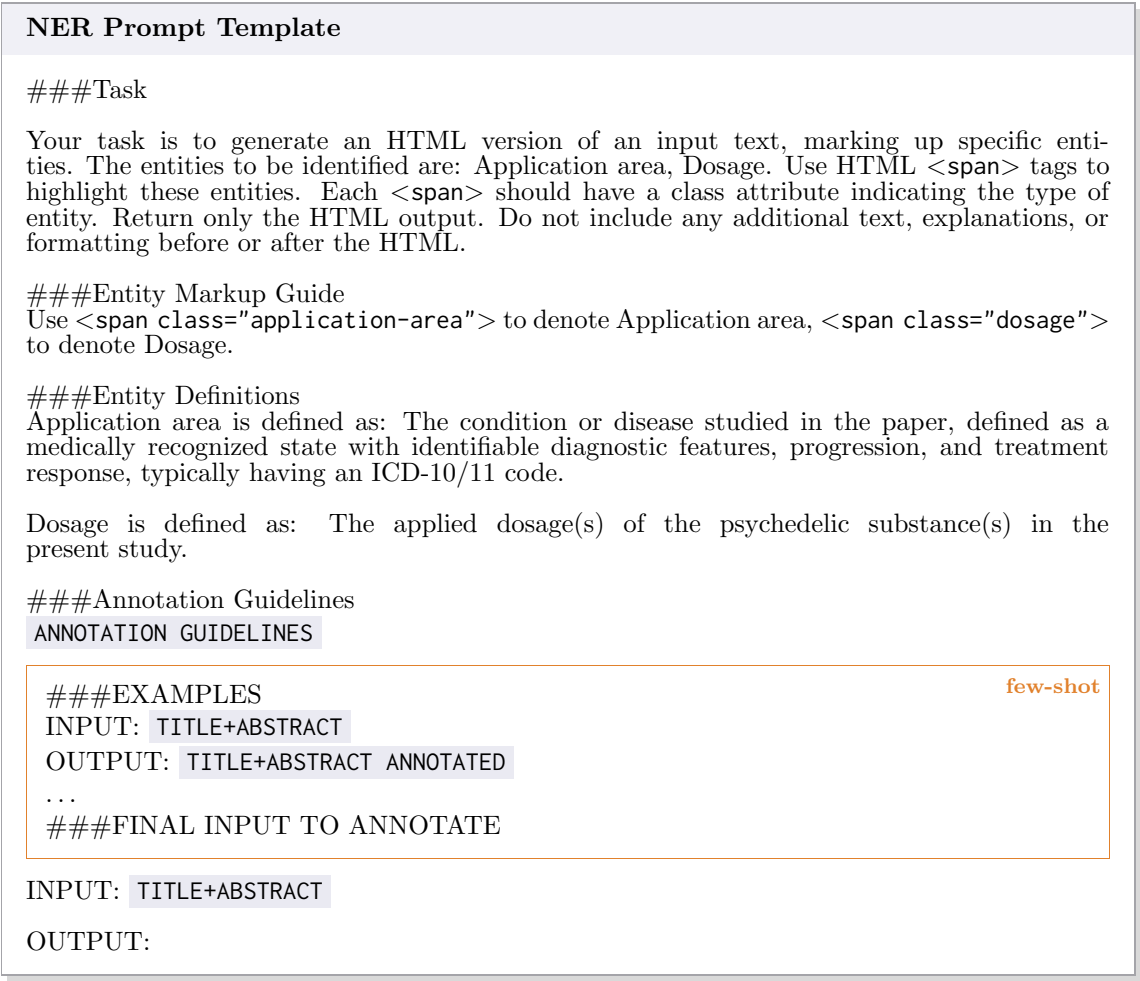


Figure 5.2.: NER prompt template. Orange box is added in few-shot experiments.

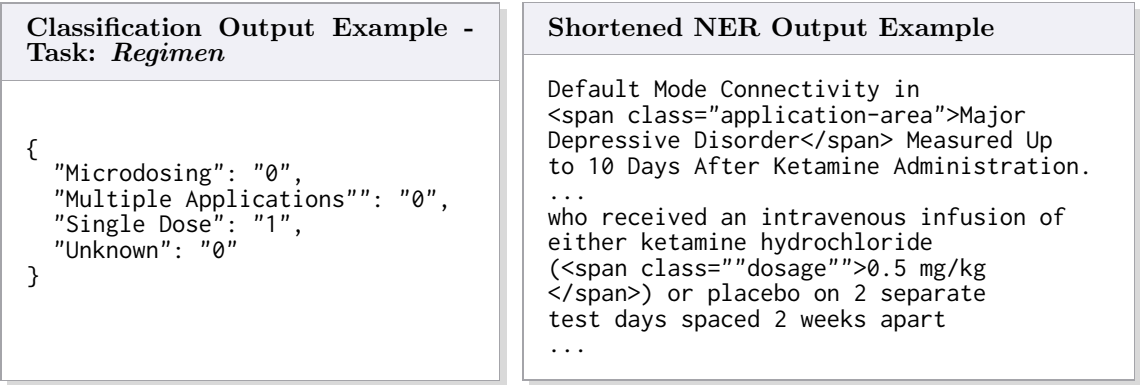


Figure 5.3.: Example outputs for classification and NER.

the 1-shot setting, and for multilabel classification tasks, examples with an average number of labels were selected.

Following prior findings that substantial performance gains can already be achieved with a small number of examples (s. Section 2.2.2.1), we experiment with 1-, 3-, and 5-shot prompting. For the Llama-2-based models used in this thesis (Llama-2-13B, Me-LLaMA-13B, and Llama-2-70B), only 1-shot could be used due to the 4096-token context window limit. Figure 5.1 and Figure 5.2 illustrate how the zero-shot prompt templates are adapted for the few-shot setting. For the few-shot examples, classification outputs are provided in the same JSON format as required during inference, while NER examples consist of the full input text with entities annotated using HTML `<span>` tags (s. Figure 5.3). Manual testing showed that adding a final heading (*Final Input to Annotate*) was important for the model to know what to annotate or classify, so it was included in all few-shot prompts.

5.2.5. Instruction Fine-Tuning (IFT) Experiment

Due to computational limitations, the best performing 8B LLM in the few-shot setting, namely Llama-3.1, was chosen to be instruction fine-tuned (s. Section 2.2.2.2 for more information on IFT).

Dataset Construction The training splits of the PsyNamic dataset, covering all 18 classification and NER tasks, were used to compose the instruction tuning dataset. Each sample was converted into an instruction-response pair: The instruction followed the same task-specific prompt template as in the zero-shot experiments (s. Figure 5.1 and s. Figure 5.2). The response contained the gold-standard annotation, serialized as structured JSON output for classification tasks and HTML-tag-enriched input text for the NER task. No few-shot examples were included. All samples were then randomly shuffled using a fixed seed to prevent task-order bias during training.

Training Llama-3.1 was instruction fine-tuned employing the PEFT-method QLoRA (s. Section 2.2.2.4). Training was implemented using the `transformers` (Wolf et al., 2020), `peft` (Mangrulkar et al., 2022), and `trl` (von Werra et al., 2020) libraries. The original model parameters were loaded in 4-bit precision using NF4 quantization with double quantization enabled, while LoRA adapter parameters were maintained in 16-bit precision. The LoRA rank was set to $r = 64$. Training was performed for three epochs, with a batch size of 8 and gradient accumulation over four steps,

using the CLM loss computed exclusively over the assistant responses. Refer to Appendix D for the complete set of hyperparameters.

5.2.6. Label-Supervised Fine-tuning (LST) Experiment

Like for the IFT experiment, the best performing 8B LLM in the few-shot setting, namely Llama-3.1, was chosen to be label-supervised fine-tuned (s. Section 2.2.2.3 for more information on LST). We apply LST exclusively to the NER task.

Following the original implementation of Li et al. (2023), training was performed using the PEFT-method LoRA (s. Section 2.2.2.4) and was implemented with the `transformers` (Wolf et al., 2020), `peft` (Mangrulkar et al., 2022), and `billm` (Li et al., 2023) libraries. The training dataset consisted of the BIO-formatted training split of the NER task. The base model was loaded in full precision, while LoRA parameters were kept in 16-bit precision. The LoRA rank was set to $r = 32$. Fine-tuning was performed for five epochs using mixed-precision training (FP16) to optimize memory usage. Refer to Appendix E for the complete set of hyperparameters.

5.3. Postprocessing of LLM Outputs

Although the outputs of the LLMs are prompted to resemble Figure 5.3, in practice they frequently deviate from the expected format and must be postprocessed to allow for a consistent evaluation. The manual effort needed to inspect the model outputs and devise the parsing rules should not be underestimated (Ashok and Lipton, 2023; Wang et al., 2025), and it may introduce errors.

5.3.1. Postprocessing for Classification

For classification, the expected output is a JSON object (see Figure 5.3). Despite explicit instructions to produce only the JSON, the model often includes additional explanations or text. These explanations are removed during postprocessing to isolate the intended prediction. Predictions sometimes contain malformed JSON, such as unclosed brackets. In other cases, the output is provided in a one-hot encoded format or as plain label strings, which are also converted into the appropriate JSON representation. The postprocessing pipeline handles all of these cases, extracting the intended prediction while flagging partially faulty outputs for further inspection.

However, in a few cases, the output remains unparsable. These samples are treated as empty predictions (no classes assigned), so they are still included in the evaluation and contribute to the performance metrics.

5.3.2. Postprocessing for NER

For NER, the expected output is the sample text annotated using HTML-style `span` tags, where each span encompasses a named entity and specifies its type (s. Figure 5.3). As the first postprocessing step, the predicted entities are extracted from the spans. Following the examples of Jimenez Gutierrez et al. (2022) and Keloth et al. (2024), any hallucinated entities, either entity strings not present in the input text or entities with an unknown entity type, are removed. This provides us with a set of entity strings with the corresponding entity type that are used to evaluate NER under the extraction task definition (i.e., where positional information is disregarded, s. Section 4.1).

The extracted entities are then converted into token-level BIO tags. To ensure accurate evaluation, the tokenization of the model output must correspond exactly to the tokens in the gold annotation (Obeidat et al., 2025). When LLMs produce the expected output, the entities are directly aligned with the tokens based on their positions and are converted into BIO labels. In some cases, when the model output consists of extracted entities only or an incomplete version of the input text, the positional information is missing or incomplete. In these instances, the entity strings are matched to the tokenized text using string matching. While this approximation may introduce minor errors, such cases are the minority, and this approach allows evaluation on BIO labels in all samples. Moreover, string-based alignment to obtain BIO labels is common in prior research (Xie et al., 2025; Lu et al., 2024).

5.4. Evaluation

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}, \quad (5.1)$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}, \quad (5.2)$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (5.3)$$

$$\text{Weighted F1} = \sum_{i=1}^N \frac{\text{support}_i}{\sum_{j=1}^N \text{support}_j} \times F1_i \quad (5.4)$$

To evaluate all experiments, we rely mainly on variants of the F1 score (5.3), and in some cases also report precision (5.1) or recall (5.2). For all evaluation metrics, we follow the example of Jeong et al. (2024) and compute 95% confidence intervals via non-parametric bootstrap resampling, drawing 1,000 samples with replacement from the predicted and gold-standard labels. This procedure estimates metric variability and accounts for uncertainty in the results.

5.4.1. Evaluation for Classification

For the classification task, we report weighted F1 (5.4), which computes the F1 score for each class i and averages them according to the number of true instances per class. In other words, classes with more examples contribute more to the overall score, but performance on smaller classes is still included. This makes the metric robust for datasets with imbalanced label distributions, like the PsyNamic dataset (s. Section 4.3).

5.4.2. Evaluation for NER

As described in Section 4.1, we apply two types of evaluation metrics for NER:

5.4.2.1. BIO-based Metrics: F1 BIO - Strict, F1 BIO - Partial

Firstly, we evaluate on BIO labels to assess how well the LLMs perform on NER while preserving positional information. For this, we use the `nervaluate` library (Batista and Upson, 2025; Batista, 2018). It defines a true positive at the entity level, meaning a true positive is counted when an entity is correctly identified. Metrics can be computed in two modes: *strict*, which requires an exact match of entity boundary and type, and *partial*, which allows partial overlap of the entity span. We report F1-score (5.3) in both modes, following the example of (Hu et al., 2024; Kelothe et al., 2024). In this thesis, we refer to these metrics as **F1 BIO - Strict** and **F1 BIO - Partial**. Additionally, in strict mode we make use of `nervaluate`’s detailed error analysis. It categorizes predictions as *correct* when the boundaries and type match, *incorrect* when the entity is predicted at the correct location but with the wrong type or boundaries, *missed* when a gold entity is not predicted, and *spurious* when the system produces an entity that does not exist in the gold annotation.

5.4.2.2. Extraction-based Metrics: F1 Strict

Secondly, we report an extraction-based NER metric that does not require token-level alignment and ignores entity positions. This approximates the extraction task definition of NER described in Section 4.1. While most prior work evaluates NER via the BIO format (Xie et al., 2025; Lu et al., 2024; Jimenez Gutierrez et al., 2022; Hu et al., 2024; Kelothe et al., 2024; Chen et al., 2025), Wu et al. (2025) instead evaluates model solely on the set of extracted entities. Our implementation follows this evaluation setup. Firstly, duplicated entity mentions within the predicted and gold entity set are removed. A predicted entity is counted as a true positive only if its string and type match exactly one gold entity. The remaining entities predicted are false positives and unmatched gold entities are false negatives. F1 can then be computed according to (5.3). In this thesis, we refer to this metric as **F1 - Strict**.

6. Results and Discussion

6.1. Zero-shot Experiments

In this section, we will discuss the results of the zero-shot experiments for all classification tasks and the NER task across the 11 selected LLMs. For more details on the implementation, consult Section 5.2.3.

6.1.1. Classification

BERT shows best overall performance. Table 6.1 reports the zero-shot performance of the 11 LLMs on the 18 classification tasks compared to the BERT-Baseline, measured in F1-weighted. BERT achieves the best overall rank, 1.5, indicating it was the best performing model for most classification tasks. This is reinforced in Figure 6.1, where BERT also shows the highest median performance with an F1 score of 0.87. On average, the performance gap to the best performing LLM is 0.10, but it reaches as high as 0.35 for the task *Data Collection*. In total, BERT outperforms all LLMs in 15 of the 18 tasks. However, when looking at the results in detail, there is strong evidence for BERT’s superiority in only 9 of these 15 tasks. In those tasks, the confidence intervals of BERT and the best performing LLM do not, or only marginally, overlap (s. Appendix B.1.2 for detailed results per task).

GPT-4o matches BERT’s performance in 1/3 of the tasks. GPT-4o is the best performing LLM, achieving an overall rank of 2.6. As shown in Table 6.1, it frequently delivers the second best performance on individual tasks. In about a third of the classification tasks, it matches BERT, demonstrating only minimal performance gaps. It even surpasses BERT in 3 tasks, namely *Study Purpose*, *Substances*, and *Outcome*. However, this should be interpreted with caution, because partially overlapping confidence intervals indicate only moderate confidence that these differences are not due to chance (s. Appendix B.1.2 for detailed results per task). Among the open-source models, MedGemma performs best, with an overall rank of 3.7. Med-LLaMA3 scores the

Task	Llama-2-13b-chat-hf	Me-LLaMA-13B-chat	Llama-2-70b-chat-hf	Me-LLaMA-70B-chat	Meta-Llama-3-8B-Instruct	Med-LLaMA-3-8B	Llama-3.1-8B-Instruct	gpt-4o-2024-08-06	gpt-4o-mini	gemma-3-27b-it	medgemma-27b-text-it	BERT-Baseline
Relevance	0.60	0.49	0.61	0.72	0.67	0.38	0.69	<i>0.77</i>	<i>0.75</i>	0.74	0.70	0.91
Study Type	0.54	0.44	0.62	<i>0.65</i>	0.60	0.31	0.50	0.62	0.46	0.59	<i>0.64</i>	0.87
Data Type	0.11	0.27	0.44	0.30	0.35	0.20	0.43	<i>0.58</i>	0.48	0.44	<i>0.63</i>	0.79
Data Collection	0.38	<i>0.54</i>	0.34	0.16	0.30	0.12	0.26	<i>0.47</i>	0.22	0.20	0.18	0.89
Number of Participants	0.04	0.03	0.12	0.17	0.39	0.00	0.40	<i>0.67</i>	0.45	0.46	<i>0.70</i>	0.92
Sex of Participants	0.09	0.36	0.70	0.73	0.44	0.63	0.67	<i>0.85</i>	0.65	0.75	<i>0.81</i>	0.94
Study Conclusion	0.20	0.41	0.13	0.56	0.61	0.58	0.40	<i>0.73</i>	0.55	0.51	<i>0.66</i>	0.79
Study Purpose	0.73	0.63	0.78	0.72	0.75	0.51	0.74	0.82	0.79	0.80	<i>0.81</i>	<i>0.81</i>
Study Control	0.31	0.31	0.58	0.57	0.58	0.18	0.62	<i>0.70</i>	<i>0.65</i>	0.64	0.63	0.75
Age of Participants	0.84	0.79	0.84	0.84	0.84	0.48	<i>0.86</i>	0.84	<i>0.84</i>	0.84	0.84	0.93
Substances	0.75	0.60	0.83	0.81	0.85	0.52	0.85	0.92	0.87	<i>0.90</i>	<i>0.91</i>	0.87
Application Form	0.35	0.42	0.34	0.40	0.33	0.12	0.30	<i>0.64</i>	0.36	0.35	<i>0.48</i>	0.85
Regimen	0.31	0.32	0.53	0.69	0.65	0.28	<i>0.69</i>	<i>0.73</i>	0.66	0.52	0.59	0.73
Setting	0.50	0.56	0.56	<i>0.73</i>	0.61	0.47	0.59	<i>0.86</i>	0.67	0.71	0.72	0.87
Substance Naivety	0.12	0.35	0.14	0.29	0.23	0.08	0.20	<i>0.78</i>	0.19	0.39	<i>0.64</i>	0.86
Condition	0.73	0.60	0.66	<i>0.83</i>	0.74	0.35	0.70	0.82	0.74	0.77	<i>0.83</i>	0.93
Outcome	0.51	0.35	0.70	0.72	0.77	0.08	0.77	<i>0.86</i>	0.80	<i>0.83</i>	0.86	0.80
Clinical Trial Phase	0.24	0.54	0.24	0.33	0.42	0.35	0.51	<i>0.69</i>	0.53	0.53	<i>0.59</i>	0.79
Overall Rank	9.7	9.1	8.3	6.8	7.2	11.2	7.1	<i>2.6</i>	5.7	5.3	<i>3.7</i>	1.5

Table 6.1.: Zero-shot performance in F1-weighted across all models and classification tasks. Green+bold marks the best, yellow+cursive the second best and orange+underlined the third best performance per task. Overall Rank represents the average ranking of each model across task, the lower the value the better.

lowest, often ranking last across tasks.

BERT is consistently strong, while LLMs vary widely across tasks. In Figure 6.1, one can see the distribution of performance over all classification tasks. The BERT-baseline exhibits the smallest variability across tasks, with most classification tasks scoring between 0.8 and 0.9 F1 score. Among the LLMs, MedGemma and GPT-4o show the most stable performance across tasks. However, these models show notable drops on specific tasks: MedGemma has a median performance around 0.7 F1 score but falls as low as 0.18 on *Data Collection* and 0.59 on *Regimen*. GPT-4o is somewhat more consistent, without extreme outliers, yet it also shows task-specific weaknesses: The performance for *Data Collection* drops to 0.47, which is 28 percentage points below the median performance of GPT-4o. The most extreme task performance difference can be seen with Llama-2-13B, with F1-scores ranging from 0.11 for *Data Type* to 0.84 for *Age Participants*. Further evidence for inconsistency can be seen in Table 6.1, where Me-LLaMA-70B or Llama-3.1 occasionally achieve

the second or third best result, despite overall only reaching a mediocre rank.

Older and small models produce unparsable results. Most models required some post-processing because the outputs did not conform perfectly to the desired format (s. Section 5.3). With a few heuristics, the outputs were generally parsable. Exceptions were observed with Llama-2-13B, Me-LLaMA-13B, and Med-LLaMA3. On average, they produce 2-7% unparsable outputs. Although these numbers are small, they are enough to render the models unreliable to be part of an automated pipeline.

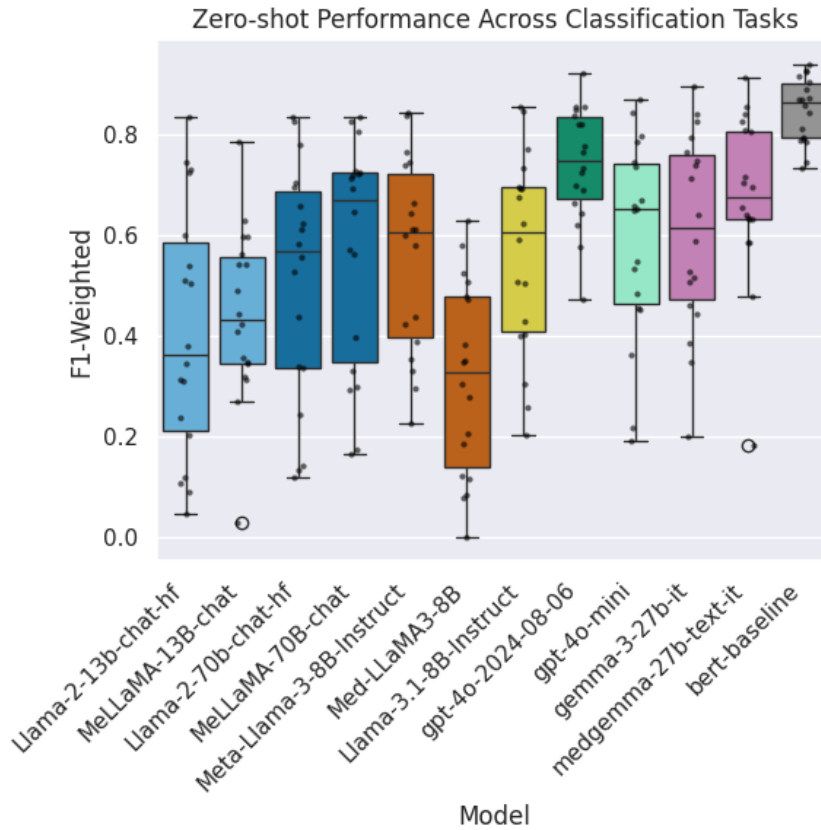


Figure 6.1.: Zero-shot performance in F1-weighted across all classification tasks. Coloring shows the relatedness of the models.

6.1.2. Named Entity Recognition

		Llama-2-13b-chat-hf	Me-LLaMA-13B-chat	Llama-2-70b-chat-hf	Me-LLaMA-70B-chat	Meta-Llama-3-8B-Instruct	Med-LLaMA3-8B	Llama-3.1-8B-Instruct	gpt-4o-2024-08-06	gpt-4o-mini	gemma-3-27b-it	medgemma-27b-text-it	biomedbert-abstract
Dosage	F1 BIO - Strict	0.04	0.19	0.09	0.15	0.26	0.09	0.26	<i>0.48</i>	0.13	0.17	<u>0.33</u>	0.84
	F1 BIO - Partial	0.04	0.19	0.09	0.15	0.26	0.09	0.26	<i>0.48</i>	0.13	0.17	<u>0.33</u>	0.84
	F1 Strict	0.04	0.18	0.11	0.20	0.30	0.11	0.26	<i>0.53</i>	0.15	0.22	<u>0.40</u>	0.86
Application Area	F1 BIO - Strict	0.10	0.13	0.12	0.28	0.27	0.11	0.14	<u>0.29</u>	0.15	0.17	<u>0.29</u>	0.75
	F1 BIO - Partial	0.10	0.13	0.12	0.28	0.27	0.11	0.14	<u>0.29</u>	0.15	0.17	<u>0.29</u>	0.75
	F1 Strict	0.07	0.11	0.13	0.26	0.26	0.04	0.17	<u>0.36</u>	0.21	0.24	<u>0.38</u>	0.77
Overall	F1 BIO - Strict	0.07	0.15	0.10	0.24	0.27	0.10	0.17	<i>0.33</i>	0.14	0.17	<u>0.30</u>	0.78
	F1 BIO - Partial	0.08	0.15	0.10	0.24	0.27	0.10	0.17	<i>0.34</i>	0.15	0.17	<u>0.30</u>	0.78
	F1 Strict	0.06	0.14	0.12	0.25	0.28	0.07	0.19	<i>0.41</i>	0.19	0.23	<u>0.38</u>	0.80
Overall Rank		12	8.3	9.6	5	4	11	6.7	2	8.6	6.7	3	1

Table 6.2.: Zero-shot performance across all models for NER overall and per entity type, measured in F1 BIO strict and partial and F1 Strict. Green+bold marks the best, yellow+cursive the second best and orange+underlined the third best performance. Overall rank represents the average ranking of each model across the different overall performance metrics, the lower the value the better.

BERT outperforms all LLMs by a large margin. Table 6.2 reports the zero-shot performance of the 11 LLMs on the NER task, measured using three evaluation metrics (s. Section 5.4.2 for details). The **BERT-baseline** outperforms all LLMs by a large margin: in F1 Bio - strict, it exceeds the best LLM by 45 percentage points, and in F1 Strict by 39 percentage points. Figure 6.2 shows non-overlapping confidence intervals for BERT and GPT-4o, the best performing LLM. This reinforces the reliability of the observed gap. MedGemma follows closely behind GPT-4o, with very similar overall scores differing by only 3-4 percentage points. However, when taking entity types into consideration, GPT-4o performs markedly better at predicting the entity type *Dosage*. Since *Dosage* entities are less frequent than *Application Area* entities in the dataset (s. Section 4.3), this difference has limited impact on the overall F1 and explains the similar level of overall performance scores. The poorest performing models are Llama-2-13B and Med-LLaMA3, both of which achieve F1 scores of 0.1 or lower.

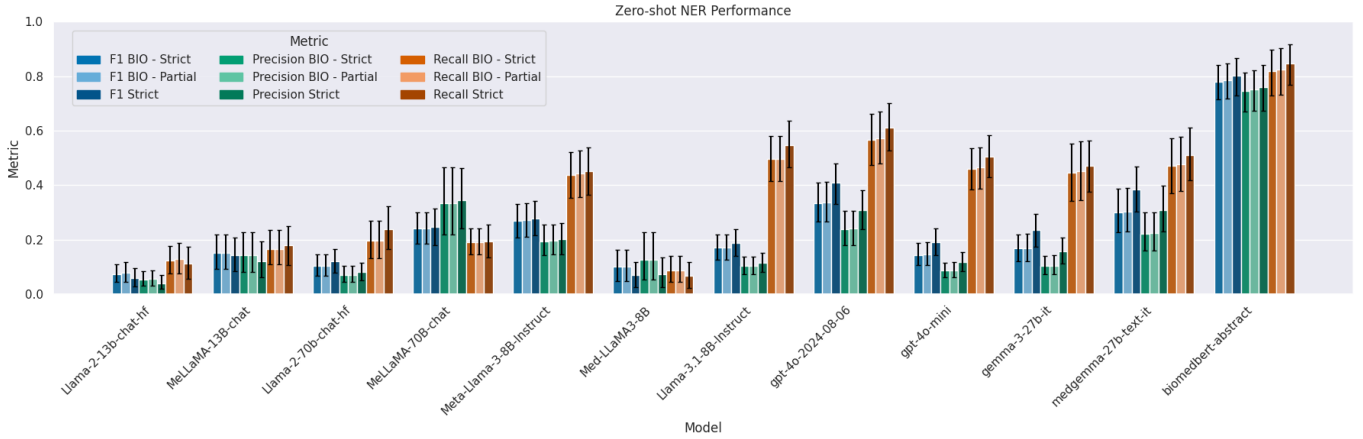


Figure 6.2.: Zero-shot performance across all models for NER, measured variants of F1, precision and recall. Error bars represent 95% bootstrap confidence intervals.

LLMs achieve high recall at the cost of spurious entities. When juxtaposing the F1, precision, and recall metrics in Figure 6.2, an interesting pattern emerges: For most LLMs, recall is notably higher than precision, with the exception of Me-LLaMA-70B and Med-LLaMA3-8B. This implies that the models successfully identify many true entities but also predict a substantial number of additional, incorrect ones. Consistent with this, Figure 6.3 shows that *spurious* entities are the most common error type across the LLMs. Although *spurious* predictions are also the most frequent error type for BERT, the effect is more pronounced with LLMs: On average, they produce approximately seven times as many spurious entities as correct ones. One contributing factor for this behavior is that 46% of the samples in the test set contain no entities (s. Section 4.3), which LLMs struggle with: On average, they falsely predict entities in 80% of these empty samples, despite the prompt explicitly stating that the prediction can be empty. A similar behavior was reported by Jimenez Gutierrez et al. (2022), who observed that GPT-3 has difficulty with entity-less NER examples.

F1 strict scores are slightly higher with all models without affecting model ranking. Across all models, including BERT, F1 Strict overall (s. Section 5.4.2) generally reports higher scores than the BIO-based metrics. This is expected, as F1 Strict does not take into account positional information and duplicate predictions, which makes it a more forgiving metric. The difference is most notable for GPT-4o and MedGemma, with increases of 7-8 percentage points. Differences between F1 BIO - Strict and F1 BIO - partial are minimal, typically not more than one percentage point. This indicates that partial overlaps of predicted entities and the true labels are rare. This might be explained by the fact that entities in the PsyNamic dataset

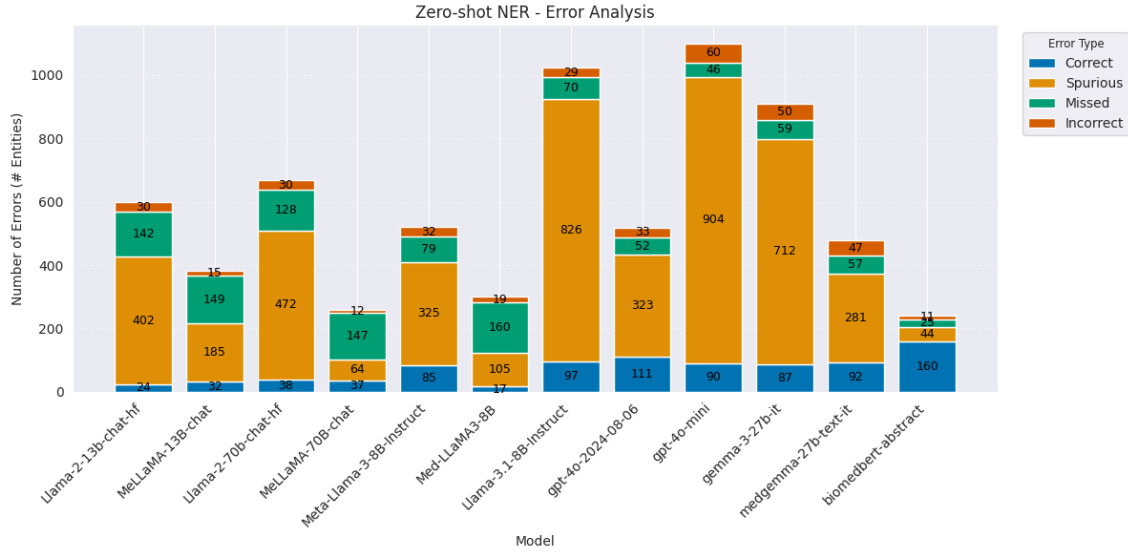


Figure 6.3.: Error analysis of zero-shot performance on NER. Incorrect: wrong entity predicted; Spurious: predicted but not in gold; Missed: gold entity not detected.

are relatively short, mostly between 2-3 tokens. Consequently, in the rest of the experiments, we will focus on F1 BIO-Strict and F1 Strict metrics and disregard the partial metric. Overall, the ranking of the models remains largely consistent across metrics and entities. Top and bottom performers are stable, while minor shifts occur among mid-tier models.

6.2. Few-shot Experiment

In this section, we discuss the results of the few-shot experiments exploring 1-shot, 3-shot, and 5-shot prompting. For the classification tasks, all 11 LLMs were evaluated; for the NER task, the 70B models, Llama-2-70B and Me-LLaMA-3-70B, had to be omitted due to limited computational resources (s. Section 5.2.2 for more details). Also, due to the limited context size of the Llama-2-based models (i.e., Llama-2-13B, Me-LLaMA-13B, Llama-2-70B, and Me-LLaMA-70B), they were only evaluated in 1-shot prompting. For more details on the implementation, consult Section 5.2.4.

6.2.1. Classification

Few-shot prompting narrows LLM versus BERT gap but BERT remains overall best. Table 6.3 and Figure 6.4 present an overview of the few-shot performance across all tasks and models. BERT remains the overall best performing model, achiev-

	Llama-2-13b-chat-hf			Me-Llama-13B-chat			Llama-2-70b-chat-hf			Me-Llama-70B-chat			Meta-Llama-3-8B-Instruct			Med-LLaMA3-8B			Llama-3.1-8B-Instruct			gpt-4o-2024-08-06			gpt-4o-mini			gemma-3-27b-it			medgemma-27b-text-it			BERT-baseline			
	0	1	5	0	1	5	0	1	5	0	1	5	0	1	5	0	1	5	0	1	5	0	1	5	0	1	5	0	1	5	0	1	5	0	1	5	
Rela- nce	60	45	49	40	61	55	72	50	67	42	66	59	38	3	0	0	69	58	63	74	77	70	70	79	75	71	70	75	74	68	59	67	70	72	56	72	91
Study Type	54	49	44	50	62	61	65	75	60	67	64	58	31	13	10	15	50	57	46	56	62	63	64	68	46	54	56	57	59	58	53	54	64	66	61	58	87
Data Type	11	16	27	27	44	37	30	31	35	39	45	52	20	14	14	3	43	30	45	43	58	59	62	67	48	40	53	57	44	46	48	52	63	50	58	53	79
Data Collection	38	59	54	31	34	56	16	31	30	23	37	50	12	7	2	6	26	18	35	46	47	54	56	72	22	19	21	29	20	20	22	38	18	24	28	43	89
Number of Participants	4	0	3	4	12	6	17	19	39	22	14	8	0	1	4	5	40	49	42	27	67	74	72	71	45	40	44	41	46	59	62	52	70	62	72	66	92
Sex of Participants	9	4	36	64	70	62	73	70	44	60	23	20	63	45	2	0	67	72	63	64	85	77	78	84	65	72	64	77	75	77	78	77	81	83	84	77	94
Study Con- clusion	20	12	41	54	13	14	56	37	61	21	29	49	58	10	0	0	40	17	33	40	73	74	72	70	55	46	47	40	51	23	25	33	66	62	44	42	79
Study Purpose	73	67	63	58	78	78	72	75	75	76	77	71	51	50	39	44	74	72	75	74	82	81	83	83	79	78	79	78	80	80	80	82	81	80	80	78	81
Study Control	31	23	31	48	58	63	57	60	58	56	65	59	18	18	15	22	62	65	69	72	70	71	70	70	65	66	60	63	64	64	63	67	63	65	65	70	75
Age of Participants	84	82	79	77	84	83	84	84	84	86	63	73	48	64	23	0	86	86	83	82	84	85	87	82	84	84	84	84	84	84	84	84	84	85	89	80	93
Substances	75	71	60	69	83	82	81	84	85	83	80	80	52	55	19	0	85	85	84	84	92	91	92	92	87	89	88	86	90	90	90	91	92	90	87	87	87
Applica- tion Form	35	26	42	18	34	39	40	33	33	37	34	46	12	31	1	11	30	34	32	41	64	61	72	70	36	38	47	50	35	42	63	66	48	47	61	65	85
Regimen	31	39	32	41	53	60	69	70	65	70	63	62	28	19	19	19	69	74	69	63	73	76	79	77	66	73	74	74	52	51	54	53	59	61	63	61	73
Setting	50	54	56	57	56	61	73	69	61	63	58	65	47	33	33	26	59	63	61	58	86	88	88	86	67	70	76	74	71	78	79	83	72	81	82	87	87
Substance Naivety	12	13	35	20	14	13	29	18	23	24	28	23	8	12	44	18	20	21	35	29	78	86	84	83	19	27	50	59	39	35	66	67	64	33	69	74	86
Condition	73	62	60	65	66	66	83	80	74	76	80	83	35	51	6	15	70	73	76	76	82	86	88	88	74	77	80	82	77	74	83	82	83	82	86	84	93
Outcome	51	56	35	51	70	73	72	80	77	76	72	64	8	10	4	0	77	75	75	74	86	86	84	86	80	79	79	77	83	83	80	79	86	87	85	83	80
Clinical Trial Phase	24	40	54	41	24	29	33	57	42	47	52	53	35	44	4	2	51	57	62	60	69	59	71	71	53	60	52	55	53	36	62	57	59	60	65	63	79
Best Shot Rank	10.2	9.7	8.8	7.1	6.8	10.9	6.5	1.9	5.6	5.0	3.7	1.8																									

Table 6.3.: Zero-shot, 1-shot, 3-shot, 5-shot performance in F1-weighted (reported as percentages, F1x100) across all models and tasks. Green+bold marks best, yellow+cursive second best, orange+underlined third best. Best Shot Rank shows each model’s average rank based on strongest shot. Framed numbers mark best shot configuration for a model on a task.

ing an average rank of 1.8, closely followed by GPT-4o with a rank of 1.9. In contrast to the zero-shot setting, where only 1/3 of the tasks showed comparable performance between LLMs and BERT, few-shot prompting narrows this gap: 11 tasks now exhibit similar performance, as indicated by overlapping confidence intervals (s. Appendix C.1.1 for detailed results per task). This closer performance can be largely attributed to GPT-4o, with MedGemma, Me-LLaMA-70B, and Llama-3.1 matching BERT in occasional tasks. In Table 6.3, it appears at first glance that LLMs even outperform BERT in 5 tasks. However, when accounting for result variability, they only clearly surpass BERT in one task, namely *Outcome*, as indicated by a minimal overlap in confidence intervals. Hence, LLMs do not show a clear performance advantage. On the contrary, for six tasks, *Relevance*, *Study Type*, *Data Type*, *Data Collection*, *Number of Participants*, and *Application Form*, the performance gap remains greater than 10 percentage points in favor of BERT.

Few-shot prompting yields small and inconsistent gains across tasks and models.

In 17 of 18 tasks, the best-performing LLM result is obtained using a few-shot (1-, 3-, or 5-shot) prompting, as shown in Table 6.3 and Figure 6.4. The only exception is the task *Sex of Participants*, where GPT-4o achieves the highest score in the zero-shot setting. However, when looking at Figure 6.5 which shows the average improvement, the impact of ICL seems less convincing: Across all models, the magnitude of improvement with adding 1-5 examples compared to zero-shot is small, often within ± 0.05 F1 score. Also, the large error bars relative to bar heights imply substantial variation across tasks. In many cases, the error bars extend to both sides of 0. This suggests that adding in-context examples improves performance for some tasks while degrading it for others. For Llama-3 and Med-LLaMA3, the average effect is even negative, although the wide intervals again point to strong task-level variability. The only two models that show somewhat consistent improvements with more examples are GPT-4o, gpt-4o-mini, and Llama-3.1, reflected by narrower error bars that lie mostly above zero. Nevertheless, the error bars of the different few-shot configurations still overlap suggesting that the performance differences between 1-, 3-, and 5-shot prompting are difficult to distinguish. The marking of the best-performing setting per task and model in Table 6.3, along with the task-specific improvement plots in Appendix C.1.2, again underlines that the effect of few-shot prompting is highly variable, and adding more examples does not consistently boost performance.

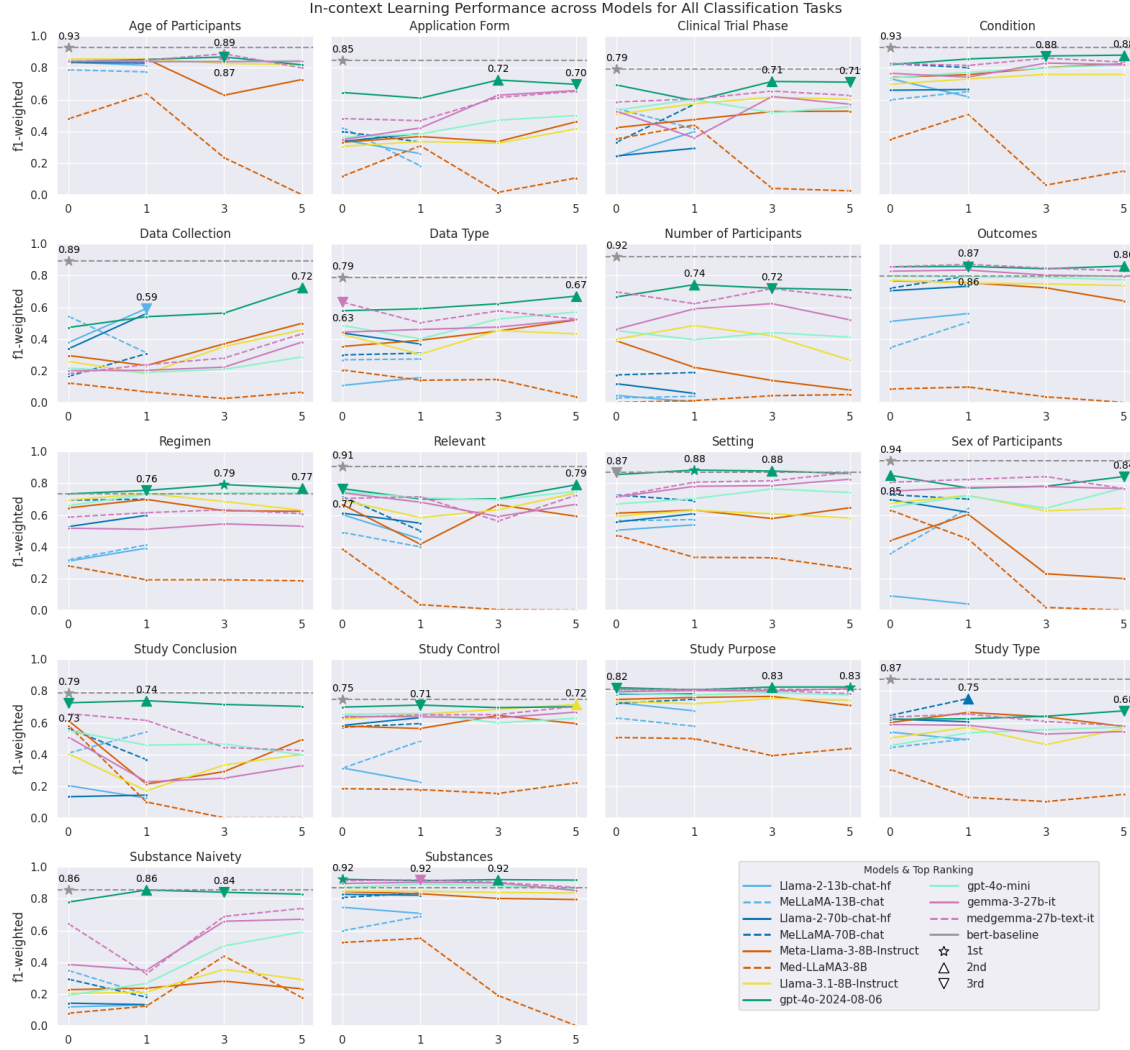


Figure 6.4.: Performance across all models in zero-shot, 1-shot, 3-shot and 5-shot scenario for all classification task. Star marking the best, triangle up 2nd best, triangle down 3rd best performance for a classification tasks. Color shows relatedness of models, dashed lines marking medical models. BERT-baseline marked as gray dashed horizontal line.

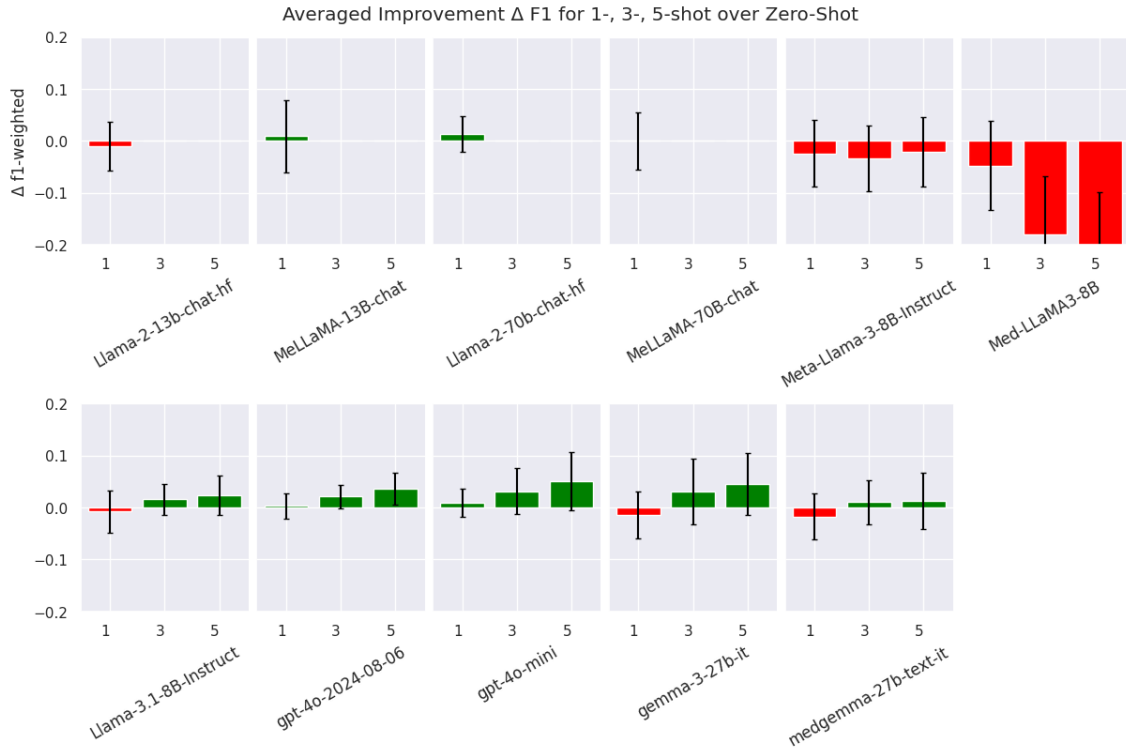


Figure 6.5.: Improvement of 1-shot, 3-shot and 5-shot over zero-shot prompting averaged over all tasks. Error bars represent 95% t-distribution confidence intervals based on task-level variance. Green bars indicate improvement in performance, red bars indicate decline in performance.

6.2.2. Named Entity Recognition

		Llama-2-13B-chat-hf				Me-LLaMA-13B-chat				Meta-Llama-3-8B-Instruct				Med-LLaMA3-8B				Llama-3.1-8B-Instruct				gpt-4o-2024-08-06				gpt-4o-mini				gemma-3-27b-it				medgemma-27b-text-it				BERT-baseline
		0	1			0	1			0	1	3	5	0	1	3	5	0	1	3	5	0	1	3	5	0	1	3	5	0	1	3	5	0	1	3	5	
Dosage	F1 BIO - Strict	4	7			19	0			26	16	13	16	9	0	0	6	26	28	43	20	<u>48</u>	<u>46</u>	46	40	13	15	19	8	17	15	12	18	33	29	29	26	84
	F1 Strict	4	9			18	0			30	31	24	31	11	0	0	4	26	28	45	20	<u>53</u>	47	<u>48</u>	41	15	16	22	8	22	14	13	20	40	31	34	30	86
Application Area	F1 BIO - Strict	10	15			13	2			27	27	31	19	11	6	0	0	14	17	25	27	29	33	34	36	15	24	31	25	17	20	26	38	29	<u>39</u>	47	<u>51</u>	75
	F1 Strict	7	17			11	3			26	29	34	23	4	7	0	0	17	25	35	35	36	42	45	43	21	34	43	36	24	32	36	45	38	48	<u>52</u>	<u>51</u>	77
Overall	F1 BIO - Strict	7	<u>13</u>	<u>15</u>	2	<u>27</u>	23	23	18	<u>10</u>	5	0	2	17	19	<u>28</u>	26	33	36	36	<u>37</u>	14	22	<u>28</u>	22	17	19	23	<u>33</u>	30	37	<u>42</u>	<u>44</u>	78				
	F1 Strict	6	<u>15</u>	<u>14</u>	2	28	30	<u>30</u>	26	<u>7</u>	5	0	1	19	25	<u>38</u>	31	41	43	<u>46</u>	43	19	28	<u>36</u>	29	23	28	30	<u>37</u>	38	43	<u>47</u>	44	80				
Best Shot Overall Rank		8.5				8.5				7				10				5				3				5.5				4.5				2				1

Table 6.4.: Zero-shot, 1-shot, 3-shot and 5-shot performance across all models for NER overall and per entity type, measured in F1 BIO strict and F1 Strict (reported as percentages, F1x100). Green+bold marks the best, yellow+cursive the second best and orange+underlined the third best performance. Overall rank represents the average ranking of each model across the different overall performance metrics, the lower the value the better.

Few-shot prompting improves LLMs but a large gap remains. Table 6.4 reports the few-shot performances on the NER task. Few-shot prompting leads to modest improvements (with overlapping uncertainty) for the best-performing LLM, with a gain of 14 percentage points in F1 BIO - Strict and 9 percentage points in F1 Strict. Although GPT-4o outperformed all other models in the zero-shot setting, MedGemma takes the lead with few-shot prompting. Compared to classification tasks, GPT-4o seems to benefit less from the added examples in NER. However, the overlapping confidence intervals indicate no substantial difference between GPT-4o and MedGemma (s. Appendix C.2 for detailed results). Despite these gains, there remains a substantial performance gap: MedGemma lags behind BERT by 34 percentage points in F1 BIO - Strict and 33 in F1 Strict. Med-LLaMA3 remains the worst performing model, closely followed by Llama-2-13B and Me-LLaMA-13B.

Few-shot prompting leads to improvements in most LLMs. Figure 6.6 summarizes the performance changes of 1-, 3-, and 5-shot relative to zero-shot prompting, evaluated using F1 BIO - Strict and F1 Strict. In contrast to the classification tasks, larger performance changes are observed for NER: improvements frequently exceed 0.1. For example, MedGemma improves by 14 percentage points in the 5-shot setting, and Gemma improves by 17 percentage points in the same setting when measured using F1 BIO - Strict. This is in line with previous work like (Wu et al., 2025), which

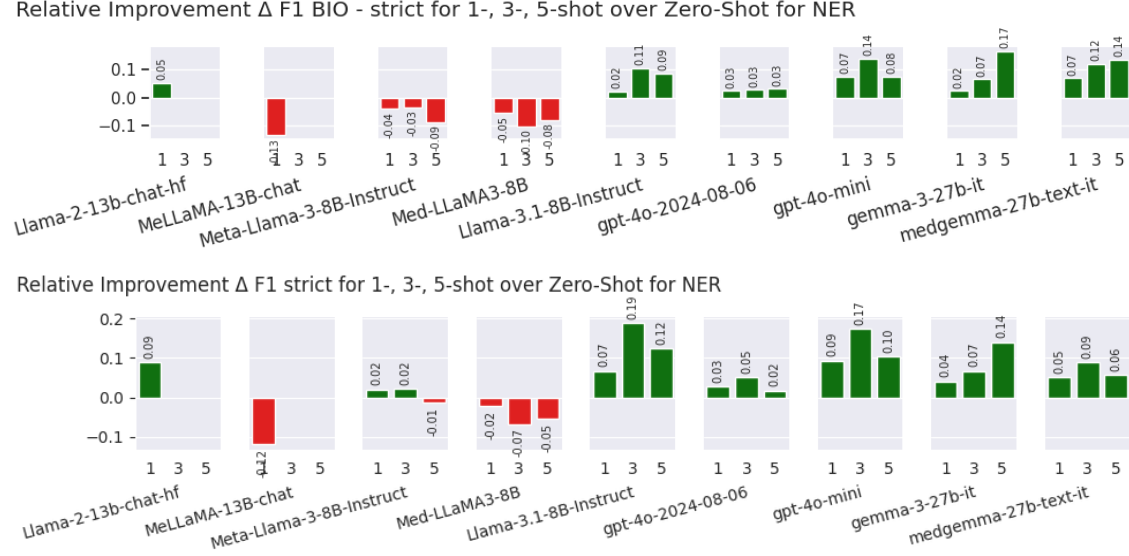


Figure 6.6.: Improvement from 1-, 3- and 5-shot prompting over zero-shot prompting for NER.

states that NER benefits more from in-context examples than other tasks. Overall, the improvements tend to be more pronounced when measured in F1 Strict than in F1 BIO - Strict. Most models benefit from few-shot prompting on NER, with the exception of Me-LLaMA-13B and Med-LLaMA3, which generally perform poorly on NER. As observed for the classification tasks, the performance of these two models degrades when examples are provided. Although few-shot prompting improves performance for most models, increasing the number of examples does not consistently lead to better results: only Gemma and MedGemma show a continuous increase with more examples when evaluated using F1 BIO - Strict. Consistent with this observation, Table 6.4 shows that the best performance for many models is achieved with 3-shot prompting just as often as with 5-shot prompting.

Few-shot prompting reduces the number of spurious entities. Figure 6.7 shows the distribution of error types for all models and few-shot settings. For most models, few-shot prompting leads to a substantial reduction in spurious entities. For example, Llama-3.1 exhibits a 42% reduction from zero-shot to 5-shot prompting. This trend is visible in the models’ behavior on samples without annotated entities: while in zero-shot LLMs, entities are predicted in 80% of such samples on average, this proportion drops to 54% (taking the best few-shot setting per model into consideration). These results indicate that few-shot prompting generally increases precision by reducing over-predictions.

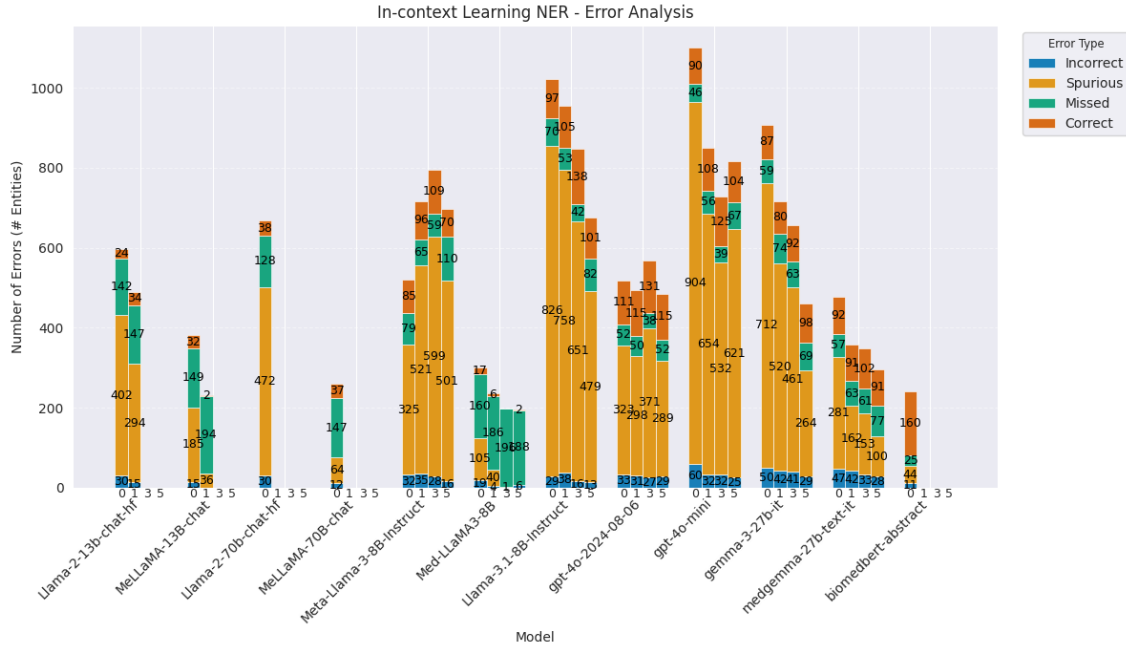


Figure 6.7.: Error analysis performance on NER for all models in zero-shot, 1-shot, 3-shot and 5-shot prompting. Incorrect: wrong entity predicted; Spurious: predicted but not in gold; Missed: gold entity not detected.

6.3. Fine-tuning Experiments

As Llama-3.1 was the best performing small (8B) open-source model both for the classification task (rank 6.5) and NER (rank 5), it was chosen to be used for the fine-tuning experiments. Llama-3.1 was instruction fine-tuned using a dataset built from the training split of all classification and NER tasks. For NER only, Llama-3.1 was additionally label-supervised fine-tuned. For more details on the implementation, consult Section 5.2.5 and Section 5.2.6.

6.3.1. Classification

IFT greatly improves a model’s zero-shot abilities. Figure 6.8 summarizes the performance distribution in all classification tasks; Figure 6.9 provides task-level results. Both figures clearly show that Llama-3.1 benefits strongly from the IFT: the performance improves in 17 of the 18 classification tasks. The only exception is the task *Substance*, where performance remains unchanged at an already high level. On average, IFT yields an increase of 27 percentage points. The improvements are particularly visible in the tasks where Llama-3.1 performed poorly in the zero-shot setting, such as *Substance Naivety* (+0.68), *Data Collection* (+0.64), or *Application Form* (+0.57).

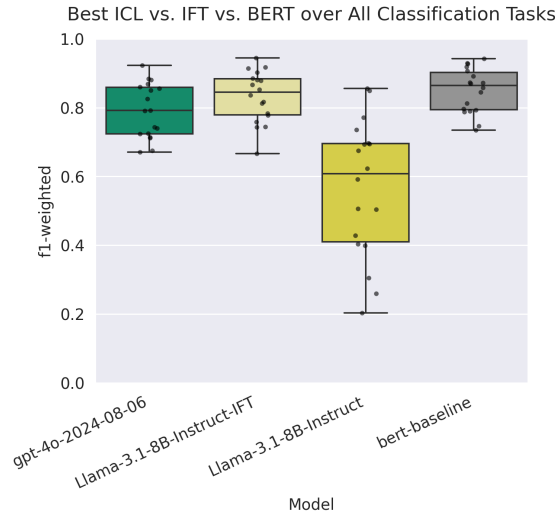


Figure 6.8.: Distribution of performance across all tasks for: (1) highest ranking LLM with ICL, GPT-4o (best ICL setting per task) (2) instruction fine-tuned Llama-3.1, (3) Llama-3.1 zero-shot before fine-tuning, and (4) BERT baseline.

IFT yields BERT-like consistency and performance across tasks. Figure 6.8 illustrates that Llama-3.1 with zero-shot prompting shows large variability across classification tasks, with F1 scores ranging from 0.20 to 0.86. After IFT, the performance becomes markedly more consistent across tasks: most tasks cluster within a comparable range, resembling the level of consistency that BERT demonstrates. In 15 of 18 classification tasks, Llama-3.1-IFT reaches performance comparable to BERT (s. overlapping confidence intervals in Figure 6.9). Only three tasks remain—namely, *Data Type*, *Number of Participants*, and *Study Type*—where BERT achieves higher performance.

Small IFT models can rival large models using few-shot prompting. In Figure 6.8, one can compare the instruction-tuned Llama-3.1 with the best ranked LLM using ICL, GPT-4o. Overall, Llama-3.1-IFT attains a median F1 score of 0.85 across classification tasks, which is considerably higher than the 0.80 median of GPT-4o. In combination with the comparable performance spread, this suggests that instruction-tuning a small LLM can match or even slightly exceed the performance of the much larger GPT-4o using ICL. A closer look at the task-level results (s. Figure 6.9) shows that Llama-3.1-IFT outperforms GPT-4o in four tasks, namely *Application Form*, *Data Collection*, *Relevance*, and *Sex of Participants*. In these cases, performance differences show little to no overlap in confidence intervals, which supports the performance advantage of the IFT model. Only the task *Substances* remains where the IFT model is outperformed by an LLM using prompting only.

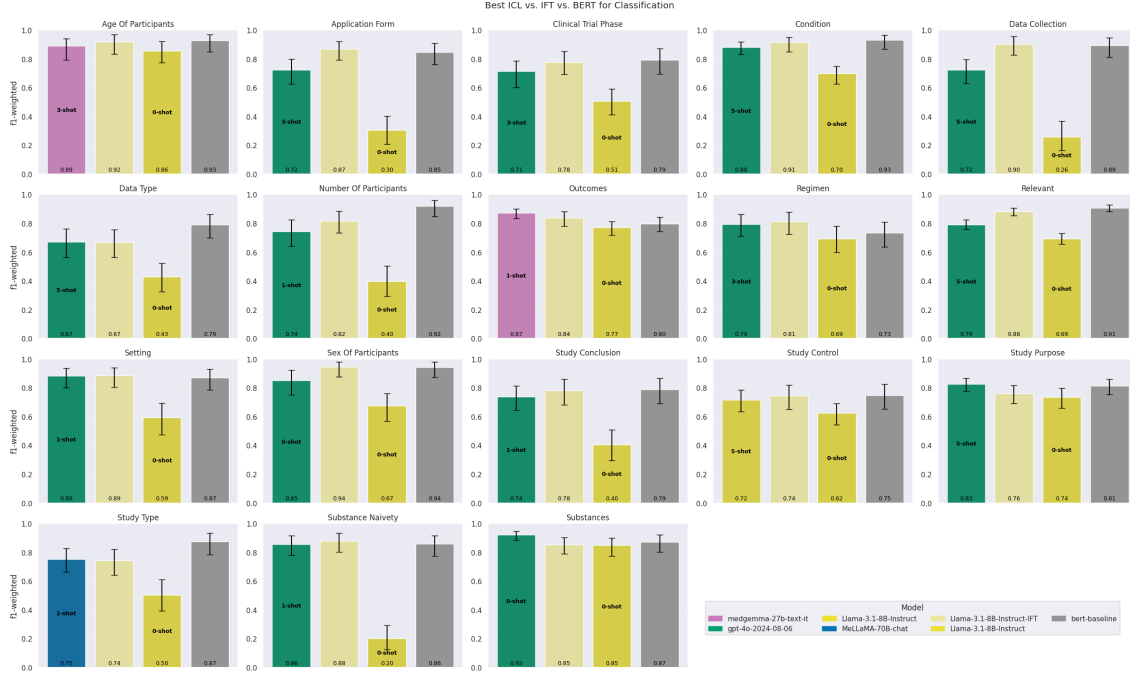


Figure 6.9.: Per task, performance is shown for: (1) best ICL result (best across zero-, 1-, 3-, and 5-shot and models), (2) instruction fine-tuned Llama-3.1, (3) Llama-3.1 zero-shot before fine-tuning, and (4) BERT baseline. Error bars represent 95% bootstrap confidence intervals.

6.3.2. Named Entity Recognition

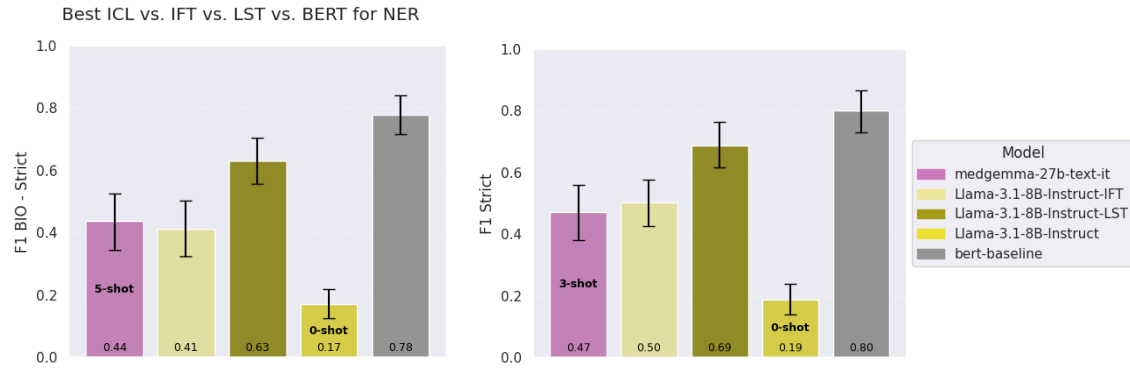


Figure 6.10.: Performance of NER is shown for: (1) best ICL result (best across zero-, 1-, 3-, and 5-shot and models), (2) instruction-fine-tuned Llama-3.1, (3) label-supervised fine-tuned Llama-3.1, (4) Llama-3.1 zero-shot before fine-tuning, (5) BERT baseline. Error bars represent 95% bootstrap confidence intervals.

Small IFT models rival few-shot larger models. Figure 6.10 shows that instruction fine-tuning greatly improves the performance of Llama-3.1: When measured using F1 BIO-strict, performance increases by 24 percentage points, and by 31

percentage points when measured using F1 strict. This results in a performance level similar to what **MedGemma** achieves using few-shot prompting, as indicated by overlapping confidence intervals. Overall, this suggests that IFT can help smaller models, such as the 8B **Llama-3.1**, achieve the performance of larger models, such as the 27B **MedGemma**, that rely on ICL.

LST clearly outperforms IFT and ICL but BERT remains best performing on NER.

Figure 6.10 shows that LST can improve **Llama-3.1** performance even more than IFT, with increases of 46 percentage points in F1 BIO-strict and 50 percentage points in F1 strict. The non-overlapping confidence intervals indicate that LST clearly outperforms the best few-shot ICL results and **Llama-3.1-IFT**. Despite these gains, there remains a performance gap of 15 percentage points in F1 BIO-strict and 11 percentage points in F1 strict compared to **BERT**. Thus, while LST substantially improves smaller models, it neither matches nor exceeds **BERT** in NER.

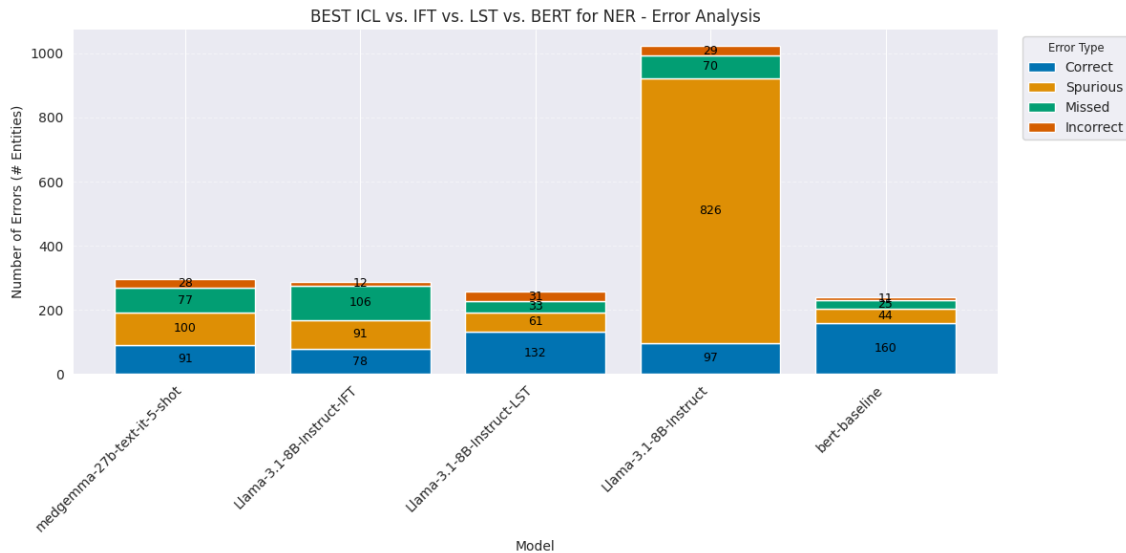


Figure 6.11.: Error analysis for (1) best ICL result (best across zero-, 1-, 3-, and 5-shot), (2) instruction-fine-tuned **Llama-3.1**, (3) label-supervised fine-tuned **Llama-3.1**, (4) **Llama-3.1** zero-shot before fine-tuning, (5) **BERT** baseline. Incorrect: wrong entity predicted; Spurious: predicted but not in gold; Missed: gold entity not detected

IFT and LST reduce spurious entities; LST reduces missed entities. Figure 6.11 shows the error analysis of **Llama-3.1-IFT** and **Llama-3.1-LST**. In both fine-tuning scenarios, the number of *spurious* entities is drastically reduced compared to the untuned model. This improvement is at least partially due to the behavior on samples without annotated entities: the untuned **Llama-3.1** falsely predicts entities in all empty samples, whereas **Llama-3.1-IFT** does so in only 13% and **Llama-3.1-LST** in

22% of cases. LST seems to perform particularly well because it produces not only very few spurious entities, which improves precision, but also a very low number of missed entities, indicating better recall.

6.4. Model Considerations: Size and Medical vs. General Domain

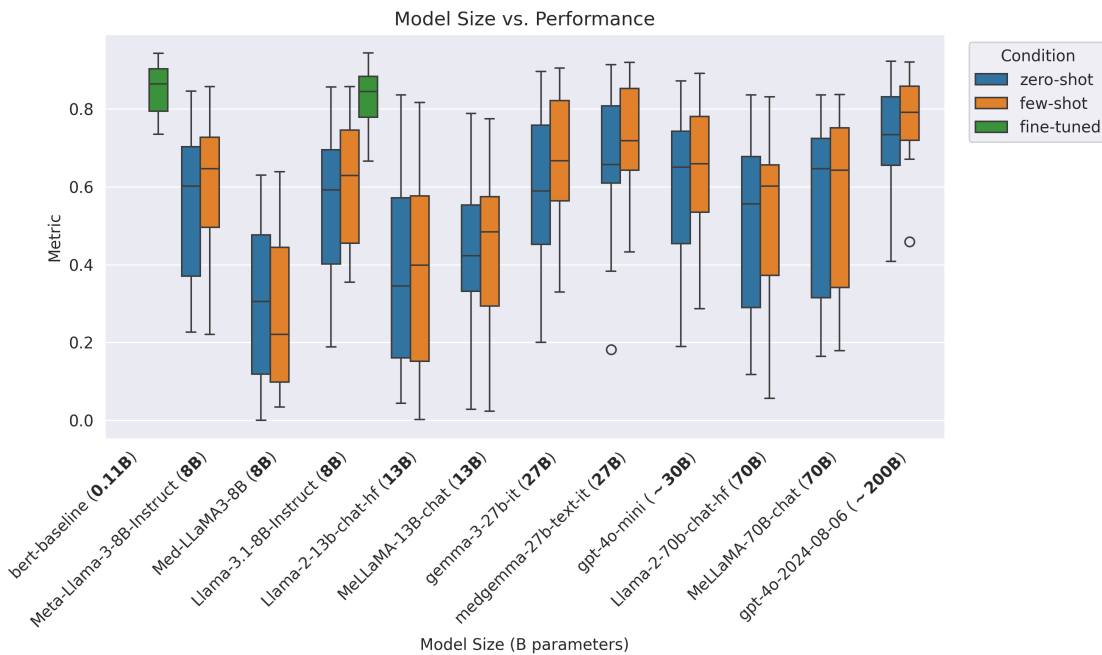


Figure 6.12.: Performance of all models ordered according to size (left to right, small to large) across all tasks (F1 Strict for NER, F1-weighted for classification) in zero-shot, few-shot and fine-tuned condition. For few-shot, the highest performance across 1-, 3-, and 5-shot settings is shown. LST is omitted as it was only evaluated for NER.

Model size does not compete with model fine-tuning. In Figure 6.12, the performance of the LLMs across all classification tasks and the NER task is plotted against model size. As described in the previous results section, GPT-4o, by far the largest model, achieves the best performance over all classification tasks and performs on par with MedGemma in NER under both zero- and few-shot settings. This aligns with previous research, such as Wu et al. (2025), which also found that larger models generally outperform smaller ones when relying on prompting strategies only. However, the relationship between size and performance does not hold consistently for other models: For example, the second largest models, Me-LLaMA-70B

and Llama-70B, show greater variability and lower median performance than the 27B MedGemma. Similarly, Llama-3.1 with 8B parameters performs on average better than Llama-2-13B. Furthermore, fine-tuning either a BERT model or a smaller LLM, such as Llama-3.1 with IFT, often matches or exceeds the performance of even very large proprietary models like GPT-4o. Overall, these results suggest that model size alone is not a reliable predictor of performance and that fine-tuning can allow smaller models to compete with much larger ones.

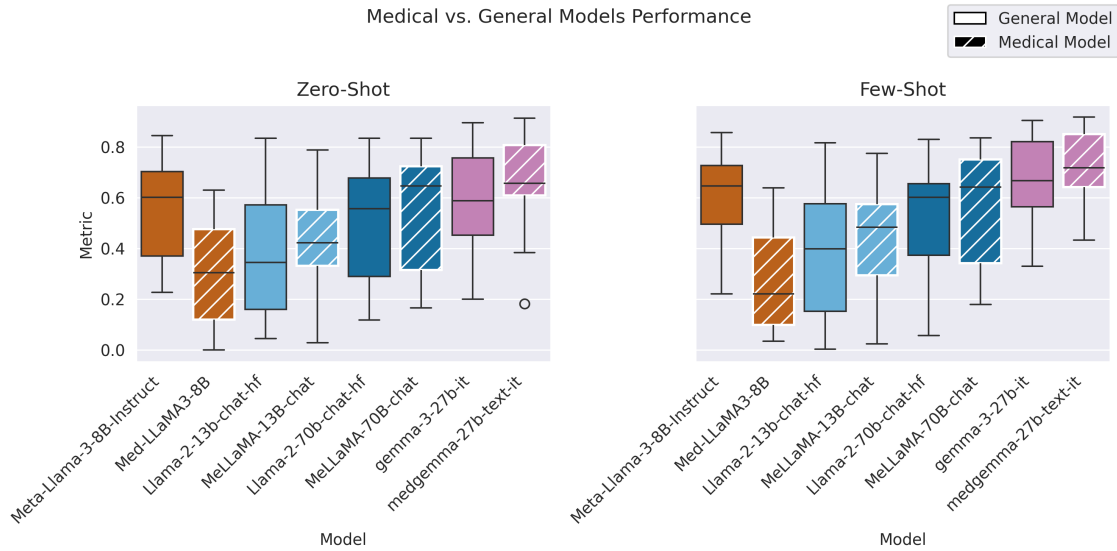


Figure 6.13.: Performance of medical models compared to their general base models across all tasks (F1 Strict for NER, F1-weighted for classification) in zero-shot and few-shot conditions. For few-shot, the highest performance across 1-, 3-, and 5-shot settings is shown. Hatched boxes indicate medical models, and color represents model relatedness.

Medical models do not systematically outperform the general baseline. Figure 6.13 compares four model pairs, each consisting of a general-domain baseline and its domain-adapted medical counterpart. In three of the four model pairs, the medical LLM shows a higher median score, both in zero-shot and few-shot settings. However, this should be interpreted with caution, as the differences are small and highly variable across tasks. Also, this pattern does not hold for Llama-3 and Med-LLaMA3: the latter not only fails to outperform its general-domain backbone but actually performs substantially worse and also ranks last across all PsyNamic tasks¹³. Me-LLaMA-13B and MedGemma show more stable performance across tasks than their general-domain counterparts. This is not the case for the other two model pairs. In sum, there is no consistent evidence that medical models outper-

¹³ Its poor performance raises the question whether the model was trained in the same way as its predecessor, Me-LLaMA, or e.g., with less or no IFT, as documentation is very limited.

form their backbones, which matches the observation in previous research like Wu et al. (2025).

7. Conclusion

This chapter first summarizes the contributions and experiments presented in this thesis. Then it revisits the research questions and answers them on the basis of the obtained results. Finally, it outlines directions for future work and discusses the limitations of this thesis.

7.1. Summary of Contributions

In this thesis, we conduct an extensive comparison of BERT-base models and LLMs for biomedical text classification and NER using the PsyNamic dataset. We systematically examine whether ICL-based and fine-tuning-based LLM approaches can outperform established BERT baselines.

We perform zero-shot and few-shot experiments with 11 mostly small LLMs, spanning proprietary and open-source models as well as general-domain and medical-domain variants. For ICL, we design task-specific prompts informed by prior work and keep the prompt structure consistent across models to ensure fair comparisons. For NER, we adopt an inline annotation strategy using HTML markup, which preserves positional information and enables reliable conversion to BIO labels. Few-shot prompting is evaluated with fixed, manually selected in-context examples (1-, 3-, and 5-shot) that reflect the label distribution of each task. The diversity of evaluated models further allows us to analyze the impact of model size and medical domain adaptation on performance.

Beyond prompting, we investigate two fine-tuning strategies for LLMs, both applied to the strongest 8B open-source model identified in the ICL experiments. First, we construct a multi-task instruction dataset by combining all training splits of the PsyNamic dataset and train an instruction fine-tuned model on it. This model is then evaluated across all classification tasks as well as the NER tasks. For NER specifically, we perform LST using the NER training split. Both fine-tuning strategies are implemented with parameter-efficient methods (QLoRA and LoRA) to reflect realistic compute constraints.

7.2. Answers to Research Questions

RQ1. Can LLMs outperform BERT-based models in zero-shot settings?

In zero-shot settings, LLMs did not surpass BERT-based models for classification or NER. For classification, the best performing LLM, GPT-4o, matched the performance of BERT in 1/3 of the 18 tasks. However, GPT-4o and all other LLMs exhibited high variability across tasks, with F1 scores dropping below 0.5. BERT, on the other hand, maintained consistently strong performance in all tasks. For NER, BERT clearly dominated, outperforming all LLMs by a substantial margin: GPT-4o, also the best-performing LLM in NER, lagged behind by 39-45 percentage points in F1 scores. The poor performance of LLMs can be attributed to overprediction, particularly the prediction of spurious entities in empty samples. This results in higher recall but reduces precision. In sum, these findings suggest that although the performance gap is smaller in classification, LLMs fall short in zero-shot classification and NER.

RQ2. To what extent does few-shot prompting improve LLM performance compared to BERT-based models?

Few-shot prompting provides modest improvements, yet overall BERT-based models remain superior to LLMs. In classification, providing 1-5 in-context examples allows LLMs to reach BERT-level performance in 11 out of 18 tasks and to exceed it in one task, when considering the best performing LLM per task. However, across models, the improvement from few-shot prompting varies considerably and is typically within ± 0.05 F1 score, sometimes even leading to a performance decrease. For NER, few-shot prompting yields more reliable improvements across models. For example, MedGemma improves F1 scores by 9-14 percentage points, largely by reducing spurious entities. Despite these improvements, a substantial gap remains: the best-performing LLM in few-shot NER, MedGemma, still scores 33-34 F1 points below BERT. Overall, few-shot prompting can moderately enhance the performance of LLMs, especially for NER, but it does not enable LLMs to consistently surpass BERT in either classification or NER.

RQ3. Can IFT and LST, implemented with PEFT, enable LLMs to surpass BERT-based models?

An instruction fine-tuned LLM delivers results similar to BERT in text classification, and a label-supervised fine-tuned LLM narrows the performance gap in NER, but LLMs still do not outperform BERT. In classification, IFT elevates the mid-tier model Llama-3.1 to perform on par with BERT and achieve similar consistency across all tasks. Llama-3.1-IFT even slightly exceeds the performance of large, proprietary

models like GPT-4o using ICL. For NER, Llama-3.1-IFT also matches the few-shot performance of larger LLMs with prompting only. In addition, the best performing LLM for NER is the label-supervised fine-tuned Llama-3.1, which narrows the performance gap to BERT to 11–15 F1 points. Overall, even with marked performance boosts from IFT and LST, LLMs only match BERT in classification and still fall short in NER.

RQ4. What roles do model size and medical domain adaptation play in the performance of LLMs?

Overall, performance is highly variable across tasks, and fine-tuning is a more reliable predictor of performance than size or domain adaptation. In the ICL setting for classification, the largest model, GPT-4o, achieves the highest performance, but in NER it performs on par with the much smaller MedGemma. Among the other models, performance does not reliably scale with size and shows high variability across tasks. Fine-tuning a small 8B model, with IFT or LST, can match or even surpass larger models. Medical adaptation yields inconsistent effects: some models, like MedGemma, slightly outperform their general-domain counterparts, while others, such as Med-LLaMA3, underperform.

RQ0. Can LLMs outperform BERT-based models in biomedical classification and NER?

In conclusion, for the PsyNamic dataset and the models evaluated in this thesis, LLMs do not consistently outperform BERT and therefore cannot replace the existing BERT-baseline. While IFT allows LLMs to achieve comparable performance in classification, no setting was able to match BERT’s performance in NER.

7.3. Outlook

Although this thesis provides an elaborate comparison of LLMs to a baseline BERT in biomedical classification and NER tasks, several directions remain for future work.

First, due to computational limitations, few-shot experiments for NER could not be conducted with the largest open-source models, namely Llama-2-70B and Me-LLaMA-70B. Including these models would complete the few-shot NER evaluation. However, given that both models performed below average in zero-shot NER as well as in all ICL experiments for classification, it is unlikely that their inclusion would affect the overall conclusions of this thesis.

Second, few-shot prompting could be explored in greater depth. In this work, in-context examples were manually selected to reflect the label distribution of the training data and were fixed across all inputs. Future experiments could compare this strategy with random example selection or with dynamic example selection, where the examples are chosen individually for each input sample. Previous work has shown that similarity-based retrieval methods can boost few-shot performance (Labrak et al., 2024; Lu et al., 2024).

Third, LST was applied only to the NER task in this thesis. Li et al. (2022) have shown that LST can produce consistent improvements not only for NER but also for text classification (Li et al., 2023). Hence, extending LST to the classification task also represents a promising direction for future research.

Fourth, fine-tuning experiments were limited to a single open-source model, `Llama-3.1`, due to the scope of this thesis. Applying IFT and LST to additional models could provide further insight into whether these methods consistently improve performance. Given that `Llama-3.1` demonstrated only average performance and `MedGemma` was the strongest open-source model in prompt-based experiments, fine-tuning the `MedGemma` model may result in even larger performance gains.

Finally, while this thesis evaluated a diverse set of models, the rapid evolution of LLMs means that many promising models were not included. Future work could consider additional model families, such as Qwen or DeepSeek, which have demonstrated strong performance on biomedical benchmarks like BRIDGE (Wu et al., 2025).

7.4. Limitations

This thesis has several limitations that should be considered when interpreting the results.

First, this thesis did not systematically explore prompt engineering. Instead, a fixed prompt was used for each task and model. LLMs are highly sensitive to prompt design, and a formulation that works well for one model may perform poorly for another (Keraghel et al., 2024; Jeong et al., 2024). Additionally, LLMs are not fully deterministic, producing variable outputs even with the same prompt. This variability and the unlimited search space for prompts represents an inherent limitation of this research and research on LLMs in general.

Although a diverse set of models was tested, this study cannot claim to provide com-

prehensive insights that generalize across all LLMs. In particular, due to computational limitations, only smaller LLMs could be tested (with some limited evaluation of 70B models). Nevertheless, the chosen set of models reflect a real-life scenario in which resources are not abundant.

Fine-tuning experiments were conducted with a single model, `Llama-3.1`, and no extensive hyperparameter tuning was performed due to the high computational cost. Consequently, the full potential of IFT and LST might not have been explored.

The analysis of classification tasks was limited in scope. With 18 tasks included, only a superficial evaluation was feasible. A more detailed investigation of the performance of individual labels within a classification task, the effects of task-specific class imbalance or differences across task types (binary, multiclass, multilabel) was beyond the scope of this thesis. Although some error analysis was performed for the NER task, a deeper qualitative analysis was also omitted.

7.5. Acknowledgments

The experiments with the commercial GPT models, accessed via the API were financed by STRIDE Lab¹⁴, the medical data science group of the University of Bern.

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¹⁴ <https://stride-lab.pages.uzh.ch/website/>

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A. Further PsyNamic Dataset Analysis

The following includes additional information and visualizations that provide deeper insights into the PsyNamic dataset.

A.1. Sample Length

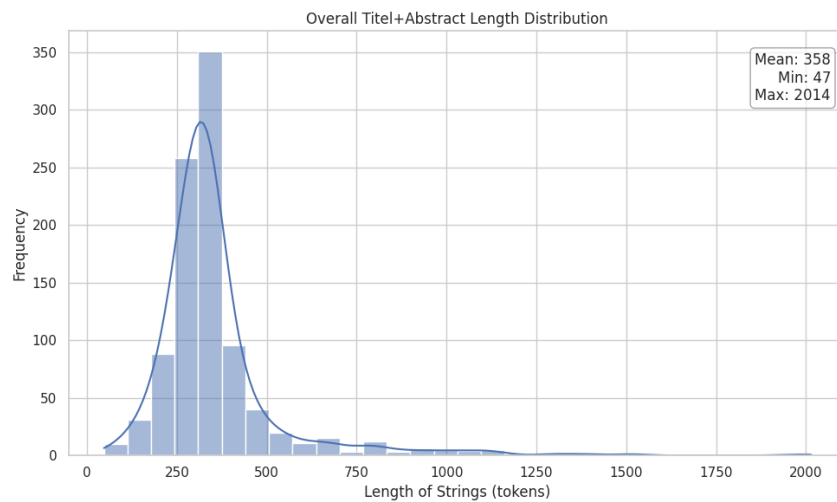


Figure A.1.: Distribution of samples length over relevant studies.

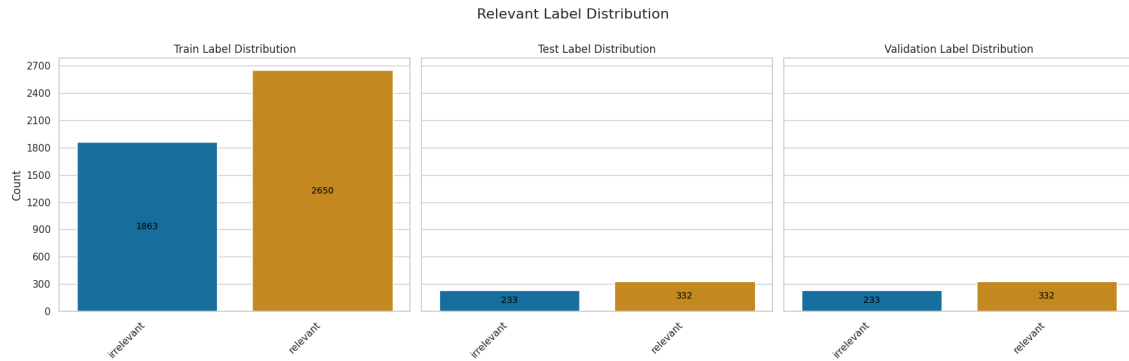
A.2. Classification Task Description

Task	Task Description
Study type	Methodological classification of the study based on its design, purpose, and approach.
Study purpose	Defines the overall purpose of the study, such as assessing efficacy, safety, pharmacokinetics, or mechanisms of action.
Study control	Describes the type of control used in the study (e.g., placebo, active control, non-controlled).
Data type	Specifies whether data were collected longitudinally or cross-sectionally.
Data collection	Indicates whether data were collected prospectively or retrospectively.
Number of participants	Categorizes studies by the total number of participants included.
Sex of participants	Reports the sex distribution of study participants.
Age of participants	Indicates the age group of participants (pediatric or adult).
Substances	Lists the psychedelic substance(s) studied.
Application form	Describes how the substance was administered (e.g., oral, nasal, intravenous).
Regimen	Defines the dosing regimen used (e.g., single dose, multiple applications, microdosing).
Setting	Identifies the context in which the substance was applied (e.g., clinical, naturalistic, party).
Substance naivety	Specifies whether participants were psychedelic-naïve or experienced users.
Condition	Indicates the medical or psychological condition of participants under study.
Outcomes	Lists the outcomes measured or reported in the study (e.g., fMRI, physiological functions).
Clinical trial phase	Identifies the trial phase (I-IV) or whether it is non-applicable.
Study conclusions	Summarizes the main finding of the study (positive, negative, neutral, mixed).

Table A.1.: Classification task descriptions.

A.3. Classification Label Distribution

A.3.1. Binary Classification



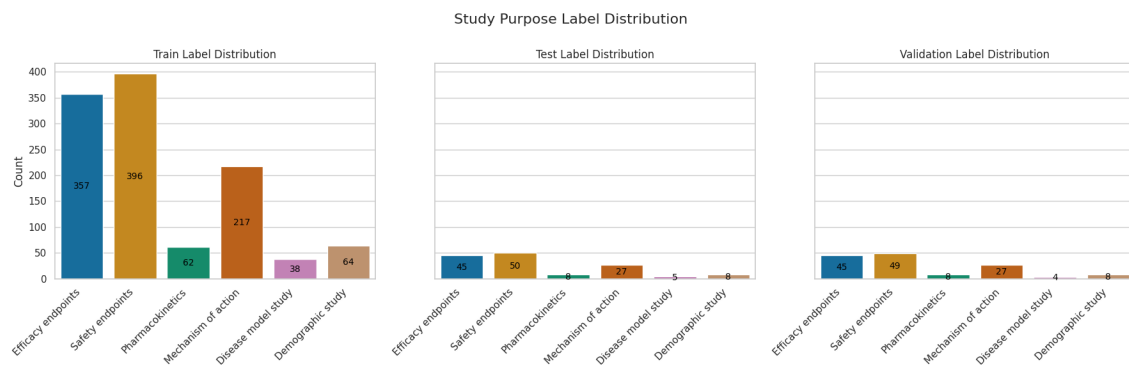
A.3.2. Multiclass Classification





Figure A.2.: Label distributions from all multiclass classification task over all data splits.

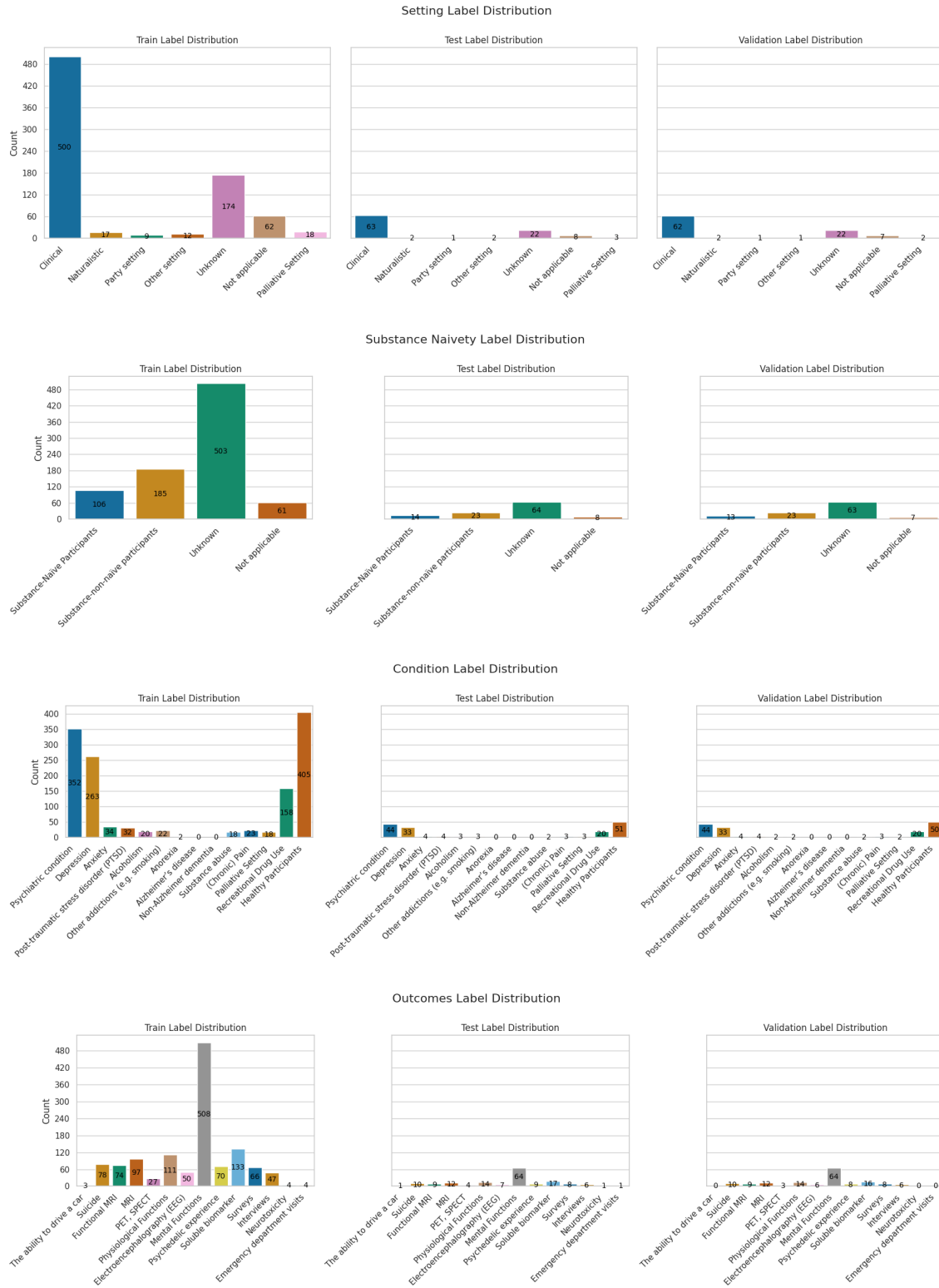
A.3.3. Multiclass, Multilabel Classification



Appendix A. Further PsyNamic Dataset Analysis



Appendix A. Further PsyNamic Dataset Analysis



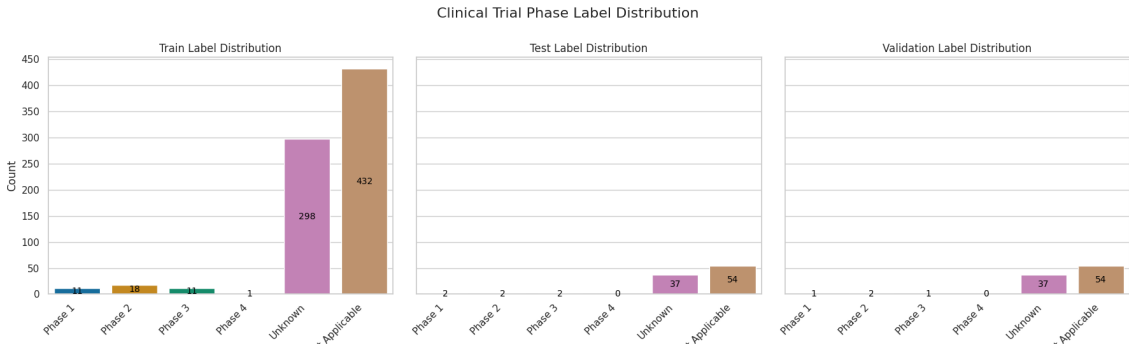


Figure A.3.: Label distributions from all multiclass, multilabel classification task over all data splits.

B. Zero-shot Experiments

B.1. Classification

B.1.1. Prompt Example

Here, is an example template for the classification task *Regimen*. The other task specific prompt components can be found in github.com/vera-bernhard/PsyNamic-BERTvsLLMs/prompts.

Classification Prompt Template for *Regimen*

TASK

The task is to classify which therapeutic regimen did the study use based on the title and abstract of a research paper. Only annotate groups actually receiving the substance (not healthy controls). If ‘Microdosing’ is selected, do not also select ‘Multiple Applications’. Apply per substance (i.e. once DMT and once ketamine in the same study is single dose). For systematic reviews, select ‘Unknown’ unless a specific regimen is assessed. Very low placebo doses do not count as one application.

INPUT

The input is the title and abstract text.

DOCUMENTATION

There are 4 Regimen options. The followings are the options and their definitions:

The followings are the options and their definitions:

Microdosing: As defined by the study authors.

Multiple Applications: Application at multiple times points (e.g., LSD given on at least 2 time points or ketamine weekly for 3 months; or regular MDMA-users for recreational drug use).

Single Dose: Substance is only applied once. Also counts as single dose if substance is applied more than once over up to 1 hour.

Unknown: E.g., in a cross-sectional study on psychedelic use among the general population and it is not clear how many times the participants consumed psychedelic substances.

OUTPUT

The output must be exactly one JSON object, and nothing else. Do not include reasoning, the input or multiple JSONs. The JSON should have relevant values for each option: Microdosing, Multiple Applications, Single Dose, Unknown

Put value 1 if the option applies to the research paper, 0 if it does not apply.

Please note again that multiple options can be selected for each research paper.

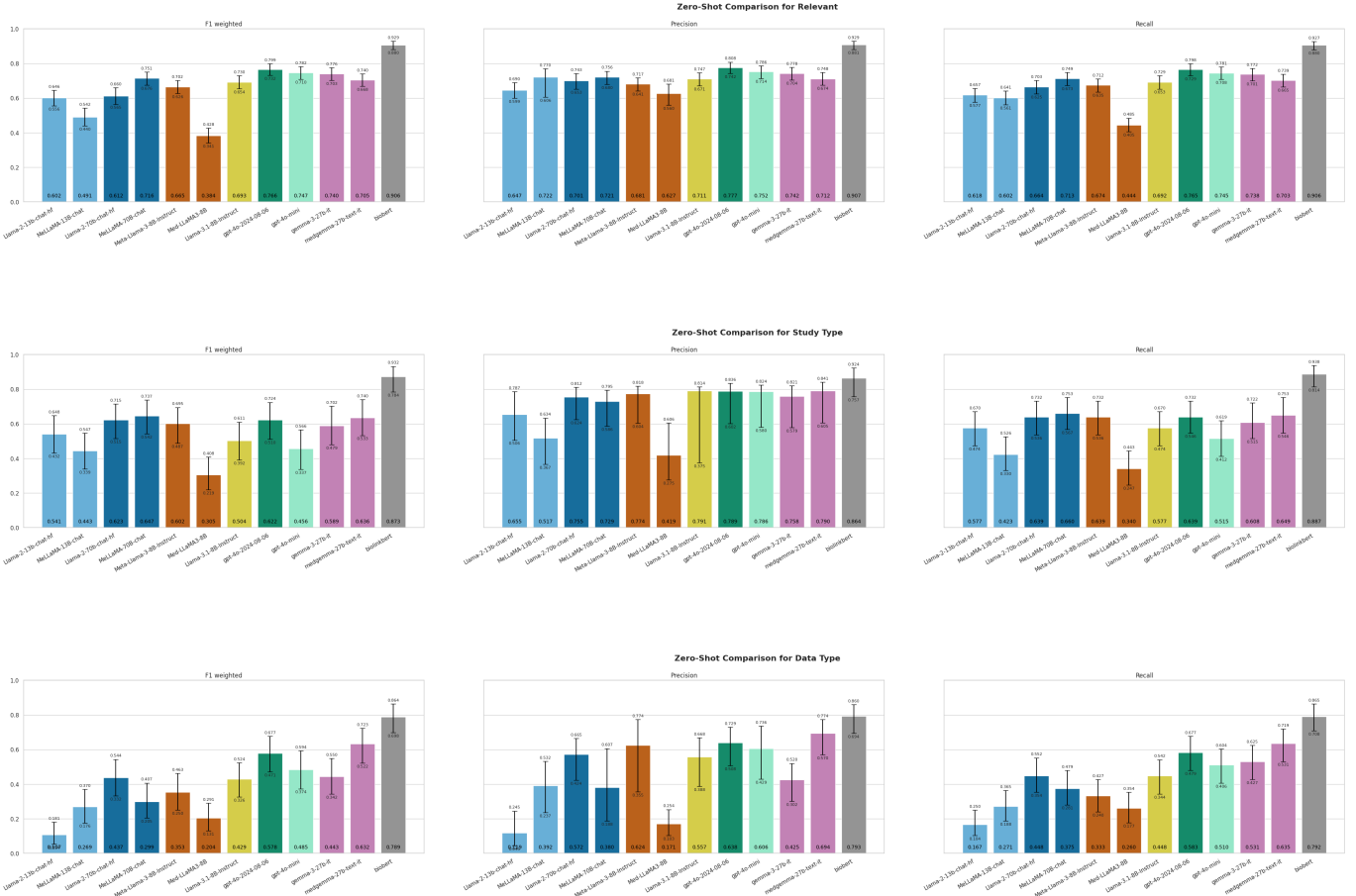
Example output format:

```
{
  "Microdosing": "",
  "Multiple Applications": "",
  "Single Dose": "",
  "Unknown": ""
}
```

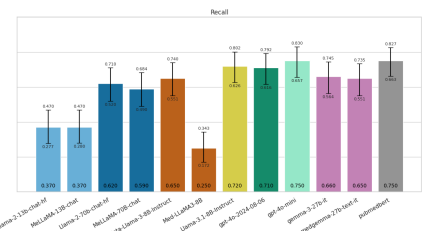
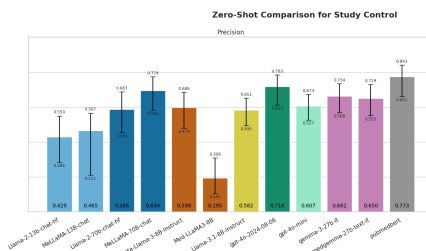
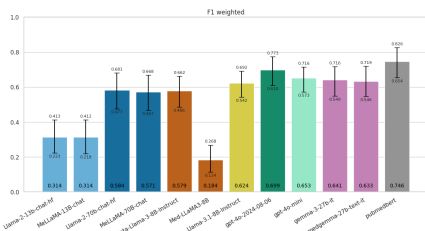
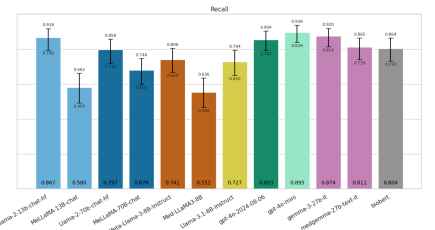
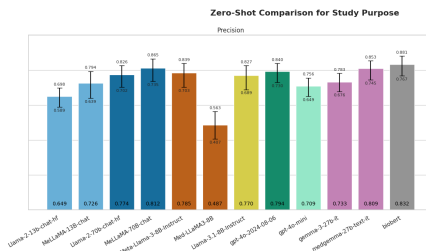
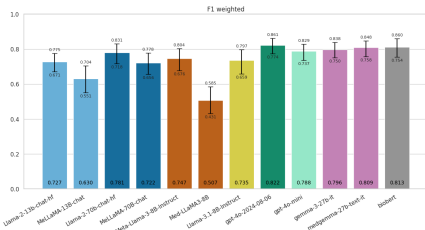
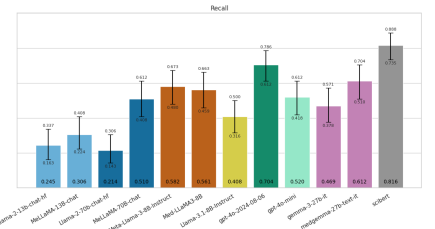
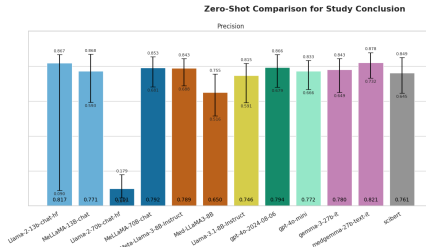
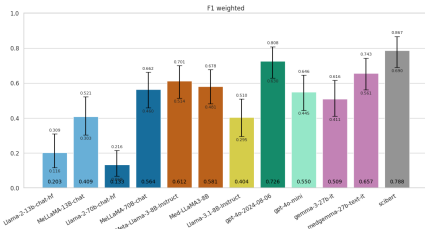
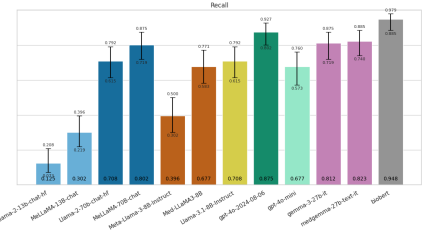
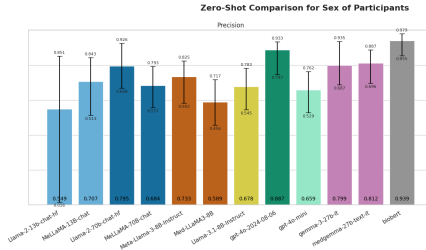
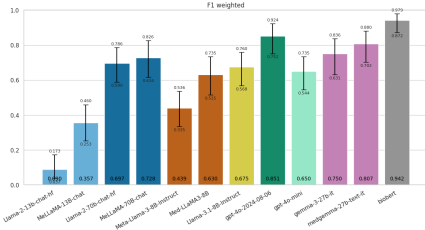
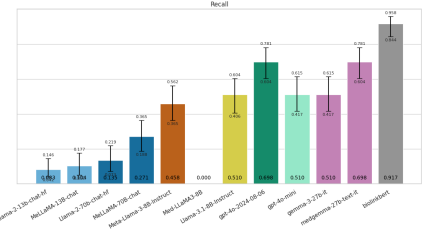
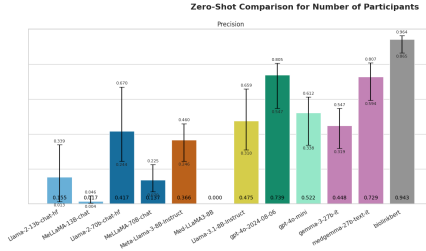
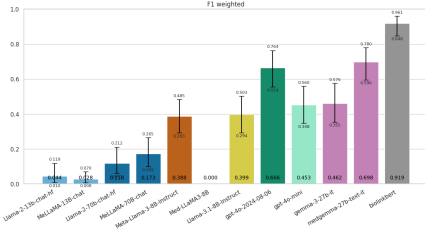
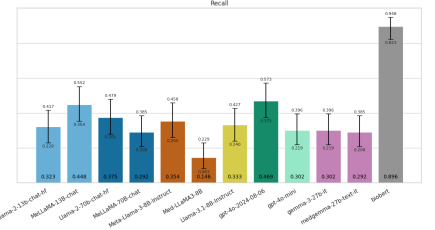
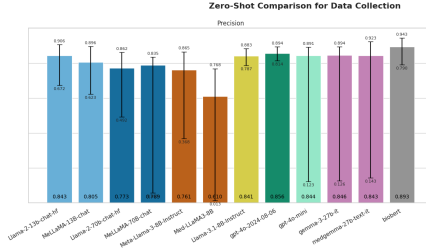
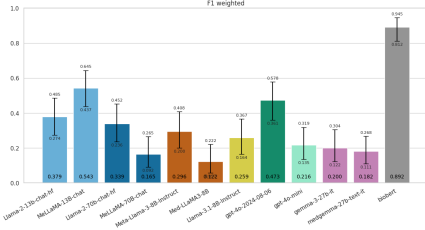
INPUT: TITLE+ABSTRACT

OUTPUT:

B.1.2. Detailed Result Per Classification Task



Appendix B. Zero-shot Experiments







B.2. Named Entity Recognition

B.2.1. Prompt Example

Here, is the full prompt for the NER task:

NER Prompt Template

###Task

Your task is to generate an HTML version of an input text, marking up specific entities. The entities to be identified are: Application area, Dosage. Use HTML `<span>` tags to highlight these entities. Each `<span>` should have a class attribute indicating the type of entity. Return only the HTML output. Do not include any additional text, explanations, or formatting before or after the HTML.

###Entity Markup Guide

Use `<span class="application-area">` to denote Application area, `<span class="dosage">` to denote Dosage.

###Entity Definitions

Application area is defined as: The condition or disease studied in the paper, defined as a medically recognized state with identifiable diagnostic features, progression, and treatment response, typically having an ICD-10/11 code.

Dosage is defined as: The applied dosage(s) of the psychedelic substance(s) in the present study.

Annotation Guidelines

Application area should be annotated according to the following criteria:

- * Can be empty, i.e., no annotation per abstract.
- * Annotate only in title and/or introductory abstract parts discussing the problem or objective (including abbreviations).
- * Annotate sweeping statements like “major depressive and bipolar disorder” as one entity.
- * Annotate full descriptive phrases (e.g., “active suicidal ideation”); annotate adjectives like “Depressed” but not generic words like “patients”.
- * Include defining features such as Acute/Chronic, Mild/Severe, Drug-resistant, Post-stroke, etc.
- * Annotate abbreviations separately, e.g., “Huntington disease” and “HD” in “Huntington disease (HD)”.
- * Exclude study targets not treatment targets (e.g., “dissociative symptoms” in some contexts).
- * Annotate disorders like “cigarette addiction” but not concepts like “smoking cessation”.
- * Exclude targets of study that are not treatment targets (e.g., “dissociative symptoms” in some contexts).
- * Do NOT annotate generic concepts like “psychiatric condition”.
- * Do NOT annotate outcomes or measures like “Hamilton Depression Scale”, “antisuicidal response”, or phrases like “depression was reduced”.
- * Do NOT annotate treatments like “antidepressant”.

Dosage should be annotated according to the following criteria:

- * Annotate the number and the corresponding unit (e.g., 2mg/kg).
- * Can be empty, i.e., no annotation per abstract.
- * Annotate only doses of the psychedelic substance in the current study (e.g., exclude doses mentioned from other studies).
- * Annotate all reported doses for the respective psychedelic substance.
- * Annotate also the term “between” if pertinent, similarly “or” and “and” should be annotated. Do not annotate “per device”.
- * Annotate also relative dosages (e.g., 1 mg/kg), do NOT annotate the terms “bw” or “body weight” or similar if pertinent
- * Do NOT annotate concentrations of solutions (e.g., 500mg/10ml) or doses measured in plasma (4 mg/ml).

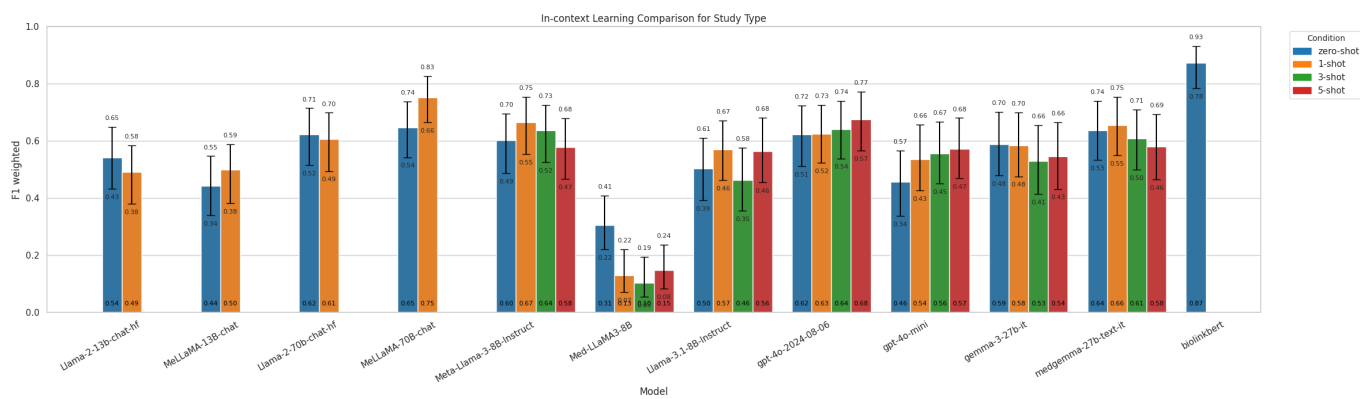
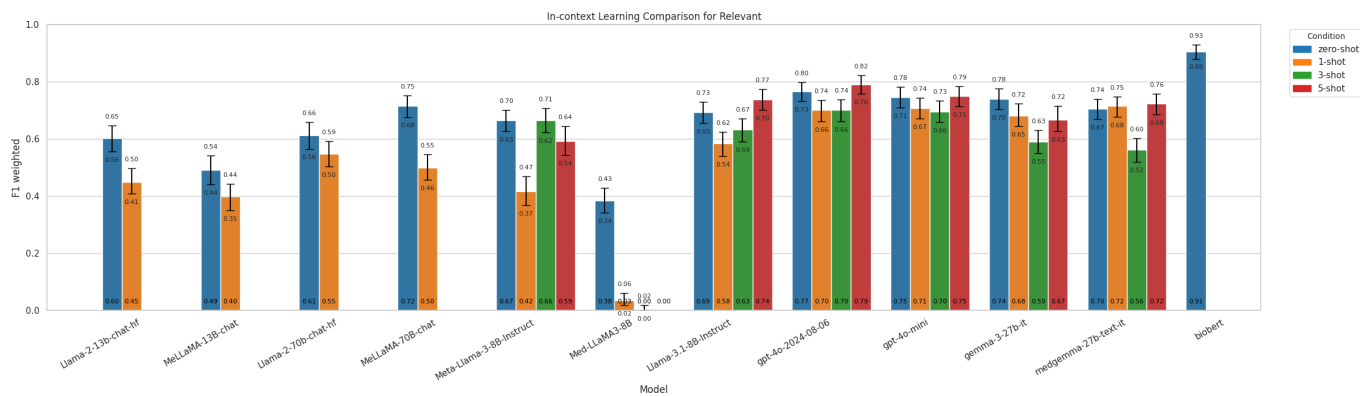
INPUT: TITLE+ABSTRACT

OUTPUT:

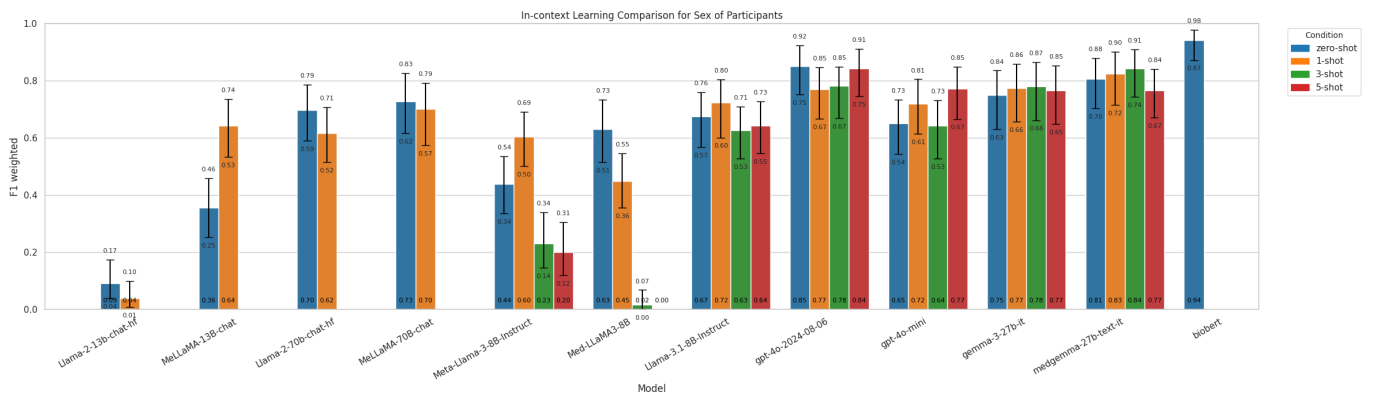
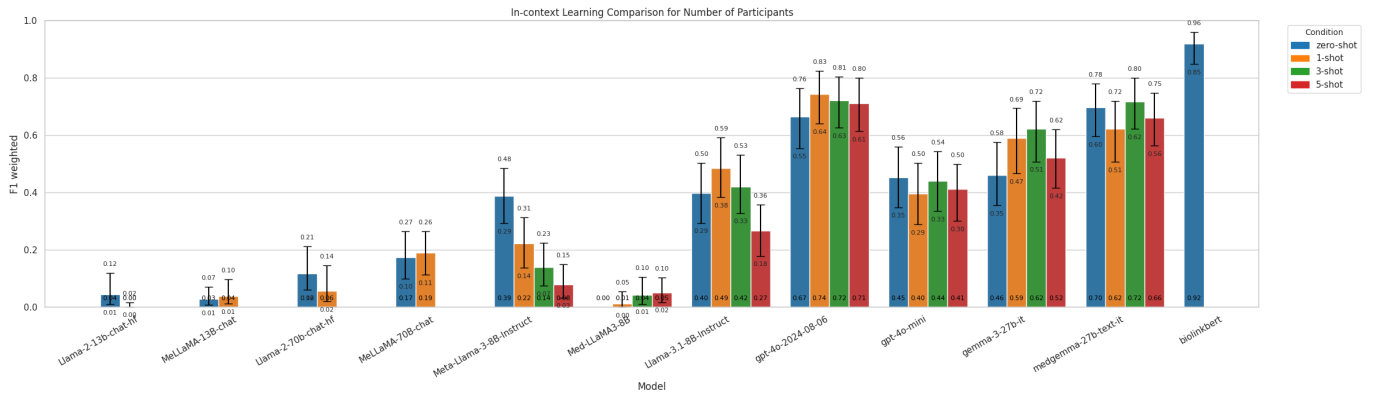
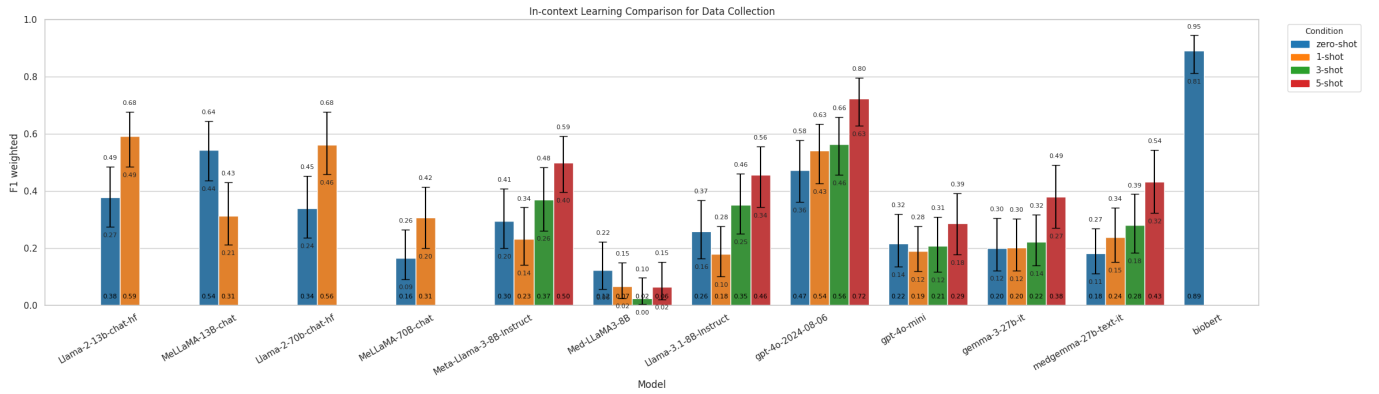
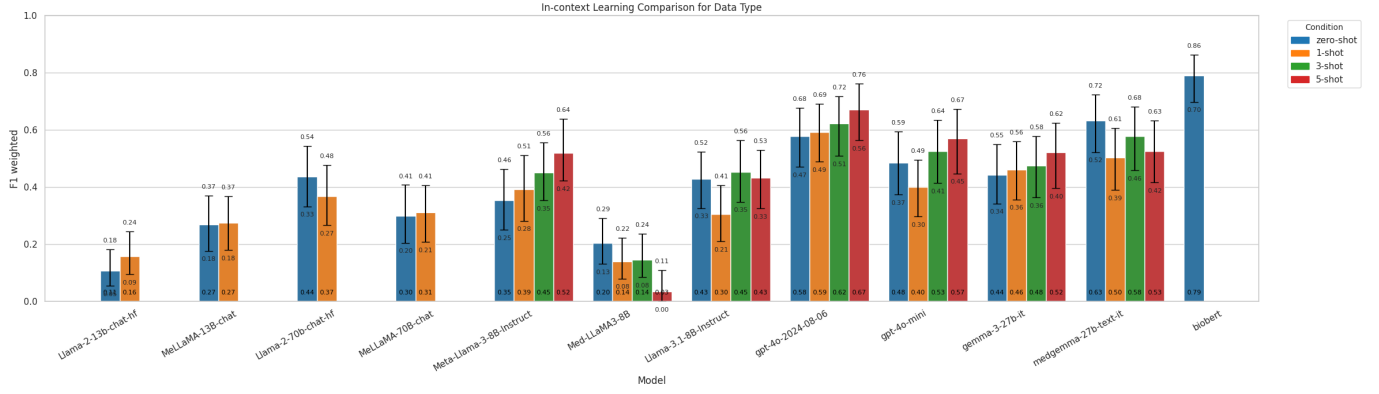
C. Few-shot Experiments

C.1. Classification

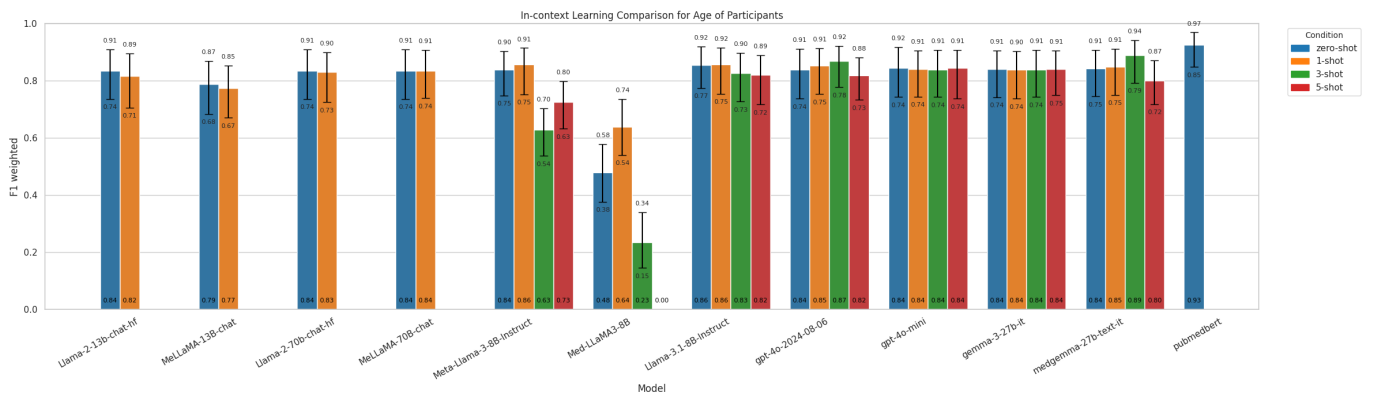
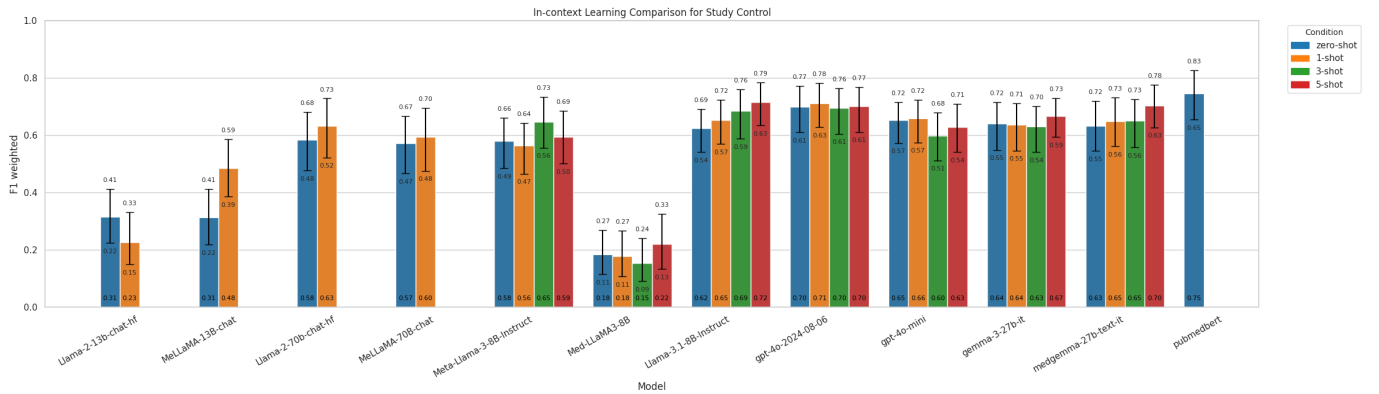
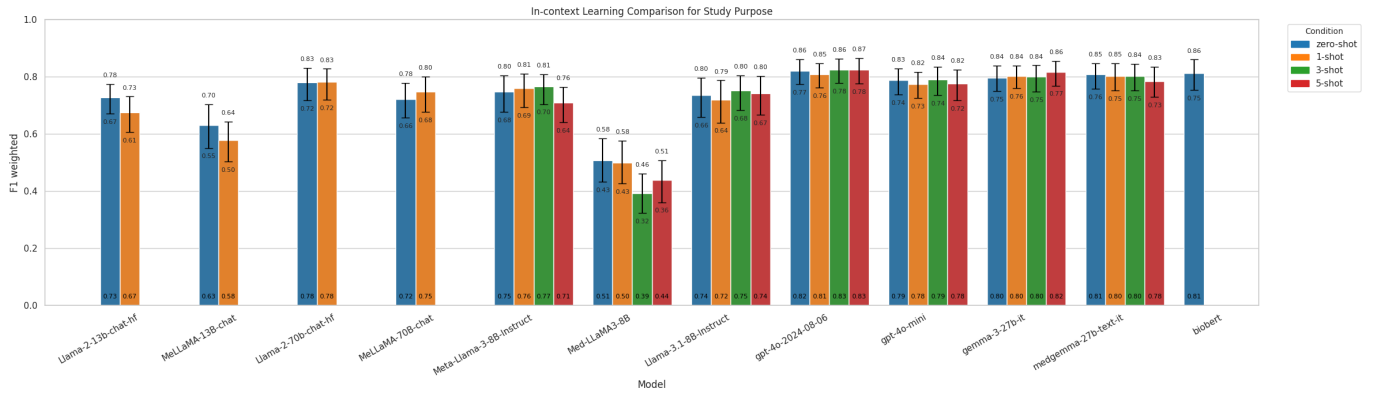
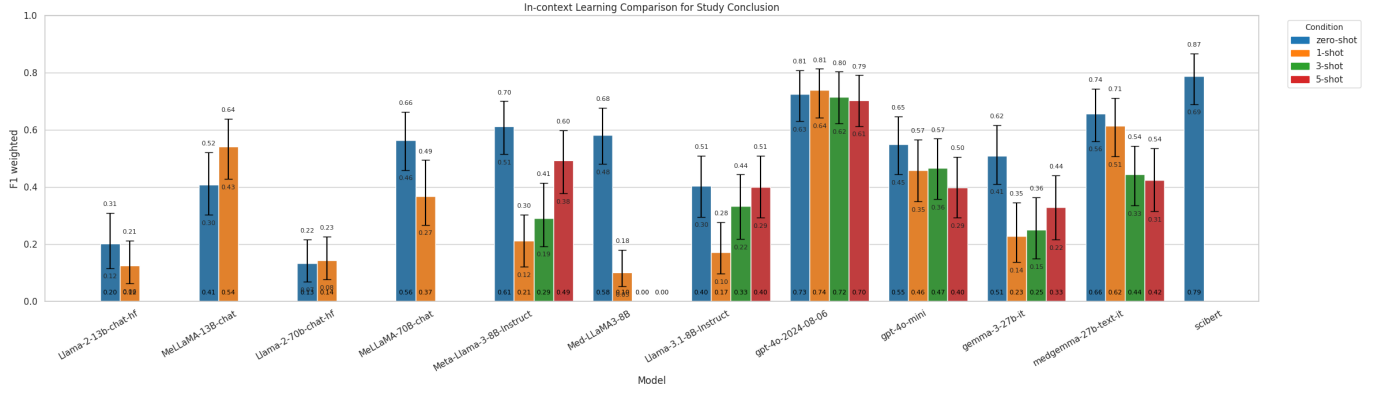
C.1.1. Detailed Result Per Classification Task



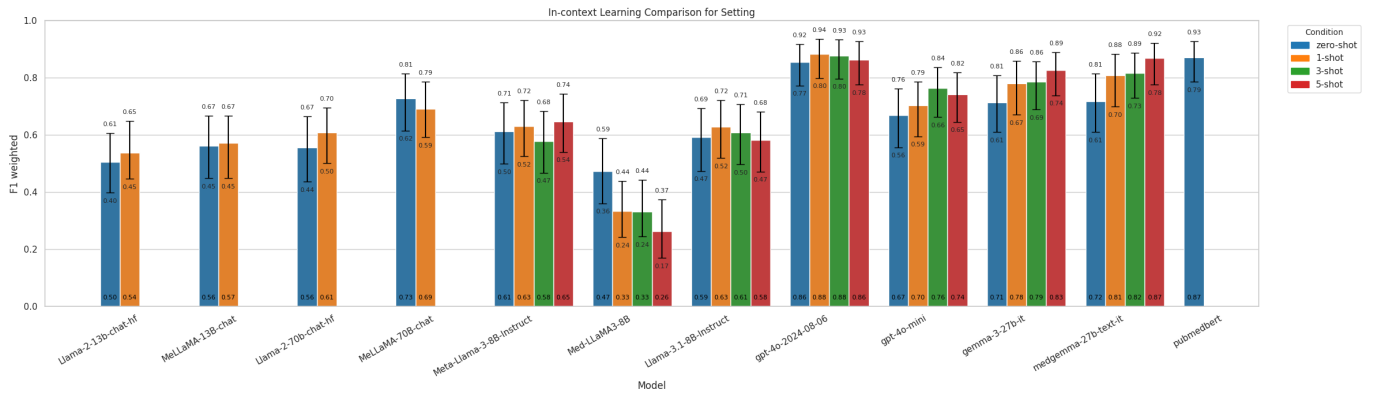
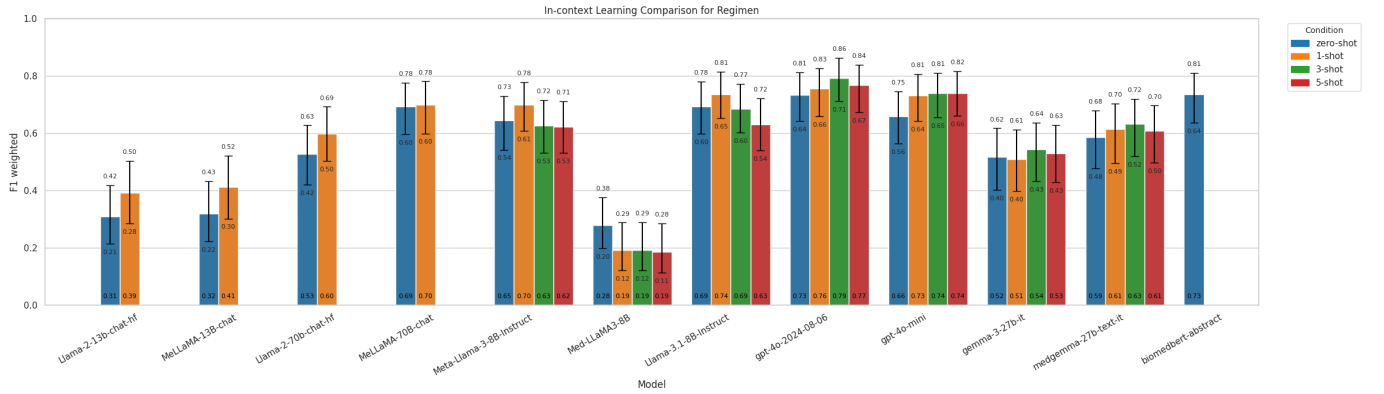
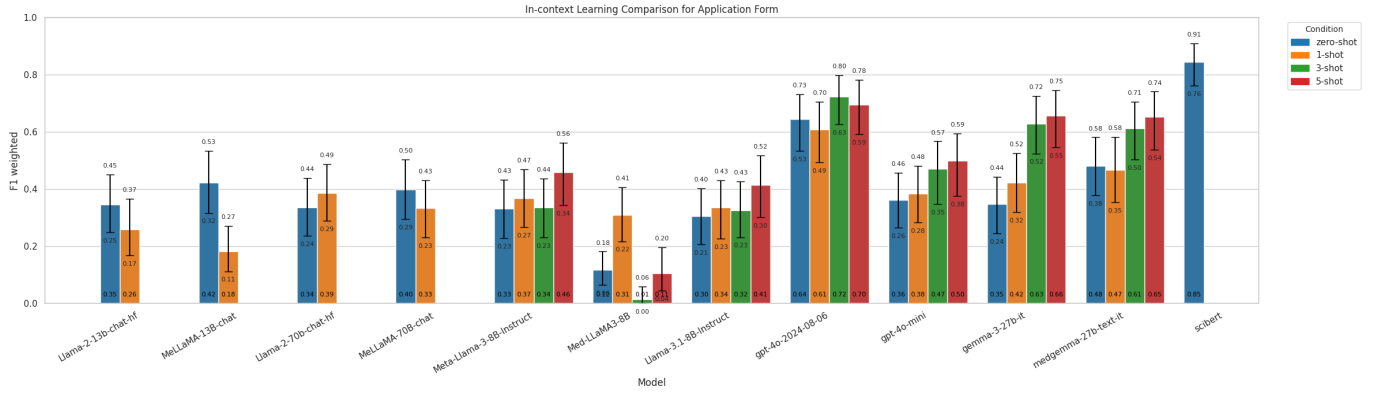
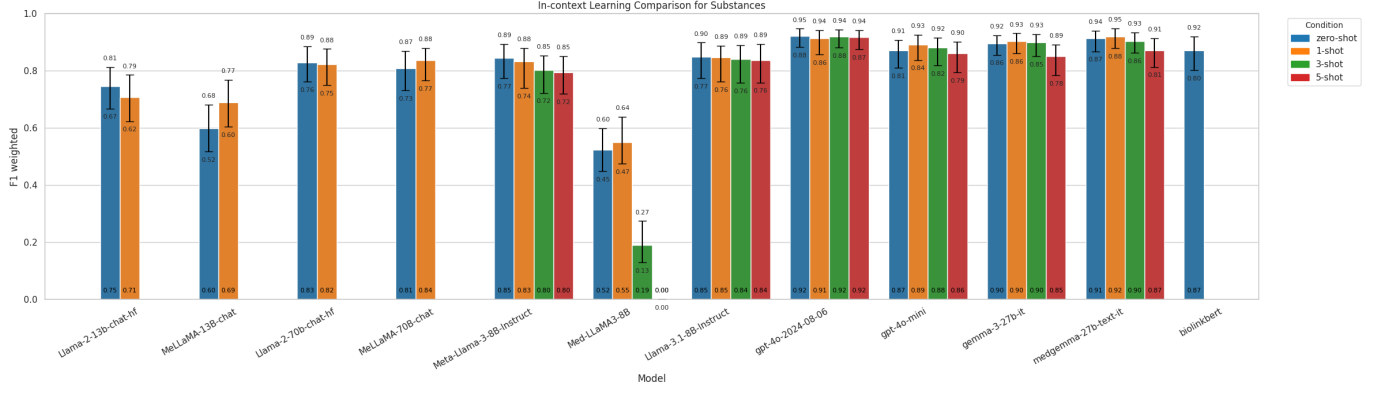
Appendix C. Few-shot Experiments



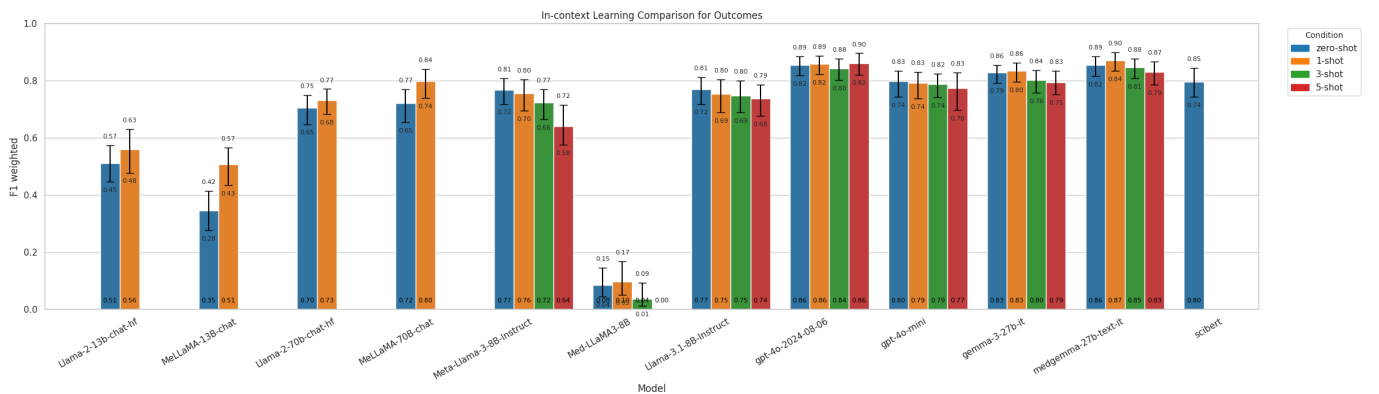
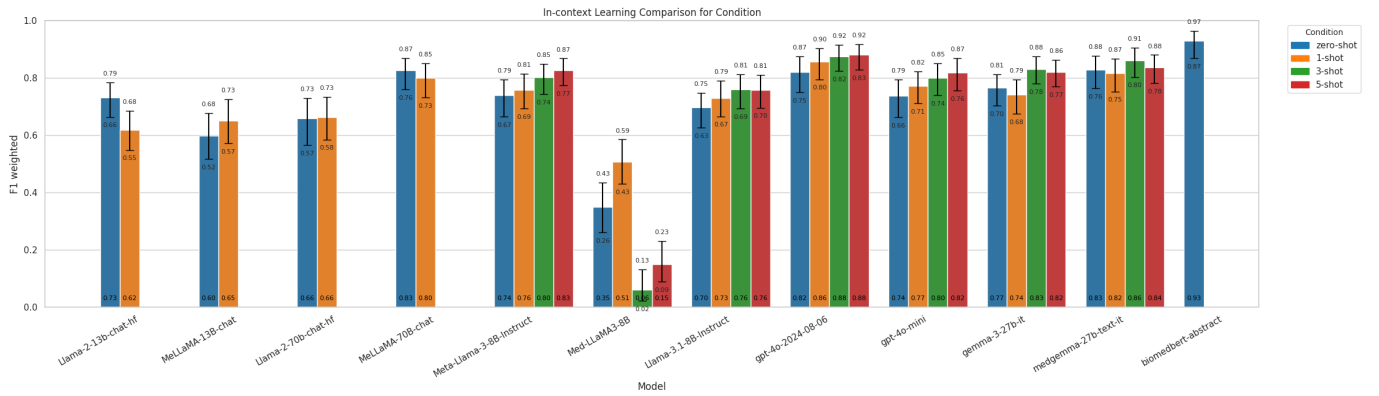
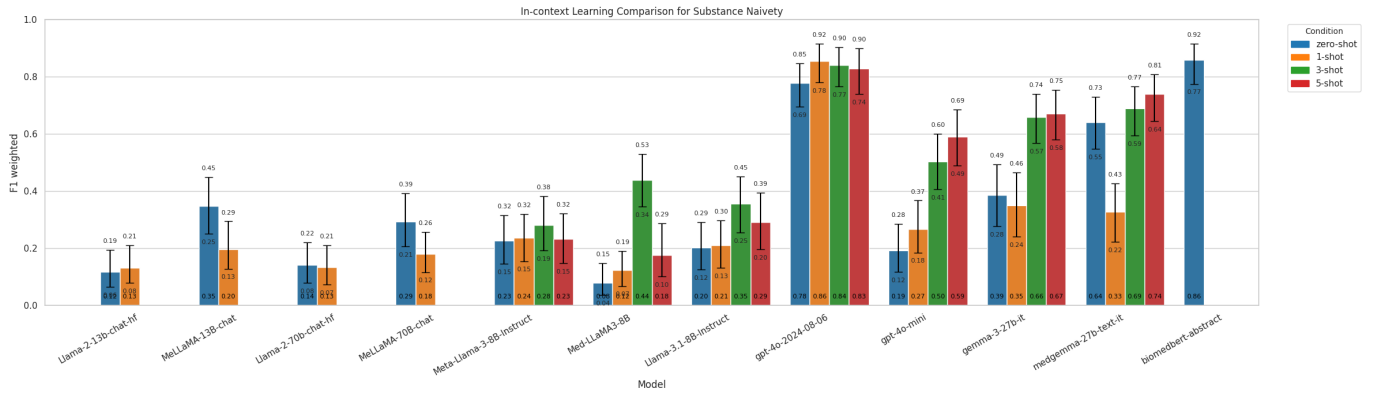
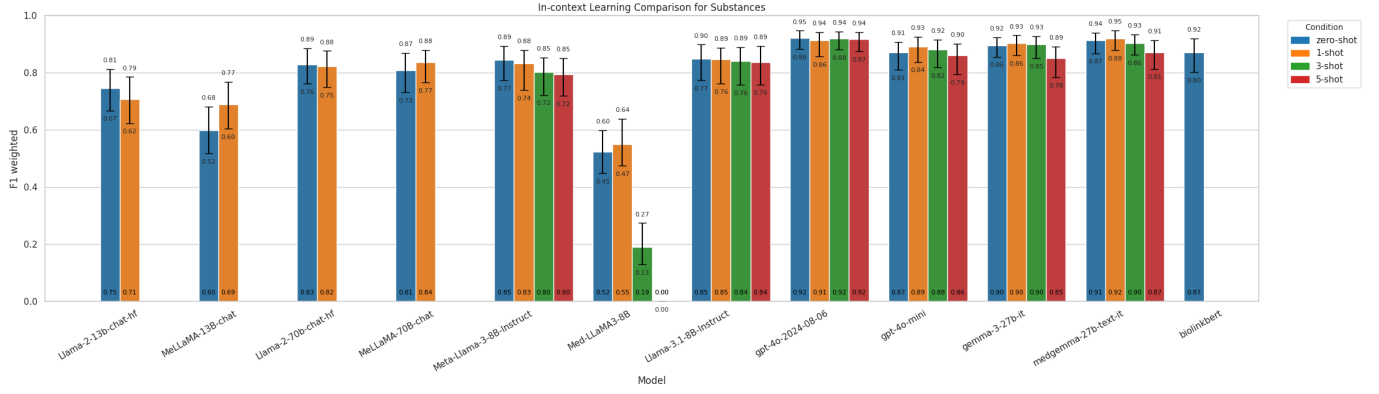
Appendix C. Few-shot Experiments



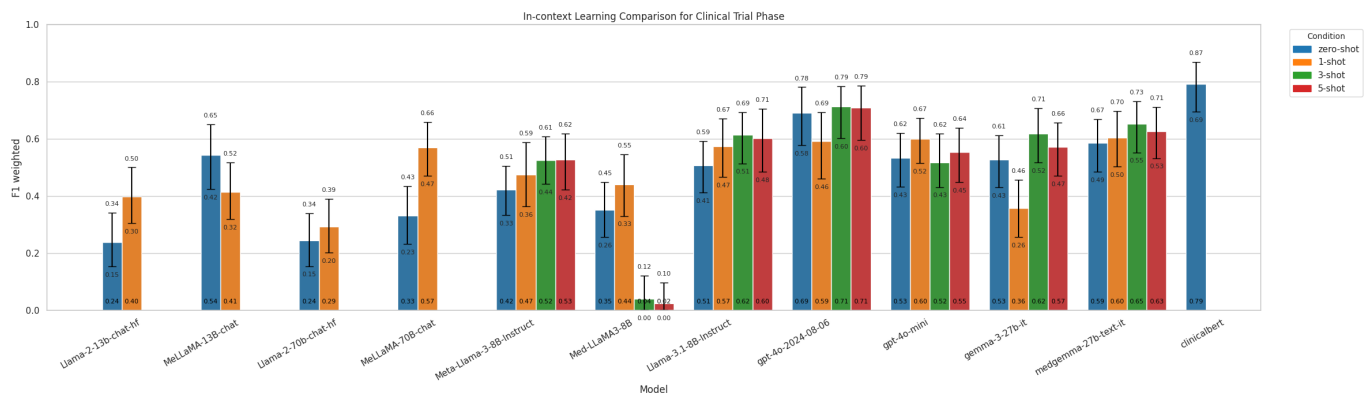
Appendix C. Few-shot Experiments



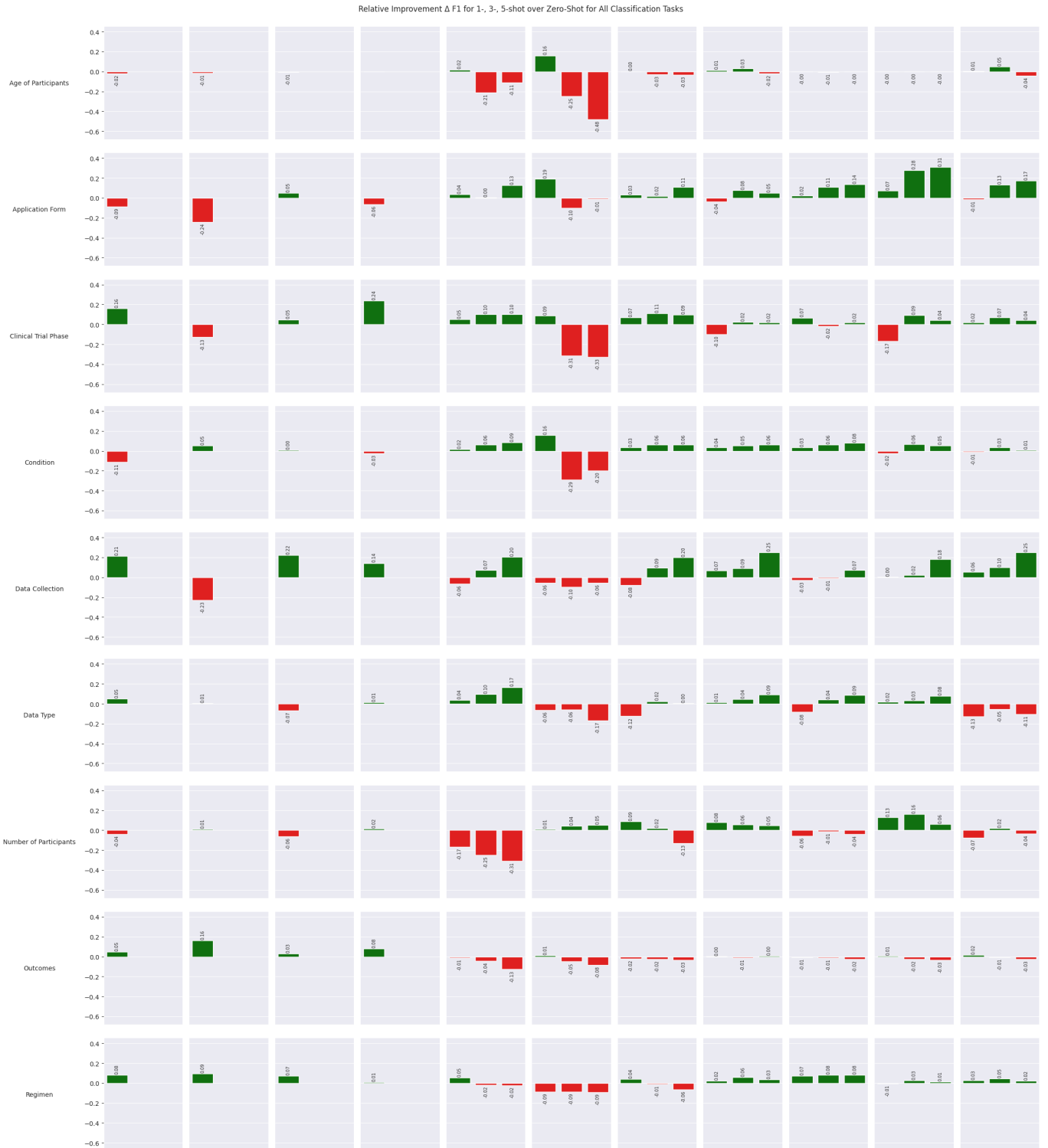
Appendix C. Few-shot Experiments



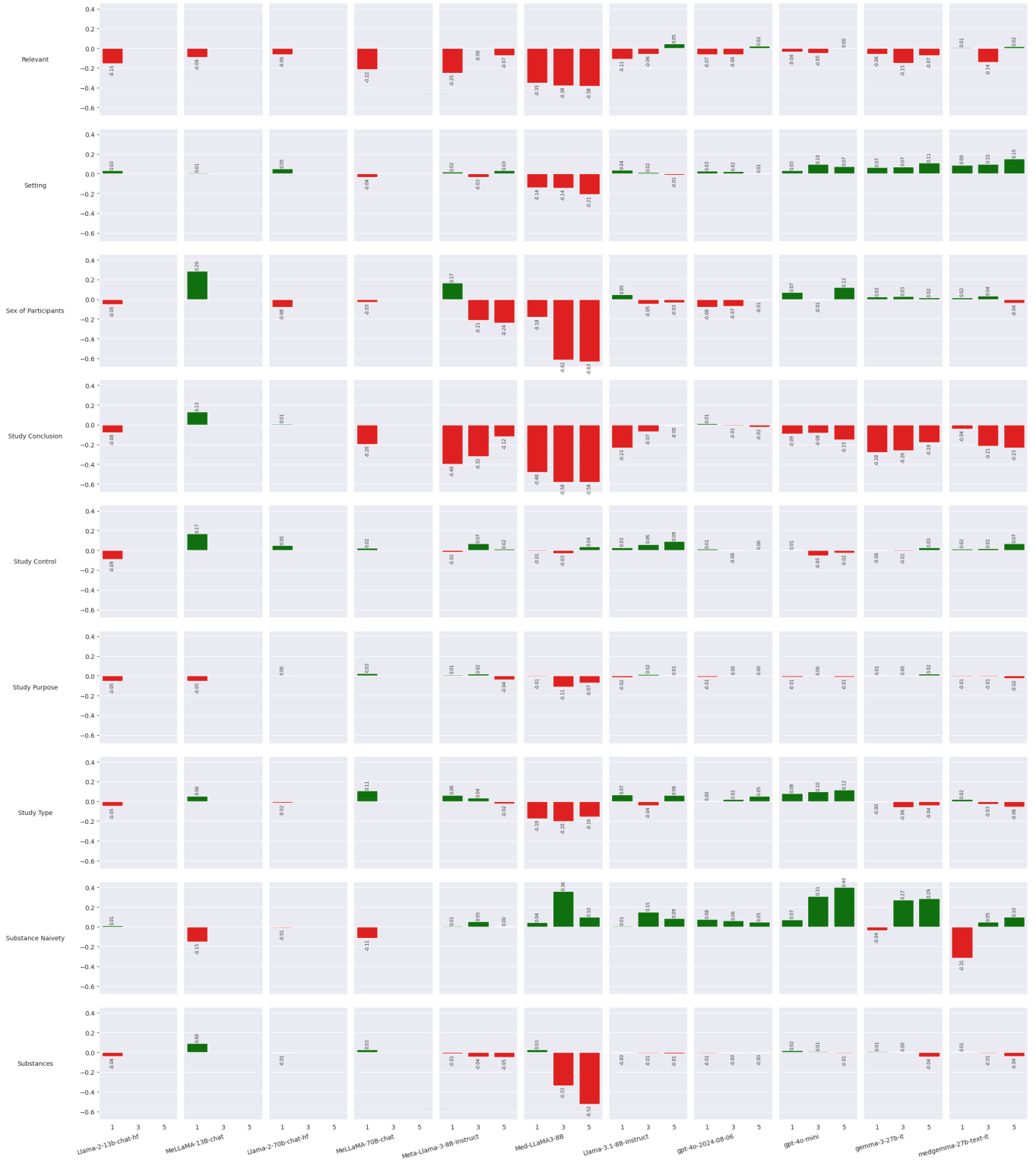
Appendix C. Few-shot Experiments



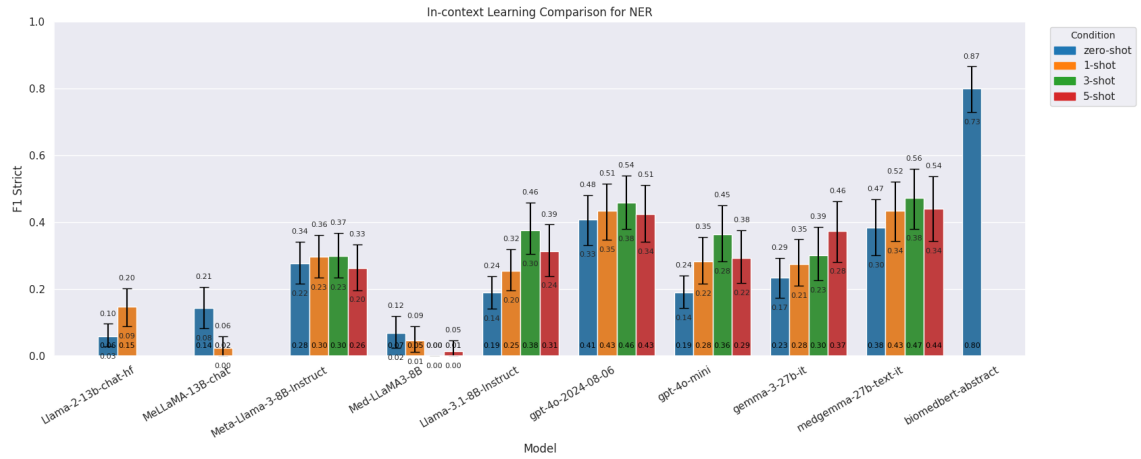
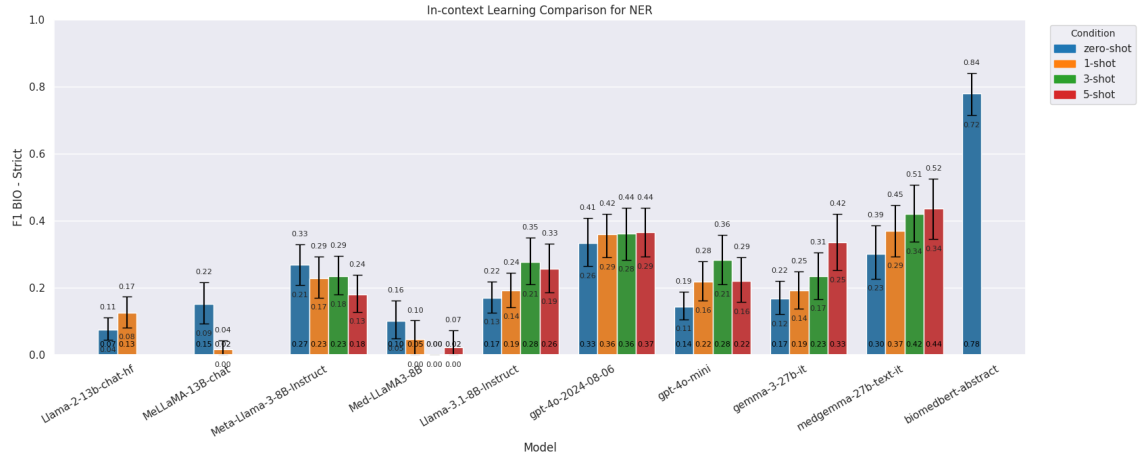
C.1.2. Relative Improvement Per Classification Task



Appendix C. Few-shot Experiments



C.2. Named Entity Recognition



D. IFT Experiment: Hyperparameters

Category	Hyperparameter
Model and Tokenizer	Base model: meta-llama/Llama-3.1-8B-Instruct Tokenizer: Fast tokenizer, EOS token used as padding
Quantization	Precision: 4-bit (NF4) Double quantization: Enabled LoRA parameters precision: 16-bit
LoRA	Rank (r): 64 Scaling factor (α): 16 Dropout: 0.1 Bias: None Task type: Causal LM
Training	Epochs: 3 Batch size per device: 8 Gradient accumulation steps: 4 Effective batch size: 32 Learning rate: 1×10^{-4} Mixed precision: FP16 Loss: Assistant-only / completion-only Checkpointing: End of each epoch Logging steps: Every 100 steps
Reproducibility	Random seed: 42 CuDNN deterministic: Enabled

Table D.1.: Complete set of hyperparameters for IFT of Llama-3.1-8B-Instruct using QLoRA.

E. LST Experiment: Hyperparameters

Category	Hyperparameter
Model and Tokenizer	Base model: meta-llama/Llama-3.1-8B-Instruct Tokenizer: Fast tokenizer, EOS token used as padding
LoRA	Rank (r): 32 Scaling factor (α): 32 Dropout: 0.1 Task type: Token classification Adapter precision: 16-bit
Training	Epochs: 5 Batch size per device: 4 Gradient accumulation: 1 Learning rate: 1×10^{-4} Weight decay: 0.01 Max sequence length: 1200 Mixed precision: FP16 Loss: Cross-entropy over first token of each word Checkpointing: End of each epoch
Reproducibility	Random seed: 42 CuDNN deterministic: Enabled

Table E.1.: Complete set of hyperparameters for IFT of Llama-3.1-8B-Instruct using LoRA.

F. Declaration of Contribution

The following lists which parts of this work were completed as part of my employment within the *PsyNamic* project and which were completed independently for this master’s thesis.

- **Employment (PsyNamic Project)**
 - **PsyNamic Dataset**
 - * Data collection
 - * Creation of data splits
 - * Basic data analysis for classification tasks
 - **BERT Baseline**
 - * Fine-tuning of BERT models for classification tasks
 - * Fine-tuning of BERT models for NER
 - * Evaluation of BERT models for classification tasks
- **Thesis Work**
 - **PsyNamic Dataset**
 - * Detailed dataset analysis for classification tasks
 - * Detailed dataset analysis for NER tasks
 - **BERT Baseline**
 - * Evaluation of BERT models for NER
 - * Retraining BERT models using chunked samples (previously, longer than 512 token samples were simply truncated)
 - * Retraining BERT models for *Study Type* and *Study Conclusion* as task definition changed from mutliclass, multilabel to multiclass only

- **Everything related to LLM Experiments**

G. Use of LLMs

Prompts To formulate the task description based on the annotation guidelines used for the ICL and IFT experiments, ChatGPT¹⁵ was used to shorten and summarize.

Writing No content of this thesis has been generated by an LLM. However, various ChatGPT¹⁶ models have been consulted for corrective purposes such as checking for grammatical mistakes and to find synonyms or more fitting alternative formulations. Also, since the authors of this thesis are not English natives, DeepL¹⁷ was occasionally used to translate from German to English. Lastly, this thesis was written in LaTeX using the platform Overleaf¹⁸ which comes with free limited usages of its own AI assist tool¹⁹. This tool is mostly used to debug latex errors but also used to correct spelling and grammatical errors. In sum, this thesis therefore complies with the Guidelines on the Use of Text Generation Models for Seminar Papers and Theses published by the Department of Computational Linguistics.

Code GitHub Copilot²⁰ was enabled in the coding environment used for this thesis. GitHub Copilot and ChatGPT²¹ models were used for debugging purposes.

¹⁵ <https://chatgpt.com>

¹⁶ <https://chatgpt.com>

¹⁷ <https://www.deepl.com/en/translator>

¹⁸ <https://www.overleaf.com>

¹⁹ <https://www.overleaf.com/about/ai-features>

²⁰ <https://github.com/features/copilot>

²¹ <https://chatgpt.com>



Declaration of Independent Authorship

Original work

I expressly declare that the written work I submitted to the University of Zurich in the spring/autumn semester of 2025 with the title

Can LLMs Beat BERT in Biomedical Information Extraction?
Evaluating Prompting and Fine-Tuning Strategies for NER and Classification

is an original work written by myself, in my own words, and without unauthorized assistance. If it is a work by several authors, I confirm that the relevant parts of the work are correctly and clearly marked and can be clearly assigned to the respective author.

I also confirm that the work has not been submitted in whole or in part to receive credit for another module at the University of Zurich or another educational institution, nor will it be submitted in the future.

Use of sources

I expressly declare that I have identified all references to external sources (including tables, graphics etc.) contained in the above work as such. In particular, I confirm that, without exception and to the best of my knowledge, I have indicated the authorship both for verbatim statements (citations) and for statements by other authors reproduced in my own words (paraphrases).

Use of text generation models

I expressly declare that I have not only identified existing external sources, but also any automatically generated text that is contained in the above work. I have used the same citation style as if the text had been generated by a human to indicate the source of the automatically generated text. If the contribution of text generation models cannot be linked to specific text passages (see the associated guidelines), I have included a chapter describing the contributions of the text generation model. I acknowledge that no explicit citation is necessary where text generation models are merely used correctively (to improve grammar or idiomaticity of my own words).

Sanctions

I acknowledge that a thesis that is used to acquire credit and proves to be plagiarism with the meaning of the document [Erläuterung des Begriffs „Plagiat“](#) leads to a grade



deduction in minor cases, a grade 1 (one) in more severe cases, without the possibility of revision, and in very severe cases can have the corresponding legal and disciplinary consequences according to §§ 7ff of the "Disziplinarordnung der Universität Zürich" and § 36 of the "Rahmenordnung für das Studium in den Bachelor- und Master-Studiengängen der Philosophischen Fakultät der Universität Zürich".

I confirm with my signature that this information is correct:

Name: Bernhard

First Name: Vera Lara

Matriculation number: 17-729-583

Date: 01.12.2025

Signature:

A handwritten signature in black ink, appearing to be 'Bernhard', written over a horizontal line.